EGGs are adhesive!
An algebraic account of equality saturation

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Abstract

The use of rewriting-based visual formalisms is on the rise. In the formal methods community, this is partly due to the introduction of adhesive categories, where most properties of classical approaches to graph transformation, such as those on parallelism and confluence, can be rephrased and proved in a general and uniform way. E-graphs (EGGs) are a formalism for program optimisation via an efficient implementation of equality saturation. In short, EGGs are (acyclic) term graphs with an additional equivalence on nodes that is closed under the operators of the signature. Instead of replacing the components of a program, the optimisation step adds new components and links them to the existing ones via an equivalence relation, until an optimal program is reached. This work describes EGGs via adhesive categories. Besides the benefits in itself of a formal presentation, which renders the properties of the data structure precise, the description of the addition of equivalent program components using standard graph transformation tools offers the advantages of the adhesive framework in modelling, for example, concurrent updates.

Keywords: Hypergraphs, terms graphs, e-graphs, adhesive categories.

1. Introduction

The introduction of *adhesive categories* marked a watershed moment for the algebraic approaches to the rewriting of graph-like structures [28, 16]. Until then, key results of the approaches on e.g. parallelism and confluence had to be proven over and over again for each different formalism at hand, despite the obvious similarity of the procedure. Adhesive categories provides such a disparate set of formalisms with a common abstract framework where many general results can be recast and uniformly proved once and for all.

In short, following the double-pushout (DPO) approach to graph transformation [14, 16], a rule is given by two arrows $l : K \to L$ and $r : K \to R$ and its application requires a match $m : L \to G$: the rewriting step from G to H is then given by the diagram aside, whose halves are pushouts.

$$\begin{tikzcd} L \arrow[d, "m"'] & K \arrow[l, "l"'] \arrow[r, "r"] \arrow[d, "k"'] & R \arrow[d, "h"] \\ G & C \arrow[l, "f"] \arrow[r, "g"'] & H \end{tikzcd}$$

Thus, L and R are the left- and right-hand side of the rule, respectively, while K witnesses those parts that must be present for the rule to be executed, yet that are not affected by the rule itself. If a category is adhesive, and the arrows of the rules are monomorphisms, the presence of a match ensures that if the pushout complement $C \to G$ exists then it is essentially unique, hence a rewriting step can be deterministically performed. The theory of $\mathcal{M}$-adhesivity [2, 24] extends the core framework, ensuring that if the arrows of the rules are in a suitable class $\mathcal{M}$ of monomorphisms then the benefits of adhesivity can be recovered [17, 18]. If only the left-hand side belongs to $\mathcal{M}$, the theory is still under development: It has been addressed for the as witnessed e.g. by [3, 5]. However, despite the elegance and effectiveness, proving that

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a category satisfies the conditions for being $\mathcal{M}$-adhesive can be a daunting task. For this reason, sufficient criteria have been provided for the core framework, e.g. that every elementary topos is adhesive [27], as well as for $\mathcal{M}$-adhesivity [12]. For structures such as *hypergraphs with equivalence* in [4], the question of their $\mathcal{M}$-adhesivity has not yet been settled.

E-graphs (shortly, EGGs) are an up-and-coming formalism for program optimisation and synthesis via a compact representation and efficient implementation of equality saturation. Albeit being a classical data structure [15], EGGs received new impulse after the seminal [36] and developed a thriving community, as witnessed by the official website [34]. The key idea of rewriting-based program optimisation is to perform the manipulation of a syntactical description of a program, replacing some of its components in such a way that the semantics is preserved while the computational costs of its actual execution are improved. Instead of directly removing sub-programs, EGGs just add the new components and link them to the older ones via the equivalence relation, until an optimal program is reached and extracted.

EGGs can be concisely defined as term graphs equipped with an equivalence on nodes that is closed under the operators of a signature [15, Section 4.2]. In the presentation of term graphs via string diagrams [12], EGGs are (hyper)trees whose edges are labelled by operators and with the possibility of sharing subtrees, with an equivalence relation $\equiv$ on nodes that is closed under composition. In plain words and using a toy example: if a and b are two constants such that $a \equiv b$, then $f(a) \equiv f(b)$ for every unary operator f.

Building on the criterion developed in [12], this work investigate $\mathcal{M}$-adhesivity for EGGs. The path proceeds by specialisation, starting from standard hypergraphs and exploring a family of categories of hypergraphs with equivalence, proving along the way that they form an $\mathcal{M}$-adhesive category for suitable choices of $\mathcal{M}$. The advantages from this approach are two-fold. On the one side, we put the benefits *per se* of a formal presentation, making precise the properties of the data structures and the correspondence between them. Given the interest in hypergraphical structures, the intermediate results seems of independent interest beyond the core proofs for EGGs. On the other side, describing the optimisation steps via the DPO approach offers the tools for modelling parallel and concurrent execution and for proving confluence and termination.

Synopsis The paper has the following structure. In Section 2 we recall the theory of $\mathcal{M}$-adhesive categories and of kernel pairs. In Section 3 we present the graphical structures of our interest, namely various categories of (labelled) hypergraphs, and we provide a functorial characterisation, which allows for proving their adhesivity properties. This is expanded in Section 4 for proving the $\mathcal{M}$-adhesivity of hypergraphs with equivalence and in Section 5, where equivalences are closed with respect to operator application, thus subsuming EGGs. In Section 6 we put the machinery to work, showing how the optimisation steps can be rephrased as the application of DPO rewriting rules, at the same time highlighting the issues linked to the format of such rules. Finally, in Section 7 we draw our conclusions, hint at future endeavours and offer some remarks on related works.

For the sake of readability, all the proofs appear in the appendices. This work is an extended version of [6]. Its overall structure reflects the one of the conference paper, even if we found technically convenient to present term graphs in a different, yet equivalent way as an instance of the newly introduced category of left-monogamous hypergraphs. Also the notion of separated hypergraphs is original, thus Section 3.3 and Section 3.4 are largely new, and consequently so Section 4 and Section 5. Finally, also Section 6 on the use of DPO rewriting has been rewritten and expanded, trying to highlight the nuances concerning its application.

2. Facts about $\mathcal{M}$-adhesive categories and kernel pairs

We open this background section by fixing notation. Given a category $\mathbf{X}$ we do not distinguish notationally between $\mathbf{X}$ and its class of objects, so "$X \in \mathbf{X}$" means that X is an object of $\mathbf{X}$. We let $\mathsf{Mor}(\mathbf{X})$, $\mathsf{Mono}(\mathbf{X})$ and $\mathsf{Reg}(\mathbf{X})$ denote the class of all arrows, monos and regular monos of $\mathbf{X}$, respectively. Given an object X, $?_X$ denotes the unique arrow from an initial object into X and by $!_X$ that unique arrow from X into a terminal one. We also use the notation $e\colon X \twoheadrightarrow Y$ to denote that an arrow $e\colon X \to Y$ is a regular epi.

2.1. $\mathcal{M}$-adhesivity

The key property of $\mathcal{M}$-adhesive categories is the *Van Kampen condition* [8, 25, 28], and for defining it we need some notions. Let $\mathbf{X}$ be a category. A subclass $\mathcal{A}$ of $\mathsf{Mor}(\mathbf{X})$ is said to be

$$
\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "m"'] & B \arrow[d, "n"] \\
C \arrow[r, "g"'] & D
\end{tikzcd}
$$

- *stable under pushouts (pullbacks)* if for every pushout (pullback) square as the one aside, if $m \in \mathcal{A}$ ($n \in \mathcal{A}$) then $n \in \mathcal{A}$ ($m \in \mathcal{A}$);
- *closed under composition* if $h, k \in \mathcal{A}$ implies $h \circ k \in \mathcal{A}$ whenever h and k are composable;
- *closed under decomposition*, or *left-cancellable*, if whenever g and $g \circ f$ belong to $\mathcal{A}$, then $f \in \mathcal{A}$.

Definition 2.1. Let $\mathcal{A} \subseteq \mathsf{Mor}(\mathbf{X})$ be a class of arrows in a category $\mathbf{X}$ and consider the cube aside. The bottom square is an $\mathcal{A}$-*Van Kampen square* if

$$
\begin{tikzcd}[row sep=small, column sep=small]
& A' \arrow[rr, "f'"] \arrow[dl, "m'"'] \arrow[dd, "a" near start] & & B' \arrow[dl, "n'"'] \arrow[dd, "b"] \\
C' \arrow[rr, "g'" near start, crossing over] \arrow[dd, "c"'] & & D' & \\
& A \arrow[rr, "f" near start] \arrow[dl, "m"] & & B \arrow[dl, "n"] \\
C \arrow[rr, "g"'] & & D \arrow[from=uu, "d" near start, crossing over] &
\end{tikzcd}
$$

1. it is a pushout square;
2. whenever the cube above has pullbacks as back and left faces and the vertical arrows belong to $\mathcal{A}$, then its top face is a pushout if and only if the front and right faces are pullbacks.

Pushout squares that enjoy only the "if" half of item (2) above are called $\mathcal{A}$-*stable*. A $\mathsf{Mor}(\mathbf{X})$-Van Kampen square is called *Van Kampen* and a $\mathsf{Mor}(\mathbf{X})$-stable square *stable*.

We can now define $\mathcal{M}$-adhesive categories.

Definition 2.2 ([2, 17, 18, 28, 24])**.** Let $\mathbf{X}$ be a category and $\mathcal{M}$ a subclass of $\mathsf{Mono}(\mathbf{X})$ including all isos, closed under composition, decomposition, and stable under pullbacks and pushouts. The category $\mathbf{X}$ is said to be $\mathcal{M}$-*adhesive* if

1. it has $\mathcal{M}$-*pullbacks*, i.e. pullbacks along arrows of $\mathcal{M}$;
2. it has $\mathcal{M}$-*pushouts*, i.e. pushouts along arrows of $\mathcal{M}$;
3. $\mathcal{M}$-pushouts are $\mathcal{M}$-Van Kampen squares.

A category $\mathbf{X}$ is said to be *strictly* $\mathcal{M}$-*adhesive* if $\mathcal{M}$-pushouts are Van Kampen. We write $m\colon X \rightarrowtail Y$ to denote that an arrow $m\colon X \to Y$ belongs to $\mathcal{M}$.

Remark 2.3. Our notion of $\mathcal{M}$-adhesive category corresponds to what in [16] is called *weak adhesive HLR category*, while a strict $\mathcal{M}$-adhesive categories corresponds to *adhesive HLR* ones. Finally, *adhesivity* and *quasiadhesivity* [28, 20] coincide with strict $\mathsf{Mono}(\mathbf{X})$-adhesivity and strict $\mathsf{Reg}(\mathbf{X})$-adhesivity, respectively.

$\mathcal{M}$-adhesivity is well-behaved with respect to the construction of (co-)slice and functor categories [29], as well with respect to subcategories, as shown by the following properties, taken from [16, Thm. 4.15], [28, Prop. 3.5] and [12, Thm. 2.12].

Theorem 2.4. *Let $\mathbf{X}$ be an (strict) $\mathcal{M}$-adhesive category. Then the following facts hold*

1. *if $\mathbf{Y}$ is an (strict) $\mathcal{N}$-adhesive category, $L\colon \mathbf{Y} \to \mathbf{A}$ a functor preserving $\mathcal{N}$-pushouts and $R\colon \mathbf{X} \to \mathbf{A}$ one preserving $\mathcal{M}$-pullbacks, then $L{\downarrow}R$ is (strictly) $\mathcal{N}{\downarrow}\mathcal{M}$-adhesive, where*
$$\mathcal{N}{\downarrow}\mathcal{M} := \{(h, k) \in \mathsf{Mor}(L{\downarrow}R) \mid h \in \mathcal{N}, k \in \mathcal{M}\}$$
2. *for every object X the categories $\mathbf{X}/X$ and $X/\mathbf{X}$ are, respectively, (strictly) $\mathcal{M}/X$-adhesive and (strictly) $X/\mathcal{M}$-adhesive, where*
$$\mathcal{M}/X := \{m \in \mathsf{Mor}(\mathbf{X}/X) \mid m \in \mathcal{M}\} \qquad X/\mathcal{M} := \{m \in \mathsf{Mor}(X/\mathbf{X}) \mid m \in \mathcal{M}\}$$
3. *for every small category $\mathbf{Y}$, the category $\mathbf{X}^{\mathbf{Y}}$ of functors $\mathbf{Y} \to \mathbf{X}$ is (strictly) $\mathcal{M}^{\mathbf{Y}}$-adhesive, where $\mathcal{M}^{\mathbf{Y}} := \{\eta \in \mathsf{Mor}(\mathbf{X}^{\mathbf{Y}}) \mid \eta_Y \in \mathcal{M}$ for every $Y \in \mathbf{Y}\}$;*
4. *if $\mathbf{Y}$ is a full subcategory of $\mathbf{X}$ closed under pullbacks and $\mathcal{M}$-pushouts, then $\mathbf{Y}$ is (strictly) $\mathcal{N}$-adhesive for every class of arrows $\mathcal{N}$ of $\mathbf{Y}$ contained in $\mathcal{M}$ that is stable under pullbacks and pushouts, contains all isos, and is closed under composition and decomposition.*

We briefly list some examples of $\mathcal{M}$-adhesive categories.

Example 2.5. **Set** is adhesive and, in fact, every topos is adhesive [27]. By the closure properties above, every presheaf $[\mathbf{X}, \mathbf{Set}]$ is adhesive, thus the category $\mathbf{Graph} = [E \rightrightarrows V, \mathbf{Set}]$ is adhesive where $E \rightrightarrows V$ is the two objects category with two morphisms $s, t \colon E \to V$. Similarly, various categories of hypergraphs can be shown to be adhesive, such as term graphs and hierarchical graphs [12]. Note that the category **sGraphs** of simple graphs, i.e. graphs without parallel edges, is $\mathsf{Reg}(\mathbf{sGraphs})$-adhesive [5] but not quasiadhesive.

We can state some useful properties of $\mathcal{M}$-adhesive category (see, for instance, [16, Thm. 4.26] or [2, Fact 2.6]). A proof is provided in Appendix A.1.

Proposition 2.6. *Let $\mathbf{X}$ be an $\mathcal{M}$-adhesive category. Then the following facts hold*

1. *every $\mathcal{M}$-pushout square is also a pullback;*
2. *every arrow in $\mathcal{M}$ is a regular mono.*

2.2. Some properties of kernel pairs and regular epimorphisms

In this section we recall the definition and some properties of *kernel pairs.*

Definition 2.7. A *kernel pair* for an arrow $f \colon A \to B$ is an object K_f together with two arrows $\pi^1_f, \pi^2_f \colon K_f \rightrightarrows A$, denoted as (K_f, π^1_f, π^2_f), such that the square aside is a pullback.

$$\begin{tikzcd} K_f \arrow[r, "\pi^2_f"'] \arrow[d, "\pi^1_f"'] & A \arrow[d, "f"] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

Remark 2.8. If (K_f, π^1_f, π^2_f) is a a kernel pair for $f \colon X \to Y$ and a product of X with itself exists, then the canonical arrow $\langle \pi^1_f, \pi^2_f \rangle \colon K_f \to X \times X$ is a mono.

Remark 2.9. An arrow $m \colon M \to X$ is a mono if and only if it admits $(M, \mathsf{id}_M, \mathsf{id}_M)$ as a kernel pair.

Together with Lemma A.1, the previous remarks allow us to prove the following result.

Proposition 2.10. *Let $f \colon X \to Y$ be an arrow and $m \colon Y \to Z$ a mono. If (K_f, π^1_f, π^2_f) is a kernel pair for $f \colon X \to Y$, then it is also a kernel pair for $m \circ f$.*

Regular epis are particular well-behaved with respect to their kernel pairs.

Proposition 2.11. *Let $e \colon X \twoheadrightarrow Y$ be a regular epi in a category $\mathbf{X}$ with a kernel pair (K_e, π^1_e, π^2_e). Then e is the coequalizer of π^1_e and π^2_e.*

We conclude this section exploring some properties of kernel pairs in an $\mathcal{M}$-adhesive category. The results below are simple, yet they appear to be original, and we give their proofs in Appendix A.1.

Lemma 2.12. *Let $f \colon X \to Y$ and $g \colon Z \to W$ be arrows admitting kernel pairs and suppose that the solid part of the four squares below is given. If the leftmost square is commutative, then there is a unique arrow $k_h \colon K_f \to K_g$ making the other three commutative.*

$$\begin{tikzcd} X \arrow[r, "h"] \arrow[d, "f"'] & Z \arrow[d, "g"] \\ Y \arrow[r, "t"'] & W \end{tikzcd} \qquad \begin{tikzcd} K_f \arrow[r, dotted, "k_h"] \arrow[d, "\pi^1_f"'] & K_g \arrow[d, "\pi^1_g"] \\ X \arrow[r, "h"'] & Z \end{tikzcd} \qquad \begin{tikzcd} K_f \arrow[r, dotted, "k_h"] \arrow[d, "\pi^2_f"'] & K_g \arrow[d, "\pi^2_g"] \\ X \arrow[r, "h"'] & Z \end{tikzcd} \qquad \begin{tikzcd} K_f \arrow[r, dotted, "k_h"] \arrow[d, "{(\pi^1_f, \pi^2_f)}"'] & K_g \arrow[d, "{(\pi^1_g, \pi^2_g)}"] \\ X \times X \arrow[r, "h \times h"'] & Z \times Z \end{tikzcd}$$

Moreover, the following facts hold

1. *if h is a mono then k_h is a mono;*
2. *if the leftmost square is a pullback then the central two are pullbacks;*
3. *if h is mono and the leftmost square is a pullback then the rightmost is a pullback.*

The previous result allows us to deduce the following lemma in an $\mathcal{M}$-adhesive context.

Proposition 2.13. *Let $\mathbf{X}$ be a strict $\mathcal{M}$-adhesive category with all pullbacks, and suppose that in the cube aside the top face is an $\mathcal{M}$-pushout and all the vertical faces are pullbacks. Then the right square is a pushout.*

$$\begin{tikzcd} & A' \arrow[rr, "f'"] \arrow[dl, tail, "m'"'] \arrow[dd, "a"' near end] & & B' \arrow[dl, tail, "n'"'] \arrow[dd, "b"] \\ C' \arrow[rr, "g'" near start, crossing over] \arrow[dd, "c"'] & & D' & \\ & A \arrow[rr, "f" near end] \arrow[dl, "m"'] & & B \arrow[dl, "n"] \\ C \arrow[rr, "g"] & & D \arrow[from=uu, "d" near start, crossing over] & \end{tikzcd} \qquad \begin{tikzcd} K_a \arrow[r, "k_{f'}"] \arrow[d, "k_{m'}"'] & K_b \arrow[d, "k_{n'}"] \\ K_c \arrow[r, "k_{g'}"] & K_d \end{tikzcd}$$

Focusing on **Set**, we can prove a sharper result (see Appendix A.1 for the proof).

Lemma 2.14. *Suppose that in* **Set** *the commuting cube in the diagram on the left is given, whose top face is a pushout, the left and bottom faces are pullbacks, and* $n\colon B \rightarrowtail D$ *is an injection. Then the following facts hold*

$$
\begin{tikzcd}[row sep=small, column sep=small]
& A' \arrow[rr, "f'"] \arrow[dl, tail, "m'"'] \arrow[dd, "a"' near end] & & B' \arrow[dl, tail, "n'"'] \arrow[dd, "b"] \\
C' \arrow[rr, "g'" near start, crossing over] \arrow[dd, "c"'] & & D' & \\
& A \arrow[rr, "f" near start] \arrow[dl, tail, "m"] & & B \arrow[dl, tail, "n"] \\
C \arrow[rr, tail, "g"'] & & D \arrow[from=uu, crossing over, "d"' near start] &
\end{tikzcd}
\qquad
\begin{tikzcd}
K_a \arrow[r, "k_{f'}"] \arrow[d, "k_{m'}"'] & K_b \arrow[d, "k_{n'}"] \\
K_c \arrow[r, "k_{g'}"'] & K_d
\end{tikzcd}
$$

1. *the right face of the cube is a pullback;*
2. *the right square, made by the kernel pairs of the vertical arrows, is a pushout.*

3. Hypergraphical structures

In this section we introduce a family of categories of hypergraphs, Most of them have been already considered in the literature, and we add the relevant references inline. Our approach offers a uniform treatment, though, and proves original results concerning their adhesivity.. Most importantly, the presentation is tailored for introducing the novel categories of hypergraphs with equivalence and EGGs in Section 4 and Section 5, respectively. However, given the many different flavours of hypergraphs we need, it is worthwhile to give a visual map of the categories that we are going to discuss and of their relationships. We use the curved arrows to denote full and faithful inclusions, while the vertical arrows are forgetful functors.

$$
\begin{tikzcd}[column sep=small]
& \mathbf{pSep}_{\mathcal{G}} \arrow[rr, hook, bend right=10, "S_{p,\mathcal{G}}"'] \arrow[dl, hook, "I_{\mathcal{G}}"'] \arrow[dd, "W_{\mathcal{G}}"' near end] & & \mathbf{pHyp}_{\mathcal{G}} \arrow[ll, bend right=10, "\mathsf{sp}_{\mathcal{G}}"'] \arrow[dr, hook, bend left=10, "P_{\mathcal{G}}"] \arrow[dd, "V_{\mathcal{G}}" near end] & \\
\mathbf{Sep}_{\mathcal{G}} \arrow[rr, hook, "S_{\mathcal{G}}"] \arrow[dd, "R_{\mathcal{G}}"'] & & \mathbf{LMH}_{\mathcal{G}} \arrow[rr, hook, "M_{\mathcal{G}}"] \arrow[dd, "O_{\mathcal{G}}"' near start] & & \mathbf{Hyp}_{\mathcal{G}} \arrow[ul, bend right=20, "\mathsf{pr}_{\mathcal{G}}"'] \arrow[dd, "U_{\mathcal{G}}"] \\
& \mathbf{pSep} \arrow[rr, hook, bend right=10, "S_p"'] \arrow[dl, hook, "I"'] & & \mathbf{pHyp} \arrow[ll, bend right=10, "\mathsf{sp}"'] \arrow[dr, hook, bend left=10, "P"] & \\
\mathbf{Sep} \arrow[rr, hook, "S"] & & \mathbf{LMH} \arrow[rr, hook, "M"] & & \mathbf{Hyp} \arrow[ul, bend right=20, "\mathsf{pr}"']
\end{tikzcd}
$$

The two bottom rows refer to categories of unlabelled structures. **Hyp** is the standard category of hypergraphs; **LMH** is the subcategory of left-monogamous hypergraphs, satisfying suitable constraints on the number of edges entering a node, and which are used here to present an alternative characterisation for term graphs; and **Sep** is the subcategory of *separated* hypergraphs, which are both left- and right-monogamous. Finally, the prefix p stands for *pruned*, meaning hypergraphs with no isolated nodes. The two top rows concern labelled hypergraphs, which are as usual obtained via the comma category construction.

3.1. Kleene star

To present the notion of *hypergraph*, a pivotal role is played by the *Kleene star monad* $(-)^{\star}\colon \mathbf{Set} \to \mathbf{Set}$, also known as *list monad*, sending a set to the free monoid on it [33, 35]. We recall some of its properties.

Proposition 3.1. *Let* X *be a set and* $n \in \mathbb{N}$*. Then the following facts hold*

1. *there are arrows* $v_n\colon X^n \rightarrowtail X^{\star}$ *such that* $(X^{\star}, \{v_n\}_{n\in\mathbb{N}})$ *is a coproduct;*
2. *for every arrow* $f\colon X \to Y$*,* $f^{\star}\colon X^{\star} \to Y^{\star}$ *is the coproduct of the family* $\{f^n\}_{n\in\mathbb{N}}$*;*
3. $(-)^{\star}$ *preserves all* connected limits *[9], in particular pullbacks and equalizers.*

Remark 3.2. Preservation of pullbacks implies that $(-)^{\star}$ sends monos to monos.

Remark 3.3. It is well known that $(\mathbb{N}, \{\delta_n\}_{n\in\mathbb{N}})$ is a coproduct for $\aleph_0$ copies of 1, where $\delta_n\colon 1 \to \mathbb{N}$ is the function picking the element n. Thus by the first point of Proposition 3.1 there is a canonical isomorphism $\Delta\colon \mathbb{N} \to 1^{\star}$. In particular, for every set X the unique arrow $!_{\mathbf{X}}\colon X \to 1$ into the terminal object induces a *length function* $\mathsf{lg}_X\colon X^{\star} \to \mathbb{N}$, which sends a word to its length, as in the diagram aside.

$$
\begin{tikzcd}
1 \arrow[r, "\mathsf{id}_1"] \arrow[d, "\delta_n"] & 1 \arrow[d, "v_n"] \\
\mathbb{N} \arrow[r, dotted, "\Delta"] & 1^{\star}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbb{N} \arrow[r, "\Delta"] & 1^{\star} \\
& X^{\star} \arrow[u, "!_X^{\star}"'] \arrow[ul, dotted, "\mathsf{lg}_X"]
\end{tikzcd}
$$

Remark 3.4. Consider a function $f\colon X \to Y$ and suppose that $(X^\star, \{x_n\}_{n\in\mathbb{N}})$ and $\{Y^\star, \{y_n\}_{n\in\mathbb{N}}\}$ are the coproducts given by the first point of Proposition 3.1. Then for every $n \in \mathbb{N}$ we have

$$\begin{aligned}\mathsf{lg}_Y \circ f^\star \circ x_n = \mathsf{lg}_Y \circ y_n \circ f^n = \Delta^{-1} \circ !_Y^\star \circ y_n \circ f^n = \Delta^{-1} \circ v_n \circ !_Y^n \circ f^n \\ = \Delta^{-1} \circ v_n \circ (!_Y \circ f)^n = \Delta^{-1} \circ v_n \circ (!_X)^n = \Delta^{-1} \circ !_X^\star \circ x_n = \mathsf{lg}_X \circ x_n\end{aligned}$$

and so $\mathsf{lg}_Y \circ f^\star = \mathsf{lg}_X$.

Remark 3.5. It is worth noticing that for X a set and every $n \in \mathbb{N}$, X^n is the preimage of n along lg_X. This amounts to say that the square aside is a pullback. This follows at once from the fact that **Set** is *extensive* (see [10] and also Appendix A.1).

$$\begin{tikzcd} X^n \arrow[r, tail, "v_n"] \arrow[d, "!_{X^n}"'] & X^\star \arrow[d, "\mathsf{lg}_X"] \\ 1 \arrow[r, tail, "\delta_n"] & \mathbb{N} \end{tikzcd}$$

Notation. Given a word $w \in X^\star$ of length l, for every $1 \le i \le l$, we denote by w_i the i^{th} letter of w. Formally, if $\mathsf{lg}_X(w) = l$, then $w = v_l(w')$ for some $w' \in X^l$ and w_i is $\pi_i(w')$, for $\pi_i\colon X^l \to X$ the i^{th}-projection.

3.2. *The category of hypergraphs*

Let us begin this section with the definition of hypergraphs [?].

Definition 3.6. An *hypergraph* is a 4-uple $\mathcal{G} := (E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G})$ made by two sets $E_\mathcal{G}$ and $V_\mathcal{G}$, called, respectively the sets, of *hyperedges* and *nodes*, plus a pair of *source* and *target* arrows $s_\mathcal{G}, t_\mathcal{G}\colon E_\mathcal{G} \rightrightarrows V_\mathcal{G}^\star$.

A *hypergraph morphism* $(E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G}) \to (E_\mathcal{H}, V_\mathcal{H}, s_\mathcal{H}, t_\mathcal{H})$ is a pair (h, k) of functions $h\colon E_\mathcal{G} \to E_\mathcal{H}$, $k\colon V_\mathcal{G} \to V_\mathcal{H}$ such that the diagrams on the right commute.

We define **Hyp** to be the resulting category.

$$\begin{tikzcd} E_\mathcal{G} \arrow[r, "s_\mathcal{G}"] \arrow[d, "h"'] & V_\mathcal{G}^\star \arrow[d, "k^\star"] \\ E_\mathcal{G} \arrow[r, "s_\mathcal{H}"] & V_\mathcal{H}^\star \end{tikzcd} \qquad \begin{tikzcd} E_\mathcal{G} \arrow[r, "t_\mathcal{G}"] \arrow[d, "h"'] & V_\mathcal{G}^\star \arrow[d, "k^\star"] \\ E_\mathcal{G} \arrow[r, "t_\mathcal{H}"] & V_\mathcal{H}^\star \end{tikzcd}$$

Let $\mathsf{prod}^\star$ be the functor $\mathbf{Set} \to \mathbf{Set}$ sending X to the product $X^\star \times X^\star$. We can use it to present **Hyp** as a comma category.

Proposition 3.7. Hyp *is isomorphic to* $\mathsf{id}_{\mathbf{Set}}{\downarrow}\mathsf{prod}^\star$

Note that by Proposition 3.1, $(-)^\star$ preserves pullbacks, while prod is continuous by definition, hence by Proposition 3.7 and Corollary B.5 we can deduce the following result.

Corollary 3.8. *A morphism* (h, k) *is a mono in* **Hyp** *if and only if its components are injective functions.*

Applying Theorem 2.4 we also get the next corollary (cf. [16, Fact 4.17]).

Corollary 3.9. Hyp *is an adhesive category.*

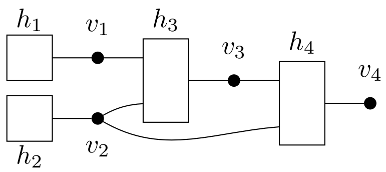
Propositons B.6 and 3.7 allow us to deduce immediately the following.

Proposition 3.10. *The forgetful functor* $U_{\mathbf{Hyp}}$ *which sends a hypergraph* $\mathcal{G}$ *to its set of nodes has a left adjoint* $\Delta_{\mathbf{Hyp}}$.

Example 3.11. Since the initial object of **Set** is the empty set, $\Delta_{\mathbf{Hyp}}(X)$ is the hypergraph which has X as set of nodes, $\emptyset$ as set of hyperedges, and $?_X$ as source and target function.

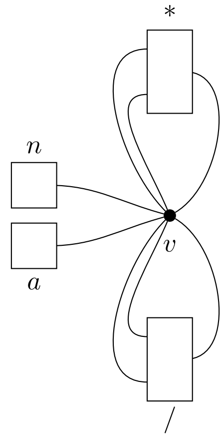
Example 3.12. We represent hypergraphs visually: bullets denote nodes and boxes hyperedges. Should we be interested in their identity, we put a name near the corresponding bullet or box. Sources and targets are represented by lines between dots and squares: the lines from the sources of a hyperedge comes from the left of the box, while the lines to the targets exit from the right of the box.

Consider, for instance, the picture below. It represents a hypergraph $\mathcal{G}$ with sets $V_\mathcal{G} = \{v_1, v_2, v_3, v_4\}$ and $E_\mathcal{G} = \{h_1, h_2, h_3, h_4\}$ of nodes and edges, respectively, such that h_1, h_2 have no source and h_3, h_4 have the lists v_1v_2 and v_3v_2 as sources, respectively.



A rich source of examples is given by *monoidal signatures*.

Definition 3.13. A *(single typed) monoidal signature* Σ is a triple $(O_\Sigma, \mathsf{ar}_\Sigma, \mathsf{co}_\Sigma)$ given by a *set of operations* O_Σ together with *arity* and *coarity* functions $\mathsf{ar}_\Sigma, \mathsf{co}_\Sigma \colon O_\Sigma \rightrightarrows \mathbb{N}$.
We say that Σ is *algebraic* if co_Σ is the constant function with value 1.
We define the *hypergraph* $\mathcal{G}_\Sigma$ *associated with* Σ taking O_Σ as set of hyperedges, 1 as set of nodes and $\Delta \circ \mathsf{ar}_\Sigma$ as the source function and $\Delta \circ \mathsf{co}_\Sigma$ as target function, where $\Delta \colon \mathbb{N} \to 1^\star$ is the isomorphism of Remark 3.3.



Example 3.14. Given a set A, let $\Sigma = (O_\Sigma, \mathsf{ar}_\Sigma, \mathsf{co}_\Sigma)$ be the algebraic signature with $O_\Sigma = \mathbb{N} + A + \{*, /\}$, and

$$\mathsf{ar}_\Sigma(n) = 0 \quad \mathsf{ar}_\Sigma(a) = 0 \quad \mathsf{ar}_\Sigma(*) = 2 \quad \mathsf{ar}_\Sigma(/) = 2$$

for every $n \in \mathbb{N}$ and $a \in A$. We depict the hypergraph $\mathcal{G}^\Sigma$ as in the picture aside.

We can interpret an arbitrary hypergaph as a *typed* or *colored* monoidal signature: the nodes provide the *types* or *colors*, while hyperedges can be thought of as operations, with arity and coarity provided by the source and target functions, respectively. This suggests the following definition, which is used later on.

Definition 3.15. A hypergraph $\mathcal{G} = (E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G})$ is *algebraic* if $t_\mathcal{G}(e)$ has length 1 for every $e \in E_\mathcal{G}$.

In order to deal with general limits of hypergraphs, it is worth to recall that, as first noted in [7], we can realize **Hyp** as a category of presheaves ([7, 11, 12]). In Appendix A.2 we report a proof of this claim for the interested reader.

Definition 3.16. We define the category $\mathbf{H}$ as the one in which the set of objects is given by $(\mathbb{N} \times \mathbb{N}) + \{\bullet\}$ and arrows are given by the identities $\mathsf{id}_{k,l}$, $\mathsf{id}_\bullet$ and exactly $k + l$ arrows $f_i \colon (k, l) \to \bullet$, where i ranges from 0 to $k + l - 1$.

Lemma 3.17. *There is an equivalence of categories* $\mathbf{Set}^\mathbf{H} \to \mathbf{Hyp}$ *sending a functor* $F \colon \mathbf{H} \to \mathbf{Set}$ *to* $\mathcal{G}_F := (E_F, F(\bullet), s_F, t_F)$*, where* $(E_F, \{\iota^F_{k,l}\})$ *is a coproduct for* $\{F(k,l)\}_{k,l\in\mathbb{N}}$ *and* $s_F, t_F \colon E_F \rightrightarrows V_F^\star$ *are the unique arrows fitting in the diagrams aside, where* $(F(\bullet), \{v^F_n\}_{n\in\mathbb{N}})$ *is the coproduct coming from the first point of Proposition 3.1 and* $s^F_{k,l}, t^F_{k,l} \colon F(k,l) \to F(\bullet)^k$*,* $t^F_{k,l} \colon F(k,l) \to F(\bullet)^l$ *are the arrows induced by, respectively,* $\{F(f_i)\}_{i=0}^{k-1}$ *and* $\{F(f_i)\}_{i=k}^{k+l-1}$*.*

$$\begin{tikzcd}
F(k,l) \arrow[r, "s^F_{k,l}"] \arrow[d, hook, "\iota^F_{k,l}"'] & F(\bullet)^k \arrow[d, hook, "v^F_k"] \\
E_F \arrow[r, dotted, "s_F"] & F(\bullet)^\star
\end{tikzcd}
\qquad
\begin{tikzcd}
F(k,l) \arrow[r, "t^F_{k,l}"] \arrow[d, hook, "\iota^F_{k,l}"'] & F(\bullet)^l \arrow[d, hook, "v^F_l"] \\
E_F \arrow[r, dotted, "t_F"] & F(\bullet)^\star
\end{tikzcd}$$

Corollary 3.18. **Hyp** *has all limits and colimits.*

Lemma 3.17 not only entails that **Hyp** is complete, but it provides us with a recipe to compute limits, even if they are not connected.

Corollary 3.19. *Let* $F \colon \mathbf{D} \to \mathbf{Hyp}$ *be a discrete diagram and let* $F(D)$ *be* (E_D, V_D, s_D, t_D)*. Suppose that* $(V_\mathcal{L}, \{\pi^D_V\}_{D\in\mathbf{D}})$ *is the product of* $\{V_D\}_{D\in\mathbf{D}}$*. For every* $D \in \mathbf{D}$ *and* $(k, l) \in \mathbb{N} \times \mathbb{N}$*, define*

$$E^D_{k,l} := \{e \in E_D \mid \mathsf{lg}_{V_D}(s_D(e)) = k \text{ and } \mathsf{lg}_{V_D}(t_D(e)) = l\}$$

with the canonical inclusion $i^D_{k,l} \colon E^D_{k,l} \rightarrowtail E_D$*. Now, consider the product* $(E_{k,l}, \{\pi^D_{k,l}\}_{D\in\mathbf{D}})$ *and take the coproduct* $(E_\mathcal{L}, \{\iota_{k,l}\}_{k,l\in\mathbb{N}})$ *of the family* $\{E_{k,l}\}_{k,l\in\mathbb{N}}$*. Then there are arrows* $s_\mathcal{L}, t_\mathcal{L} \colon E_\mathcal{L} \rightrightarrows V_\mathcal{L}^\star$ *and* $\pi^D_E \colon E_\mathcal{L} \to E_D$ *fitting in the diagram below and making* $((E_\mathcal{L}, V_\mathcal{L}, s_\mathcal{L}, t_\mathcal{L}), \{(\pi^D_E, \pi^D_V)\}_{D\in\mathbf{D}})$ *a limit for* F*.*

$$\begin{tikzcd}
E^D_{k,l} \arrow[r, tail, "i^D_{k,l}"] & E_D \arrow[r, "s_D"] & V_D^\star \\
E_{k,l} \arrow[u, "\pi^D_{k,l}"] \arrow[d, hook, "\iota_{k,l}"'] & & \\
E_\mathcal{L} \arrow[rr, dotted, "s_\mathcal{L}"'] & & V_\mathcal{L}^\star \arrow[uu, "(\pi^D_V)^\star"']
\end{tikzcd}
\qquad
\begin{tikzcd}
E^D_{k,l} \arrow[r, tail, "i^D_{k,l}"] & E_D \arrow[r, "t_D"] & V_D^\star \\
E_{k,l} \arrow[u, "\pi^D_{k,l}"] \arrow[d, hook, "\iota_{k,l}"'] & & \\
E_\mathcal{L} \arrow[rr, dotted, "t_\mathcal{L}"'] & & V_\mathcal{L}^\star \arrow[uu, "(\pi^D_V)^\star"']
\end{tikzcd}
\qquad
\begin{tikzcd}
E_{k,l} \arrow[r, "\pi^D_{k,l}"] \arrow[d, "\iota_{k,l}"'] & E^D_{k,l} \arrow[d, "i^D_{k,l}"] \\
E_\mathcal{L} \arrow[r, dotted, "\pi^D_E"'] & E_D
\end{tikzcd}$$

Example 3.20. For instance, a terminal object of **Hyp** is given by $(\mathbb{N}\times\mathbb{N}, 1, \Delta\circ\pi_1, \Delta\circ\pi_2)$, where $\pi_1, \pi_2 \colon \mathbb{N}\times\mathbb{N}\to\mathbb{N}$ are the projections and $\Delta\colon \mathbb{N}\to 1^\star$ the canonical isomorphism introduced in Remark 3.3.

3.3. Some subcategories of **Hyp**

We are now going to introduce some classes of hypergraphs, and the relative subcategories, which will provide the base for our definitions of EGGs.

3.3.1. Left-monogamous hypergraphs

We first introduce *left-monogamous* hypergraphs [31, 19], a generalization of *term graphs* [13, 32].

Definition 3.21. Let $\mathcal{H} = (E_\mathcal{H}, V_\mathcal{H}, s_\mathcal{H}, t_\mathcal{H})$ be a hypergraph. We define two functions $\mathsf{in}_\mathcal{H}, \mathsf{out}_\mathcal{H} \colon V_\mathcal{H}\times E_\mathcal{H} \rightrightarrows \mathbb{N}$ where $\mathsf{in}_\mathcal{H}(v,e)$ is the number of occurences of v in $s_\mathcal{H}(e)$ and $\mathsf{out}_\mathcal{H}(v,e)$ that of v in $t_\mathcal{H}(e)$. If v is such that $\mathsf{out}_\mathcal{H}(v,e) = 0$ for every hyperedge e we say that v is an *input* node. We will denote the set of input nodes by $\mathsf{IN}(\mathcal{H})$.

A hypergraph $\mathcal{H}$ is *left-monogamous* if for every v in $V_\mathcal{H}$, there is at most one hyperedge e_v such that $\mathsf{out}_\mathcal{H}(v,e_v)$ is non-zero and for such e_v we have $\mathsf{out}_\mathcal{H}(v,e_v) = 1$.

We will denote by **LMH** the full subcategory of **Hyp** given by left-monogamous hypergraphs. The corresponding inclusion functor will be denoted by M.

Remark 3.22. Notice that our definition, unlike the one of [19], allows for the presence of cycles in left-monogamous hypergraphs.

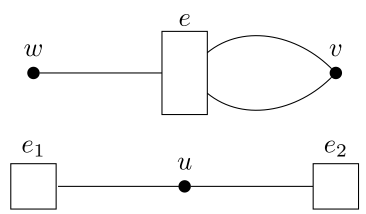
Given a set X, we say that two words $X^\star$ are *disjoint* is they do not share any letter. We can exploit this notion to rephrase the previous definition in a more elementary way (see Appendix A.2.1 for a proof).

Proposition 3.23. *Given a hypergraph $\mathcal{H}$, the following facts are equivalent*

1. *$\mathcal{H}$ is left-monogamous;*
2. *for every $e \in E_\mathcal{H}$, $t_\mathcal{H}(e)$ has no repeated letter and $t_\mathcal{H}(e)$ is disjoint from $t_\mathcal{H}(e')$ for every other hyperedge e'.*

Example 3.24. The two hypergraphs aside are examples of non left-monogamous hypergraph. In the top hypergraphs, left-monogamy fails because $\mathsf{out}_\mathcal{G}(a,e) = 2$, while in the hypergraph in the bottom we have

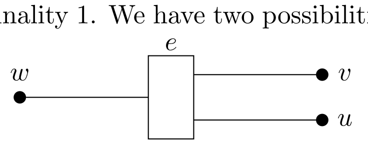
$$\mathsf{out}_\mathcal{H}(u,e_1) = \mathsf{out}_\mathcal{H}(u,e_2) = 1$$



The remainder of this section is devoted to the proof of quasiadhesivity of **LMH**. We begin by computing pullbacks and equalizers of left-monogamous hypergraphs.

Remark 3.25. Notice that **LMH** is not complete: it lacks e.g. a terminal object. To see this, notice that $\mathcal{H} := (\emptyset, 1, ?_1, ?_1)$ is left-monogamous and that $\mathbf{LMH}(\mathcal{H},\mathcal{G})$ is in bijection with $V_\mathcal{G}$ for every hypergraph $\mathcal{G}$. Thus if $\mathcal{T}$ is terminal in **LMH**, then its set of vertices $V_\mathcal{T}$ must have cardinality 1. We have two possibilities

- all the hyperedges of $\mathcal{T}$ have the empty word as target; or
- there is a (necessarily unique) $\mathcal{T}$ with a target of length one.



In both cases there is no morphism from the left-monogamous hypergraph depicted above and $\mathcal{T}$.

An elementary, but fundamental observation is stated below (see Appendix A.2.1 for a proof).

Proposition 3.26. **LMH** *is closed in* **Hyp** *under subobject: if $(f,g)\colon \mathcal{G}\to\mathcal{H}$ is a mono in* **Hyp** *and $\mathcal{H}$ is left-monogamous, then $\mathcal{G}$ is left-monogamous too.*

Since **LMH** is a full subcategory of **Hyp**, Proposition 3.26 provides equalizers of left-monogamous hypergraphs.

Corollary 3.27. **LMH** *is closed in* **Hyp** *under equalizers.*

As shown by Remark 3.25, **LMH** does not have all products, nonetheless left-monogamous hypergraphs are closed in **Hyp** under binary products (a proof can be found in Appendix A.2.1).

Lemma 3.28. **LMH** *is closed in* **Hyp** *under binary products.*

Pullbacks are built from products and equalizers, so Corollary 3.27 and Lemma 3.28 give us the following.

Corollary 3.29. **LMH** *is closed in* **Hyp** *under pullbacks.*

Definition 3.30. A morphism $(f, g)\colon \mathcal{G} \to \mathcal{H}$ of hypergraphs is *downward-closed* if $t_{\mathcal{H}}(h)$ has a letter belonging to the image of g, then h is in the image of f.

Being downward-closed is related to the preservation of input nodes, as shown by the following lemma (see Appendix A.2.1 for the proof).

Lemma 3.31. *Let $(f, g)\colon \mathcal{G} \to \mathcal{H}$ be a morphism of hypergraphs with $\mathcal{H}$ left-monogamous, then the following facts hold*

1. *if g sends input nodes to input nodes, then (f, g) is downward-closed;*
2. *if g is mono and (f, g) is downward-closed, then g sends input nodes to input nodes.*

Downward-closedness allows us to guarantee the existence of some pushouts in **LMH**, as shown by the following proposition, whose proof is postponed to Appendix A.2.1.

Proposition 3.32. *Let $(f_1, g_1)\colon \mathcal{G} \to \mathcal{H}$ and $(f_2, g_2)\colon \mathcal{G} \to \mathcal{K}$ be two morphisms in* **LMH**. *If (f_1, g_1) is mono and downward-closed, then the pushout $\mathcal{P}$ of (f_1, g_1) along (f_2, g_2) in* **Hyp** *is left-monogamous.*

Corollary 3.33. **LMH** *is closed in* **Hyp** *under pushouts along downward-closed morphisms.*

We can use Proposition 3.32 to completely characterise regular monos in **LMH**. We refer the reader to Appendix A.2.1 for the proof.

Lemma 3.34. *Let $(f, g)\colon \mathcal{G} \to \mathcal{H}$ be s monomorphism in* **LMH**. *Then the following facts are equivalent*

1. *(f, g) is a regular mono;*
2. *(f, g) sends input nodes to input nodes;*
3. *(f, g) is downward-closed.*

Combining Theorem 2.4 with Corollary 3.33 and Lemma 3.34, we get the following.

Corollary 3.35. **LMH** *is quasiadhesive.*

3.3.2. Pruned and separated hypergraphs

We can now introduce a novel subcategory of **Hyp** that is going to be used in Section 5.2 to give the underlying hypergraphs for EGGs with minimally identified tokens.

Definition 3.36. A left-monogamous hypergraph $\mathcal{G} = (E_{\mathcal{G}}, V_{\mathcal{G}}, s_{\mathcal{G}}, t_{\mathcal{G}})$ is *separated* if

1. it is *right-monogamous*, i.e. $(E_{\mathcal{G}}, V_{\mathcal{G}}, t_{\mathcal{G}}, s_{\mathcal{G}})$ is left-monogamous;
2. for every pair of, not necessarily distinct, hyperedges e and e', $s_{\mathcal{G}}(e)$ and $t_{\mathcal{G}}(e')$ are disjoint.

We denote by **Sep** the full subcategory of **Hyp** of separated hypergraphs and by S the inclusion functor.

Example 3.37. The hypergraph of Example 3.12 is left-monogamous but not separated.

Remark 3.38. Let $\mathcal{G}$ be a hypergraph. By Proposition 3.23, being right-monogamous amounts to the request that for every vertex $v \in V_{\mathcal{G}}$ there exists at most one $e_V \in E_{\mathcal{G}}$ such that $\mathsf{in}_{\mathcal{G}}(v, e) \neq 0$ and for such hyperedge we have $\mathsf{in}_{\mathcal{G}}(v, e) = 1$.

Remark 3.39. Let $\mathcal{G}$ be a separated hypergraph. By applying Proposition 3.23, this fact is equivalent to say that for every hyperedge e

- $s_{\mathcal{G}}(e)$ and $t_{\mathcal{G}}(e)$ do not contain repeated letters;
- $s_{\mathcal{G}}(e)$ is disjoint from $t_{\mathcal{G}}(e')$ for every $e' \in E_{\mathcal{G}}$;
- for every $e' \in E_{\mathcal{G}}$ different from e, $s_{\mathcal{G}}(e)$ and $t_{\mathcal{G}}(e)$ are disjoint from, respectively, $s_{\mathcal{G}}(e')$ and $t_{\mathcal{G}}(e')$.

Remark 3.40. By Remark 3.39 it is clear that separated hypergraphs enjoy a property similar to the one enjoyed by left-monogamous ones (cf. Appendix A.2.1): if $\mathcal{G}$ is separated and $(f, g) \colon \mathcal{H} \to \mathcal{G}$ is a mono, then $\mathcal{H}$ is separated too.

We can provide a construction which "splits" a given hypergraph to produce a separated one.

Definition 3.41. Given a hypergraph $\mathcal{G} = (E_{\mathcal{G}}, V_{\mathcal{F}}, s_{\mathcal{G}}, t_{\mathcal{G}})$, for every hyperedge $e \in E_{\mathcal{G}}$ with $n_e = \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ and $m_e = \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{G}}(e))$, we can consider the coproduct $(V_e, \{\iota_i^e\}_{i=1}^{n_e+m_e})$ of $n_e + m_e$ copies of 1. We can further take the coproduct $(V_{\mathsf{sp}(\mathcal{G})}, \{\iota_e\}_{e \in E_{\mathcal{G}}})$ of the family $\{V_e\}_{e \in E_{\mathcal{G}}}$ and define

$$s_{\mathsf{sp}(\mathcal{G})} \colon E_{\mathcal{G}} \to V^{\star}_{\mathsf{sp}(\mathcal{G})} \qquad\qquad t_{\mathsf{sp}(\mathcal{G})} \colon E_{\mathcal{G}} \to V^{\star}_{\mathsf{sp}(\mathcal{G})}$$
$$e \mapsto \prod_{i=1}^{n_e} \iota_e(\iota_i^e(0)) \qquad\qquad e \mapsto \prod_{i=n_e+1}^{n_e+m_e} \iota_e(\iota_i^e(0))$$

where 0 is the unique element of the terminal object 1.

We define the *separated hypergraph associated to* $\mathcal{G}$ as $\mathsf{sp}(\mathcal{G}) = (E_{\mathcal{G}}, V_{\mathsf{sp}(\mathcal{G})}, s_{\mathsf{sp}(\mathcal{G})}, t_{\mathsf{sp}(\mathcal{G})})$.

Remark 3.42. By construction $\mathsf{sp}(\mathcal{G})$ is an object of **Sep**. To see this it is enough to apply Remark 3.39

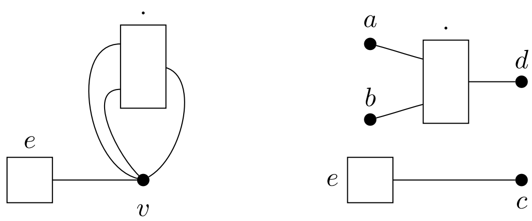
- every letter of $s_{\mathcal{G}}(e)$ and $t_{\mathcal{G}}(e)$ lies in the image of a different coprojection $\iota_e \circ \iota_i^e$, so it cannot appear more than one time;
- every letter of $s_{\mathcal{G}}(e)$ lies in the image of $\iota_e \circ \iota_i^e$ for $i \leq n_e$, while every letter of $t_{\mathcal{G}}(e)$ lies in the image of $\iota_e \circ \iota_j^e$ for $n_e < j$, guaranteeing disjointness;
- if e is different from e', then all the letters of $s_{\mathcal{G}}(e)$ and $t_{\mathcal{G}}(e)$ are in the image of ι_e, while those of $s_{\mathcal{G}}(e')$ and $t_{\mathcal{G}}(e')$ are in the image of $\iota_{e'}$ and so, again, we get the required disjointness.

Example 3.43. If $\mathcal{G}$ has no hyperedges, then $\mathsf{sp}(\mathcal{G})$ is the empty hypergraph.

Example 3.44. Let Σ be a monoidal signature, then $\mathsf{sp}(\mathcal{G}^{\Sigma})$ collects all the operations of Σ as edges with disjoint sources and targets. For instance let Σ be the algebraic signature of monoids, with $O_{\Sigma} = (\cdot, e)$ with

$$\mathsf{ar}_{\Sigma}(\cdot) = 2 \qquad \mathsf{ar}_{\Sigma}(e) = 0$$

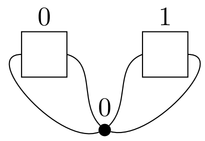
We can then depict $\mathcal{G}^{\Sigma}$ and $\mathsf{sp}(\mathcal{G}^{\Sigma})$ as below



The construction we have just introduced can be extended to a functor (see Appendix A.2.2 for a proof).

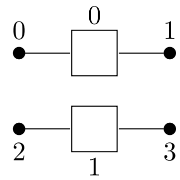
Proposition 3.45. *There exists a functor* $\mathsf{sp} \colon \mathbf{Hyp} \to \mathbf{Sep}$ *sending* $\mathcal{G}$ *to* $\mathsf{sp}(\mathcal{G})$.

Remark 3.46. Notice that sp is neither right nor left adjoint to S.



- To see that sp cannot be a right adjoint to S, consider the "double loop" $\mathcal{D} = (2, 1, !_2, !_2)$ depicted aside. Then $\mathsf{sp}(\mathcal{D})$ is isomorphic to the graph $\mathcal{G} = (2, 4, s_{\mathcal{G}}, t_{\mathcal{G}})$ with

$$s_{\mathcal{G}}(0) = 0 \quad t_{\mathcal{G}}(0) = 1 \quad s_{\mathcal{G}}(1) = 2 \quad t_{\mathcal{G}}(1) = 3$$



  Let $\mathcal{H}$ be $(\emptyset, 1, ?_{1^\star}, ?_{1^\star})$, which is clearly a separated hypergraph. Now, $\mathbf{Sep}(\mathcal{H}, \mathsf{sp}(\mathcal{D}))$ has the same cardinality as $\mathbf{Sep}(\mathcal{H}, \mathcal{G})$, which is 4, while $\mathbf{Hyp}(S(\mathcal{H}), \mathcal{D})$ is a singleton.

- Consider now the "single loop" $\mathcal{L} = (1, 1, \mathsf{id}_1, \mathsf{id}_1)$. Then $\mathsf{sp}(\mathcal{L})$ is isomorphic to $(1, 2, \mathsf{id}_2, \mathsf{id}_2)$ and so there are no arrows $\mathcal{L} \to S(\mathsf{sp}(\mathcal{L}))$. In particular, this shows that sp cannot be left adjoint to S.

If we restrict our attention to separated hypergraphs without isolated nodes, then sp becomes a right adjoint. The interested reader can consult Appendix A.2.2 for a proof.

Definition 3.47. Let $\mathcal{G}$ be a hypergraph. A node $v \in V_{\mathcal{G}}$ is *isolated* if it does not occur as a letter in either $s_{\mathcal{G}}(e)$ or in $t_{\mathcal{G}}(e)$ for any $e \in E_{\mathcal{G}}$. $\mathcal{G}$ is *pruned* if it has no isolated nodes. We denote by **pHyp** and **pSep** the full subcategories of **Hyp** given by, respectively, pruned and pruned and separated hypergraphs.

Remark 3.48. By construction, $\mathsf{sp}(\mathcal{G})$ is pruned for every hypergraph $\mathcal{G}$.

Remark 3.49. Let (f, g) and (h, k) be two morphisms $\mathcal{G} \rightrightarrows \mathcal{H}$ of **Hyp** with $\mathcal{G}$ pruned and suppose that $f = h$. We can use the absence of isolated nodes in $\mathcal{G}$ to show that $g = k$ too. Let $v \in V_{\mathcal{G}}$, we have two cases

- v occurs in $s_{\mathcal{G}}(e)$ for some $e \in E_{\mathcal{G}}$. Let $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ be such that $v = s_{\mathcal{G}}(e)_i$, then we have

$$g(v) = g(s_{\mathcal{G}}(e)_i) = (g^\star(s_{\mathcal{G}}(e)))_i = s_{\mathcal{H}}(f(e))_i = s_{\mathcal{H}}(h(e))_i = (k^\star(s_{\mathcal{G}}(e)))_i = k(s_{\mathcal{G}}(e)_i) = k(v)$$

- v occurs in $t_{\mathcal{H}}(e)$ for some $e \in E_{\mathcal{H}}$. Let $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{H}}}(t_{\mathcal{G}}(e))$ be such that $v = t_{\mathcal{G}}(e)_i$, then we have

$$g(v) = g(t_{\mathcal{G}}(e)_i) = (g^\star(t_{\mathcal{G}}(e)))_i = t_{\mathcal{H}}(f(e))_i = t_{\mathcal{H}}(h(e))_i = (k^\star(t_{\mathcal{G}}(e)))_i = k(t_{\mathcal{G}}(e)_i) = k(v)$$

In both cases $\hat{g}(v) = k(v)$ and so $\hat{g} = k$ as wanted.

Proposition 3.50. *The following properties hold*

1. *the inclusion functor* $P \colon \mathbf{pHyp} \to \mathbf{Hyp}$ *has a right adjoint* pr*;*
2. *the inclusion functor* $S_p \colon \mathbf{pSep} \to \mathbf{pHyp}$ *has a right adjoint* sp_P*.*

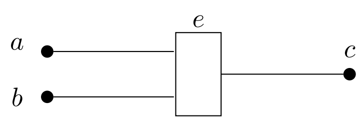
Finally, we conclude our study of separated hypergraphs proving a result regarding connected limits. A proof of it is Appendix A.2.2

Lemma 3.51. Sep *is closed in* **Hyp** *under connected limits.*

Example 3.52. In general **Sep** is not complete. For instance it does not have a terminal object. To see this notice that the $\mathcal{H} := (\emptyset, 1, ?_{1^\star}, ?_{1^\star})$ is separated and that $\mathbf{Sep}(\mathcal{H}, \mathcal{G})$ is in bijection with $V_{\mathcal{G}}$ for every hypergraph $\mathcal{G}$. Thus if $\mathcal{T}$ is terminal in **Sep**, then its set of vertices $V_{\mathcal{T}}$ must have cardinality 1.

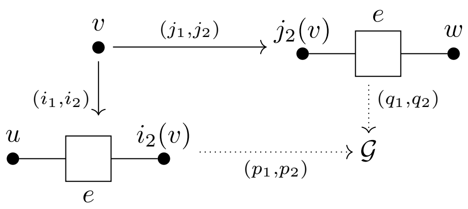
Since $\mathcal{T}$ must be separated, we have two (non disjoint) cases

- all the hyperedges of $\mathcal{T}$ have sources with length zero;
- all the hyperedges of $\mathcal{T}$ have targets with length zero.



In both cases there are no morphism from the hypergraph depicted on the right and $\mathcal{T}$.

Example 3.53. In general **Sep** does not have pushouts. For instance consider the solid part of diagram aside. Let us call the two unnamed hypergraphs on the top row $\mathcal{H}$ and $\mathcal{K}$, respectively, and call the third one $\mathcal{Q}$.



Suppose that there exists $\mathcal{G}$ and arrows $(p_1,p_2)\colon \mathcal{Q} \to \mathcal{G}$ and $(q_1,q_2)\colon \mathcal{K} \to \mathcal{G}$ which make the square a pushout in **Sep**. By construction we have

$$s_{\mathcal{G}}(q_1(e)) = q_2^\star(s_{\mathcal{K}}(e)) = q_2^\star(j_2^\star(v)) = p_2^\star(i_2^\star(v)) = p_2^\star(t_{\mathcal{Q}}^\star(e)) = t_{\mathcal{G}}(p_2(e))$$

and so $\mathcal{G}$ cannot be in **Sep**, yielding a contradiction.

If we restrict to pruned separated hypergraphs the previous issue disappeares. The crucial observation is that the a pruned separated hypergraph is characterized by its set of hyperedges and the length of their sources and targets. See Appendix A.2.2 for a proof.

Lemma 3.54. **pSep** *is equivalent to* $\mathbf{Set}/(\mathbb{N}\times\mathbb{N})$.

Corollary 3.55. **pSep** *is adhesive.*

We can use Lemma 3.54 to say something more about colimits and pullbacks in **pSep**. A proof can be found in Appendix A.2.2

Proposition 3.56. **pSep** *is closed in* **Hyp** *under colimits and pullbacks.*

3.4. Labelled hypergraphs

Our interest for hypergraphs stems from their use as a graphical representation of algebraic terms. As noted in Section 3.2, a hypergraph can be thought as a colored monoidal signature. Thus labelling corresponds to the operation of taking the slice category over a suitable hypergraph.

Definition 3.57. Let $\mathcal{G}$ be a hypergraph. The category $\mathbf{Hyp}_{\mathcal{G}}$ of hypergraphs *labelled by* $\mathcal{G}$ or of $\mathcal{G}$-*hypergraphs* is $\mathbf{Hyp}/\mathcal{G}$.

Corollary B.5 and Theorem 2.4 give an adhesivity result for $\mathbf{Hyp}_{\mathcal{G}}$ and a characterisation of its monos.

Proposition 3.58. *Let $\mathcal{G}$ be a hypergraph. Then the following facts hold*

1. *an arrow (h,k) in $\mathbf{Hyp}_{\mathcal{G}}$ is a mono if and only if h and k are injective;*
2. $\mathbf{Hyp}_{\mathcal{G}}$ *is an adhesive category.*

Example 3.59. A useful example is given by an algebraic signature Σ. Consider the hypergraph of Example 3.14, the category $\mathbf{Hyp}_{\Sigma}$ of hypergraphs labelled by Σ is the slice category $\mathbf{Hyp}/\mathcal{G}^{\Sigma}$.

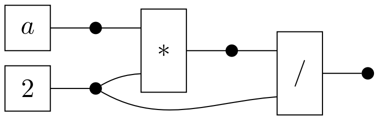
Remark 3.60. Consider an algebraic signature Σ and a hypergraph $\mathcal{H} = (E, V, s, t)$. By definition we know that $U_{\mathbf{Hyp}}(\mathcal{G}^{\Sigma})$ is the terminal object 1, so an arrow $\mathcal{H} \to \mathcal{G}^{\Sigma}$ is determined by a function $h\colon E_{\mathcal{H}} \to O_{\Sigma}$ making the two squares on the right commutative (cf. Remark 3.3).

$$\begin{tikzcd} E_{\mathcal{H}} \arrow[r, "s_{\mathcal{H}}"'] \arrow[d, "h"'] & V_{\mathcal{H}}^\star \arrow[d, "\mathsf{lg}_{V_{\mathcal{H}}}"] \\ O_{\Sigma} \arrow[r, "\Delta\circ\mathsf{ar}_{\Sigma}"] & \mathbb{N} \end{tikzcd} \qquad \begin{tikzcd} E_{\mathcal{H}} \arrow[r, "t_{\mathcal{H}}"'] \arrow[d, "h"'] & V_{\mathcal{H}}^\star \arrow[d, "\mathsf{lg}_{V_{\mathcal{H}}}"] \\ O_{\Sigma} \arrow[r, "\Delta\circ\mathsf{co}_{\Sigma}"] & \mathbb{N} \end{tikzcd}$$

Now, consider a coprojection $v_n\colon V_{\mathcal{H}}^n \to V_{\mathcal{H}}^\star$. Since $\Delta\circ\mathsf{co}_{\Sigma}$ is the constant function in 1, the second diagram aside entails, together with Remark 3.5, that $t_{\mathcal{H}}$ factors via the inclusion $v_1\colon V_{\mathcal{H}} \to V_{\mathcal{H}}^\star$ of words of length 1, i.e $\mathcal{H}$ must be algebraic.

We extend our graphical notation of hypergraphs to labelled ones putting the label of an hyperedge h inside its corresponding square.

Example 3.61. Consider again the signature Σ of Example 3.14, then the hypergraph $\mathcal{G}$ of Example 3.12 can be labeled by a morphism $(l, !_{V_{\mathcal{G}}}) : \mathcal{G} \to \mathcal{G}^{\Sigma}$ that is characterised by the image of the edges. If $l(h_1) = a$, $l(h_2) = 2$, $l(h_3) = *$, and $l(h_4) = /$, we represent it visually by putting the labels of the edges in $\mathcal{G}^{\Sigma}$ inside the boxes representing the edges of $\mathcal{G}$



3.4.1. Labelled left-monogamous and separated hypergraphs

As done in Section 3.3, we can restrict our attention to two subcategories of labelled hypergraphs.

Definition 3.62. Let $\mathcal{G}$ be a hypergraph and $(h,k)\colon \mathcal{H} \to \mathcal{G}$ a $\mathcal{G}$-hypergraph. We say that (h,k) is a *left-monogamous* (resp. *separated, pruned*) $\mathcal{G}$-hypergraph if $\mathcal{H}$ is left-monogamous (resp. separated, pruned).

We will denote by $\mathbf{LMH}_{\mathcal{G}}$, $\mathbf{pHyp}_{\mathcal{G}}$, $\mathbf{Sep}_{\mathcal{G}}$ and $\mathbf{pSep}_{\mathcal{G}}$ the corresponding subcategories of $\mathbf{Hyp}_{\mathcal{G}}$.

Remark 3.63. Of the four subcategories defined above, we are interested only in $\mathbf{LMH}_{\mathcal{G}}$, $\mathbf{Sep}_{\mathcal{G}}$ and $\mathbf{pSep}_{\mathcal{G}}$. This is due to the fact that we do not have an adhesivity result for $\mathbf{pHyp}$.

Remark 3.64. The functors pr and sp_P induce right adjoints to the inclusions $P_{\mathcal{G}}\colon \mathbf{pHyp}_{\mathcal{G}} \to \mathbf{Hyp}_{\mathcal{G}}$ and $S_{p,\mathcal{G}}\colon \mathbf{pSep}_{\mathcal{G}} \to \mathbf{pHyp}_{\mathcal{G}}$.

We can easily realize $\mathbf{LMH}_{\mathcal{G}}$, $\mathbf{Sep}_{\mathcal{G}}$ and $\mathbf{pSep}_{\mathcal{G}}$ as comma categories.

Proposition 3.65. *Let $M\colon \mathbf{LMH} \to \mathbf{Hyp}$ and $S\colon \mathbf{Sep} \to \mathbf{Hyp}$ be the inclusion functors. Moreover, let $\Gamma_{\mathcal{G}}\colon 1 \to \mathbf{Hyp}$ be the constant functor in $\mathcal{G}$. Then*

1. $\mathbf{LMH}_{\mathcal{G}}$ *is equivalent to $M{\downarrow}\Gamma_{\mathcal{G}}$;*
2. $\mathbf{Sep}_{\mathcal{G}}$ *is equivalent to $S{\downarrow}\Gamma_{\mathcal{G}}$;*
3. $\mathbf{pSep}_{\mathcal{G}}$ *is equivalent to* $\mathbf{pSep}/\mathsf{sp}(\mathsf{pr}(\mathcal{G}))$.

Corollary B.4 allows us to easily characterise regular monos in $\mathbf{Hyp}_{\mathcal{G}}$.

Proposition 3.66. *A morphism (f,g) between $(h_1,k_1)\colon \mathcal{H} \to \mathcal{G}$ and $(h_2,k_2)\colon \mathcal{K} \to \mathcal{G}$ is a regular mono in $\mathbf{LMH}_{\mathcal{G}}$ if and only if $(f,g)\colon \mathcal{H} \to \mathcal{K}$ is a regular mono in $\mathbf{LMH}$.*

In turn, by Theorem 2.4, Corollaries 3.27, 3.33 and Lemma 3.34 we get the following result.

Theorem 3.67. *Let $\mathcal{G}$ be a hypergraph. Then $\mathbf{LMH}_{\mathcal{G}}$ is quasiadhesive and $\mathbf{pSep}_{\mathcal{G}}$ is adhesive.*

Remark 3.68. Let Σ be an algebraic signature. Then we know that every hypergraph labelled by Σ must be algebraic. By Proposition 3.23 this means that a hypergraph $(h,k)\colon \mathcal{H} \to \mathcal{G}^{\Sigma}$ labelled by Σ is left-monogamous if and only if $t_{\mathcal{H}}$ is injective. This is precisely the definition of *term graphs* in [32, 13, 12].

4. Adding equivalences to hypergraphical structures

The use of hypergraphs equipped with an equivalence relation over their nodes has been argued as a convenient way to express concurrency in the DPO approach to rewriting [4]. This section presents the framework through adhesive categories, including its extension to the various kinds of hypergraphs introduced in the previous section, as a stepping stone towards an analogous characterization for EGGs.

4.1. Hypergraphs with equivalence

Let us start with the case of general hypergraphs. These were introduced in [4], even if no general consideration about their structure as a category was provided, and adhesivity, which is the main focus here, was yet to be presented to the world.

Definition 4.1. A *hypergraph with equivalence* $\mathcal{G} = (E_\mathcal{G}, V_\mathcal{G}, Q_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G}, q_\mathcal{G})$ is a 6-tuple such that $\mathcal{G} = (E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G})$ is a hypergraph, $Q_\mathcal{G}$ is a set and $q_\mathcal{G} : V_\mathcal{G} \twoheadrightarrow Q_\mathcal{G}$ is a surjection called *quotient map.* A morphism $h\colon \mathcal{G} \to \mathcal{H}$ is a triple (h_E, h_V, h_Q) such that the following diagrams commute

$$\begin{tikzcd} E_\mathcal{G} \arrow[r, "s_\mathcal{G}"] \arrow[d, "h_E"'] & V_\mathcal{G}^\star \arrow[d, "h_V^\star"] \\ E_\mathcal{H} \arrow[r, "s_\mathcal{H}"'] & V_\mathcal{H}^\star \end{tikzcd} \qquad \begin{tikzcd} E_\mathcal{G} \arrow[r, "t_\mathcal{G}"] \arrow[d, "h_E"'] & V_\mathcal{G}^\star \arrow[d, "h_V^\star"] \\ E_\mathcal{H} \arrow[r, "t_\mathcal{H}"'] & V_\mathcal{H}^\star \end{tikzcd} \qquad \begin{tikzcd} V_\mathcal{G} \arrow[r, two heads, "q_\mathcal{G}"] \arrow[d, "h_V"'] & Q_\mathcal{G} \arrow[d, "h_Q"] \\ V_\mathcal{H} \arrow[r, two heads, "q_\mathcal{H}"'] & Q_\mathcal{H} \end{tikzcd}$$

The category of hypergraphs with equivalence and their morphisms is denoted $\mathbf{Hyp}_{\mathsf{eq}}$.

Remark 4.2. Notice that in **Set** the classes of surjections, epis and regular epis coincide.

Remark 4.3. Morphisms of hypergraphs with equivalences are uniquely determined by the first two components. That is, if $h_1 = (h_E, h_V, f)$ and $h_2 = (h_E, h_V, g)$ are two morphisms $\mathcal{G} \rightrightarrows \mathcal{H}$, then we have $f \circ q_\mathcal{G} = q_\mathcal{H} \circ h_V = g \circ q_\mathcal{G}$. Since $q_\mathcal{G}$ is epi, we obtain $f = g$.

Forgetting the quotient part yields a functor $T\colon \mathbf{Hyp}_{\mathsf{eq}} \to \mathbf{Hyp}$ sending a hypergraph with equivalence $(E_\mathcal{G}, V_\mathcal{G}, C_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G}, q_\mathcal{G})$ to $(E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G})$. We now explore some of its properties to deduce information on the structure of $\mathbf{Hyp}_{\mathsf{eq}}$. A proof is in Appendix A.3.

Proposition 4.4. *Consider the forgetful functor* $T\colon \mathbf{Hyp}_{\mathsf{eq}} \to \mathbf{Hyp}$*. Then the following facts hold*

1. T *is faithful;*
2. T *has a left adjoint;*
3. T *has a right adjoint.*

Corollary 4.5. *The functor* T *preserves limits and colimits.*

From the previous results we get the following characterisation of monos in $\mathbf{Hyp}_{\mathsf{eq}}$.

Corollary 4.6. *An arrow* $(h_E, h_V, h_Q) : \mathcal{G} \to \mathcal{H}$ *in* $\mathbf{Hyp}_{\mathsf{eq}}$ *is a mono if and only if* (h_E, h_V) *is a mono in* **Hyp**.

Now, we can consider the forgetful functor $U_{\mathsf{eq}}\colon \mathbf{Hyp}_{\mathsf{eq}} \to \mathbf{Set}$ obtained by composing T and $U_{\mathbf{Hyp}}$. By Proposition 3.10 and the second point of Proposition 4.4 we get the following.

Corollary 4.7. U_{eq} *has a left adjoint* $\Delta_{\mathsf{eq}}\colon \mathbf{Set} \to \mathbf{Hyp}_{\mathsf{eq}}$.

Notice that there is another functor $K : \mathbf{Hyp}_{\mathsf{eq}} \to \mathbf{Set}$ sending (E, V, Q, s, t, q) to Q, and a morphism (h_E, h_V, h_Q) to h_Q. We exploit it to compute limits and colimits in $\mathbf{Hyp}_{\mathsf{eq}}$. A full proof of the following lemma can be found in Appendix A.3.

Lemma 4.8. *Consider a diagram* $F\colon \mathbf{D} \to \mathbf{Hyp}_{\mathsf{eq}}$ *and let* $(E_D, V_D, Q_D, s_D, t_D, q_D)$ *be the image of an object* D*. Then the following facts hold*

1. F *has a colimit, which is preserved by* K*;*
2. *consider a cone* $(L, \{l_D\}_{D\in\mathbf{D}})$ *limiting for* $K \circ F$ *and let* $((E, V), \{(\pi_E^D, \pi_V^D)\}_{D\in\mathbf{D}})$ *be one for* $T \circ F$*, then* F *has a limit* (E, V, Q, s, t, q) *and there is a mono* $m\colon Q \rightarrowtail L$ *such that the diagram on the right commutes for every* $D \in \mathbf{D}$.

$$\begin{tikzcd} V \arrow[r, "\pi_V^D"'] \arrow[d, two heads, "q"'] & V_D \arrow[dd, two heads, "q_D"] \\ Q \arrow[d, tail, dotted, "m"'] & \\ L \arrow[r, "l_D"] & Q_D \end{tikzcd}$$

Corollary 4.9. *An arrow* $(h_E, h_V, h_Q) : \mathcal{G} \to \mathcal{H}$ *in* $\mathbf{Hyp}_{\mathsf{eq}}$ *is a regular mono if and only if all its components are injective functions.*

A proof is in Appendix A.3. We have now all the ingredients to study the adhesivity properties of $\mathbf{Hyp}_{\mathsf{eq}}$. As a first step we need to introduce a class of monos.

Definition 4.10. We define Pb as the class of regular monos $(h_E, h_V, h_Q)\colon \mathcal{G} \to \mathcal{H}$ in $\mathbf{Hyp}_{\mathsf{eq}}$ such that the square on the right is a pullback

$$\begin{tikzcd} V_\mathcal{G} \arrow[r, two heads, "q_\mathcal{G}"'] \arrow[d, tail, "h_V"'] & Q_\mathcal{G} \arrow[d, tail, "h_Q"] \\ V_\mathcal{H} \arrow[r, two heads, "q_\mathcal{H}"] & Q_\mathcal{H} \end{tikzcd}$$

Now, we show that Pb is a suitable class for adhesivity. A proof is in Appendix A.3.

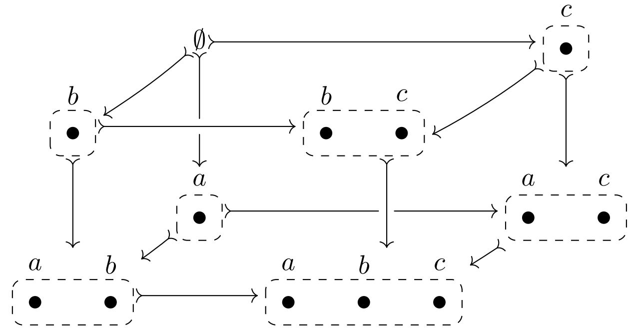
Lemma 4.11. *The class* Pb *contains all isomorphisms, it is closed under composition, decomposition and it is stable under pullbacks and pushouts.*

Finally, we show the key lemma for Pb-adhesivity of $\mathbf{Hyp}_{\mathsf{eq}}$. A proof is in Appendix A.3.

Lemma 4.12. *In* $\mathbf{Hyp}_{\mathsf{eq}}$*,* Pb*-pushouts are stable.*

Example 4.13. It is noteworthy to show that pushouts along regular monos in $\mathbf{Hyp}_{\mathsf{eq}}$ are not stable.

As a counterexample, let us consider the cube aside, where the morphisms are the inclusions defined by the labels of the nodes. The vertices are just graphs without edges, and in the graphs themselves the equivalence classes are denoted by encircling nodes with a dotted line. All the arrows are regular monos. It is immediate to see that the bottom face is a pushout and the side faces are pullback. Unfortunately, the top face fails to be a pushout.



Having proved the stability of Pb-pushouts, we now turn to prove that they are Van Kampen with respect to regular monos. A proof is found in Appendix A.3.

Lemma 4.14. *In* $\mathbf{Hyp}_{\mathsf{eq}}$*, pushouts along arrows in* Pb *are* $\mathsf{Reg}(\mathbf{Hyp}_{\mathsf{eq}})$*-Van Kampen.*

Corollary 4.15. $\mathbf{Hyp}_{\mathsf{eq}}$ *is* Pb*-adhesive.*

4.2. Some subcategories of $\mathbf{Hyp}_{\mathsf{eq}}$

We can now focus to the various kind of hypergraphs introduced in Section 3.3.

Definition 4.16. Given a hypergraph with equivalence $\mathcal{H} := (E_{\mathcal{G}}, V_{\mathcal{G}}, Q_{\mathcal{G}}, s_{\mathcal{G}}, t_{\mathcal{G}}, q_{\mathcal{G}})$, we say that $\mathcal{H}$ is *left-monogamous* (resp. *separated, pruned*) if $T(\mathcal{H})$ is left-monogamous (resp. *separated, pruned*). We will denote by $\mathbf{LMH}_{\mathsf{eq}}$, $\mathbf{Sep}_{\mathsf{eq}}$ and $\mathbf{pSep}_{\mathsf{eq}}$ the subcategories given by left-monogamous, separated, pruned and separated hypergraphs with equivalence.

Remark 4.17. By definition, there exist (forgetful???) functors O_{eq}, R_{eq} and W_{eq} as in the diagrams below, where the horizontal arrows are full and faithful inclusions

$$\begin{tikzcd}
\mathbf{LMH}_{\mathsf{eq}} \arrow[r, "M_{\mathsf{eq}}"] \arrow[d, "O_{\mathsf{eq}}"'] & \mathbf{Hyp}_{\mathsf{eq}} \arrow[d, "T"] \\
\mathbf{LMH} \arrow[r, "M"'] & \mathbf{Hyp}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbf{Sep}_{\mathsf{eq}} \arrow[r, "S_{\mathsf{eq}}"] \arrow[d, "R_{\mathsf{eq}}"'] & \mathbf{Hyp}_{\mathsf{eq}} \arrow[d, "T"] \\
\mathbf{Sep} \arrow[r, "S"'] & \mathbf{Hyp}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathbf{pSep}_{\mathsf{eq}} \arrow[r, "S_{p,\mathsf{eq}}"] \arrow[d, "W_{\mathsf{eq}}"'] & \mathbf{Hyp}_{\mathsf{eq}} \arrow[d, "T"] \\
\mathbf{pSep} \arrow[r, "S_p"'] & \mathbf{Hyp}
\end{tikzcd}$$

Let us start by dealing with some class of limits in these subcategories (see Appendix A.3 for a proof).

Lemma 4.18. *The following facts hold*

1. $\mathbf{LMH}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under equalizers, binary products and pullbacks.*
2. $\mathbf{Sep}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under connected limits;*
3. $\mathbf{pSep}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under pullbacks.*

We can now turn to colimits, applying the same strategy. A proof is deferred to Appendix A.3

Lemma 4.19. *The following facts hold*

1. $\mathbf{LMH}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under pushouts along morphisms whose image through* T *is downward-closed.*

2. $\mathbf{pSep}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under colimits.*

We are now ready to prove two adhesivity results for $\mathbf{LMH}_{\mathsf{eq}}$ and $\mathbf{pSep}_{\mathsf{eq}}$.

Definition 4.20. We define two classes of arrows

- LPb is the class of all arrows in $\mathbf{LMH}_{\mathsf{eq}}$ sent by M_{eq} to Pb;
- SPb is the class of all arrows in $\mathbf{pSep}_{\mathsf{eq}}$ sent by $S_{p,\mathsf{eq}}$ to Pb.

By definition, Lemma 4.11 and putting together Lemmas 4.18, 4.19 with Theorem 2.4 we get the following.

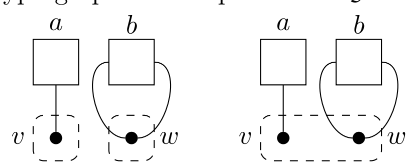
Corollary 4.21. $\mathbf{LMH}_{\mathsf{eq}}$ *is* LPb*-adhesive and* $\mathbf{pSep}_{\mathsf{eq}}$ *is* SPb*-adhesive*

4.3. Labelled hypergraphs with equivalence

We are now going to generalize the work done in the previous sections equipping hypergraphs with an equivalence relation. First of all we need a notion of labelling for $\mathbf{Hyp}_{\mathsf{eq}}$.

Definition 4.22. Let $\mathcal{G}$ be a hypergraph with equivalence. The category $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ of hypergraphs with equivalence *labelled by* $\mathcal{G}$ is $\mathbf{Hyp}_{\mathsf{eq}}/\mathcal{G}$.

Remark 4.23. The equivalence relation on the nodes of the labelling hypergraph put constraints on the relations between the labelled hypergraphs. Consider the following hypergraphs with equivalences $\mathcal{G}$ and $\mathcal{H}$.



Notice that there is no morphism $\mathcal{H} \to \mathcal{G}$. Indeed $\mathsf{id}_{T(\mathcal{G})}$ is the unique morphism $T(\mathcal{G}) \to T(\mathcal{H})$ while there are only two morphisms $f, g \colon Q_{\mathcal{H}} \rightrightarrows Q_{\mathcal{G}}$: the first one is constant in v, while the second one is constant in w.

For both cases we have

$$f(q_{\mathcal{H}}(w)) = v \neq w = q_{\mathcal{G}}(w) = q_{\mathcal{G}}(\mathsf{id}_{V_{\mathcal{G}}}(w)) \qquad g(q_{\mathcal{H}}(v)) = w \neq v = q_{\mathcal{G}}(v) = q_{\mathcal{G}}(\mathsf{id}_{V_{\mathcal{G}}}(v))$$

Intuitively, labelling with $\mathcal{G}$ force us to keep all the sorts (i.e. the labelling of the nodes) distinguished, while labelling with $\mathcal{H}$ allows us to merge different sorts.

Example 4.24. Let Σ be a single typed monoidal signature (cf. Definition 3.13). Since $\mathcal{G}_\Sigma$ has a single node, then to label another hypergraph $\mathcal{H}$ it is enough to provide a function $l \colon E_{\mathcal{H}} \to O_\Sigma$ such that the diagrams aside commute.

$$\begin{tikzcd} E_{\mathcal{H}} \arrow[r, "l"] \arrow[d, "s_{\mathcal{H}}"'] & O_\Sigma \arrow[d, "\mathsf{ar}_\Sigma"] \\ V_{\mathcal{H}}^\star \arrow[r, "\mathsf{lg}_{V_{\mathcal{H}}}"] & \mathbb{N} \end{tikzcd} \qquad \begin{tikzcd} E_{\mathcal{H}} \arrow[r, "l"] \arrow[d, "t_{\mathcal{H}}"'] & O_\Sigma \arrow[d, "\mathsf{co}_\Sigma"] \\ V_{\mathcal{H}}^\star \arrow[r, "\mathsf{lg}_{V_{\mathcal{H}}}"] & \mathbb{N} \end{tikzcd}$$

Notice that, if Σ is algebraic, then such a function exists only if $\mathcal{H}$ is algebraic too. In that case, since the image of $\lg_{V_{\mathcal{H}}}$ is simply the singleton $\{1\}$, the second diagram commutes by default.

Let $V_{\mathcal{G}} \colon \mathbf{Hyp}_{\mathsf{eq},\mathcal{G}} \to \mathbf{Hyp}_{\mathsf{eq}}$ be the functor forgetting the labelling and $\mathsf{Pb}_{\mathcal{G}}$ the class of morphisms in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ whose image in $\mathbf{Hyp}_{\mathsf{eq}}$ is in Pb. From Lemma 4.8 and Corollary B.11, the second point of Theorem 2.4 and from Corollary 4.15, we can deduce the following.

Proposition 4.25. *Consider the forgetful functor* $V_{\mathcal{G}} \colon \mathbf{Hyp}_{\mathsf{eq},\mathcal{G}} \to \mathbf{Hyp}_{\mathsf{eq}}$. *Then the following facts hold*

1. $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ *has all colimits and all connected limits, which are created by* $V_{\mathcal{G}}$*;*
2. $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ *is* $\mathsf{Pb}_{\mathcal{G}}$*-adhesive.*

Using Corollary 4.6 and Corollary 4.9 we immediately get the following result.

Corollary 4.26. *Let* $h = (h_E, h_V, h_Q)$ *be an arrow in* $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$. *Then the following facts hold*

1. *h is mono if and only if h_E and h_V are injective;*
2. *h is a regular mono if and only if h_E, h_V and h_Q are injective.*

We can now easily define left-monogamous, pruned and separated labelled hypergraphs with equivalence.

Definition 4.27. Let $\mathcal{G}$ be a hypergraph and $(\mathcal{H},(f,g))$ a labelled hypergraph with equivalence. We say that $(\mathcal{H},(f,g))$ is a *left-monogamous* (resp. *separated, pruned*) $\mathcal{G}$-hypergraph with equivalence if $\mathcal{H}$ is left-monogamous (resp. separated, pruned).

We will denote by $\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$, $\mathbf{Sep}_{\mathsf{eq},\mathcal{G}}$, $\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ the subcategories of $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ given by left-monogamous, separated, pruned and separated labelled hypergraphs with equivalence.

Clearly we can realize $\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$, $\mathbf{Sep}_{\mathsf{eq},\mathcal{G}}$, $\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ as comma categories.

Proposition 4.28. *Let $\mathcal{G}$ be an hypergraph with equivalence and consider the inclusion functors $M_{\mathsf{eq}}\colon \mathbf{LMH}_{\mathsf{eq}} \to \mathbf{Hyp}_{\mathsf{eq}}$, $S_{\mathsf{eq}}\colon \mathbf{Sep}_{\mathsf{eq}} \to \mathbf{Hyp}_{\mathsf{eq}}$ and $S_{p,\mathsf{eq}}\colon \mathbf{pSep}_{\mathsf{eq}} \to \mathbf{Hyp}_{\mathsf{eq}}$. Let $\Gamma_{\mathcal{G}}\colon 1 \to \mathbf{Hyp}_{\mathsf{eq}}$ be the constant functor in $\mathcal{G}$. Then*

1. $\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ *is equivalent to* $M_{\mathsf{eq}}{\downarrow}\Gamma_{\mathcal{G}}$;
2. $\mathbf{Sep}_{\mathsf{eq},\mathcal{G}}$ *is equivalent to* $S_{\mathsf{eq}}{\downarrow}\Gamma_{\mathcal{G}}$;
3. $\mathbf{pSep}_{\mathcal{G}}$ *is equivalent to* $S_{p,\mathsf{eq}}{\downarrow}\Gamma_{\mathcal{G}}$.

In particular, the previous proposition, together with Lemmas 4.18, 4.19, B.3 and Corollary B.4, yields the following result.

Corollary 4.29. $\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ *under equalizers, binary product and pullbacks.* $\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ *under pullbacks. Both are closed in* $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ *under pushouts along arrows in* $\mathsf{Pb}_{\mathcal{G}}$.

Definition 4.30. We define two classes of arrows

- $\mathsf{LPb}_{\mathcal{G}}$ is the class of all arrows in $\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ sent by $M_{\mathsf{eq},\mathcal{G}}$ to $\mathsf{Pb}_{\mathcal{G}}$;
- $\mathsf{SPb}_{\mathcal{G}}$ is the class of all arrows in $\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ sent by the inclusion to $\mathsf{Pb}_{\mathcal{G}}$.

We can now use Theorem 2.4 together with Lemmas 4.18 and 4.19 to get the following.

Corollary 4.31. $\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ *is* $\mathsf{LPb}_{\mathcal{G}}$*-adhesive and* $\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ *is* $\mathsf{SPb}_{\mathcal{G}}$*-adhesive*

Remark 4.32. When $\mathcal{G}$ is the hypergraph associated to an algebraic signature Σ, then left-monogamous labelled hypergraphs with equivalence are exactly the *term graphs with equivalence* of [6].

5. EGGs

The previous section proved some adhesivity property for the categories $\mathbf{Hyp}_{\mathsf{eq}}$, $\mathbf{LMH}_{\mathsf{eq}}$, $\mathbf{pSep}_{\mathsf{eq}}$ and for their labelled variants. We extend these results to encompass equivalence classes which are closed under the target arrow, i.e. under operator composition, thus precisely capturing EGGs.

5.1. E-hypergraphs

We start introducing the notion of *e-hypergraphs*, hypergraphs equipped with an equivalence relation that is closed under the target arrow: in other words, whenever the relation identifies the source of two hyperedges, it identifies their targets too.

Definition 5.1. Let $\mathcal{G} = (E, V, Q, s, t, q)$ be a hypergraph with equivalence, two hyperedges $e, e' \in E$ are *q-joined* if $q^\star(t(e')) = q^\star(t(e))$.

We will say that $\mathcal{G}$ is an *e-hypergraph* if every two hyperedges e, e' such that $q^\star(s(e')) = q^\star(s(e))$ are q-joined. We will denote by $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ the full subcategory of $\mathbf{Hyp}_{\mathsf{eq}}$ whose objects are e-hypergraphs, and by $E\colon \mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}} \to \mathbf{Hyp}_{\mathsf{eq}}$ the associated inclusion functor.

We can reformulate the definition of e-hypergraphs in two other ways. See Appendix A.4 for a proof

Proposition 5.2. *Let $\mathcal{G} = (E, V, Q, s, t, q)$ be a hypergraph with equivalence and (A, π_1^A, π_2^A), (B, π_1^B, π_2^B) kernel pairs for, respectively, $q^\star \circ s$ and $q^\star \circ t$. Then the following facts are equivalent*

1. $\mathcal{G}$ *is an e-hypergraph;*
2. $q^\star \circ t \circ \pi_1^A = q^\star \circ t \circ \pi_2^A$*;*
3. *there exists a unique arrow* $k\colon A \to B$ *such that*

$$\pi_1^B \circ k = \pi_1^A \qquad \pi_2^B \circ k = \pi_2^A$$

Example 5.3. Consider the hypergraph $\mathcal{G}$ of Example 3.12 and take as quotient the identity $\mathsf{id}_{V_\mathcal{G}} : V_\mathcal{G} : \to \mathcal{G}$. Then the kernel pair of $\mathsf{id}_{V^\star_\mathcal{G}} \circ s$ concide with the kernel pair of s, which is empty. Thus $\mathcal{G}$ is, trivially, an e-hypergraph

As a next step, we we need to prove that connected limits in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ are computed as in $\mathbf{Hyp}_{\mathsf{eq}}$. Full proofs are in Appendix A.4.

Lemma 5.4. *Let* $F\colon \mathbf{D} \to \mathbf{Hyp}_{\mathsf{eq}}$ *be a connected diagram and let* $F(D)$ *be* $(E_D, V_D, Q_D, s_D, t_D, q_D)$*. Suppose that* $(\mathcal{L}, \{(a_D, b_D, c_D)\}_{D\in\mathbf{D}})$ *is a limiting cone for* F*, then for every hyperedges* $e, e' \in E_\mathcal{L}$*, if* $a_D(e)$ *and* $a_D(e')$ *are* q_D*-joined for every* $D \in \mathbf{D}$*, then* e *and* e' *are* $q_\mathcal{L}$*-joined.*

Corollary 5.5. $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ *has all connected limits and* E *creates them.*

Corollary 5.6. *If an arrow* $h : \mathcal{G} \to \mathcal{H}$ *in* $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ *is a regular mono in* $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ *then* $E(h)$ *is a regular mono in* $\mathbf{Hyp}$.

In order to provide conditions under which pushouts in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ are computed as in $\mathbf{Hyp}_{\mathsf{eq}}$, we need to restrict ourselves to a suitable class of morphisms.

Definition 5.7. We define Pb^+ as the class of arrows $(h_E, h_V, h_Q)\colon \mathcal{G} \to \mathcal{H}$ in $\mathbf{Hyp}_{\mathsf{eq}}$ which belongs to Pb and such that the square on the right is a pullback.

$$\begin{tikzcd} E_\mathcal{G} \arrow[r, "s_\mathcal{G}"] \arrow[d, tail, "h_E"'] & V_\mathcal{G} \arrow[d, tail, "h_V"] \\ E_\mathcal{H} \arrow[r, "s_\mathcal{H}"] & V_\mathcal{H} \end{tikzcd}$$

As for Pb, we must now show that Pb^+ is a suitable class for adhesivity. A proof is in Appendix A.4.

Lemma 5.8. *The class* Pb^+ *contains all isomorphisms, it is closed under composition, decomposition and it is stable under pullbacks and pushouts.*

We can now turn to pushouts. We refer again the reader to Appendix A.4 for the details.

Lemma 5.9. *Suppose that the square on the right is a pushout in* $\mathbf{Hyp}_{\mathsf{eq}}$ *and that* m *is a mono in* Pb^+*. If* $\mathcal{G}_1$*,* $\mathcal{G}_2$ *and* $\mathcal{G}_3$ *are e-hypergraphs, then* $\mathcal{P}$ *is is an e-hypergraph.*

$$\begin{tikzcd} \mathcal{G}_1 \arrow[r, "h"] \arrow[d, tail, "m"'] & \mathcal{G}_2 \arrow[d, "n"] \\ \mathcal{G}_3 \arrow[r, "z"] & \mathcal{P} \end{tikzcd}$$

Let now ePb^+ be the class of arrows in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ that are sent to Pb^+ by E. By definition and Lemma 5.8, we have that this class is closed under composition and decomposition, it contains all isomorphisms and it is stable under pullbacks. Using Lemma 5.8 and Lemma 5.9 we can further deduce that it is stable under pushouts. Then Theorem 2.4 and Corollary 4.15 yield the following result.

Corollary 5.10. $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ *is* ePb^+*-adhesive.*

As in the previous section, we can focus ourselves to two subcategories of $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$.

Definition 5.11. We will define $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq}}$ and $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq}}$ as the full subcategories of $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ given by those e-hypergraphs that are, respectively, left-monogamous and pruned and separated. We will denote by M_e and by S_e the corresponding functors.

To study the adhesivity of these subcategories we need to introduce analogs of Pb^+.

Definition 5.12. We define two classes of arrows

- LPb^+ is the class of all arrows in $\mathbf{LMH}_{\mathsf{eq}}$ sent by M_{eq} to Pb^+;
- SPb^+ is the class of all arrows in $\mathbf{pSep}_{\mathsf{eq}}$ sent by $S_{p,\mathsf{eq}}$ to Pb^+.

Let eLPb^+ and eSPb^+ be the classes of arrows in $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq}}$ and $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq}}$ lying in LPb^+ and SPb^+, respectively. By Lemmas 4.18, 4.19 and Corollary 5.10 we obtain immediately the following result.

Corollary 5.13. $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq}}$ *is* eLPb^+*-adhesive and* $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq}}$ *is* eSPb^+*-adhesive.*

5.2. Labelled e-hypergraphs

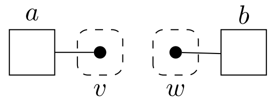
Finally, we can turn our attention to the labelled context. In such an environment, we want the equivalence relation to identify the target of two hyperedges whenever their sources are related *and* their labels are the same. This justifies the following definition.

Definition 5.14. Let $\mathcal{G}$ be a hypergraph with equivalence. We say that an object $(a, b, c)\colon \mathcal{H} \to \mathcal{G}$ of $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ is a *labelled e-hypergraph* if every two hyperedges $e, e' \in E_{\mathcal{H}}$ such that

$$q^{\star}_{\mathcal{H}}(s(e)) = q^{\star}_{\mathcal{H}}(s(e')) \qquad a(e) = a(e')$$

are $q_{\mathcal{H}}$-joined.

We define the category $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ as the full subcategory of $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ given by labelled e-hypergraphs. We denote by $E_{\mathcal{G}}$ the corresponding inclusion functor .



Example 5.15. Let $\mathcal{G}$ be the graph aside. Note that $\mathsf{id}_{\mathcal{G}}$ is a labelled e-hypergraph even if its underlying hypergraphs with equivalence $\mathcal{G}$ is not in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$.

We now provide an equivalent characterisation to labelled e-hypergraphs (see Appendix A.4 for details).

Lemma 5.16. *Let $(a, b, c)\colon \mathcal{H} \to \mathcal{G}$ be an object of $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ and (A, π^A_1, π^A_2), (B, π^B_1, π^B_2) kernel pairs for, respectively, $(q^{\star} \circ s_{\mathcal{H}}, a), (q^{\star} \circ t_{\mathcal{H}}, a)\colon E_{\mathcal{H}} \to Q_{\mathcal{H}} \times E_{\mathcal{G}}$. Then the following are equivalent*

1. *(a, b, c) is a labelled e-hypergraph;*
2. *$q^{\star} \circ t_{\mathcal{H}} \circ \pi^A_1 = q^{\star} \circ t_{\mathcal{H}} \circ \pi^A_2$;*
3. *there exists a unique arrow $k\colon A \to B$ such that*

$$\pi^B_1 \circ k = \pi^A_1 \qquad \pi^B_2 \circ k = \pi^A_2$$

Definition 5.17. Let $V_{\mathcal{G}}\colon \mathbf{Hyp}_{\mathsf{eq},\mathcal{G}} \to \mathbf{Hyp}_{\mathsf{eq}}$ be the forgetful functor, we define $\mathsf{Pb}^+_{\mathcal{G}}$ as the class of arrows in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ whose image in $\mathbf{Hyp}_{\mathsf{eq}}$ is in Pb^+. Moreover, we will denote by $\mathsf{ePb}^+_{\mathcal{G}}$ the class of morphisms in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ whose image through $E_{\mathcal{G}}$ is in $\mathsf{Pb}^+_{\mathcal{G}}$.

We want now to show that pullbacks and pushouts along arrows in $\mathsf{ePb}^+_{\mathcal{G}}$ in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ are computed as in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$. See Appendix A.4 for a proof.

Lemma 5.18. *Let $\mathcal{G}$ be a hypergraph with equivalence. Then $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ is closed in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ under connected limits and pushouts along arrows in $\mathsf{ePb}^+_{\mathcal{G}}$.*

By Theorem 2.4 we get the following result.

Corollary 5.19. *$\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ is $\mathsf{ePb}^+_{\mathcal{G}}$-adhesive.*

Now, let $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ and $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ be the full subcategories of $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ given by the objects whose underlying hypergraphs with equivalence are, respectively, left-monogamous and pruned and separated.

Remark 5.20. When $\mathcal{G}$ is the hypergraph associated to an algebraic signature, the object of $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ coincides with what in [6] are called *e-term graphs with equivalence.*

Putting together Corollary 4.29 with Lemma 5.18 we get the following result.

Lemma 5.21. *Let $\mathcal{G}$ be a hypergraph with equivalence, then $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ and $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ are closed in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ under pullbacks and pushouts along arrows in $\mathsf{Pb}_{\mathcal{G}}$.*

Define $\mathsf{eLPb}^+_{\mathcal{G}}$ and $\mathsf{eSPb}^+_{\mathcal{G}}$ as the classes of arrows whose underlying morphisms are in $\mathsf{Pb}_{\mathcal{G}}$ or in $\mathsf{Pb}_{\mathcal{G}}$. The following result now follows at once.

Corollary 5.22. *$\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\mathcal{G}}$ is $\mathsf{eLPb}^+_{\mathcal{G}}$-adhesive and $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\mathcal{G}}$ is $\mathsf{eSPb}^+_{\mathcal{G}}$-adhesive.*

6. On term representations, and their adhesive rewriting

Qua ci vorrebbe una tabella riassuntiva

The previous sections offered a tour on various categories of hypergraphs, with and without an equivalence. In particular, we have shown how they can be described as suitable $\mathcal{M}$-adhesive categories, as summarised by the table below

Qua ci vorrebbe una tabella riassuntiva

The fact holds for various classes of hypergraphs, and most importantly, for the class of e-hypergraphs, where the equivalence is closed with respect to operator composition. The overall trajectory aimed at recasting the EGGs formalism, as originally presented in [36], in the adhesive mold. However, when confronted with the many graphical presentations discussed in the previous chapters, the reader may feel overwhelmed.

6.1. Representing terms

Classically, the introduction of term graphs (that is, left-monogamous algebraic hypergraphs) was motivated by the desire to provide an alternative representation of terms, based on graphs instead of trees, to exploit the sharing of sub-terms for efficiency reasons. Consider e.g. a signature Σ with a constant a and a binary operator $/$: the term a/a admits two different representations as a term graph in $\mathbf{LMH}_\Sigma$. The image on the left below is the usual representation of the term as a tree, the image on the right is an hypergraph where the occurrence of the constant a is shared

figure 1 e 3 sotto senza le classi di equivalenza

The introduction of an equivalence relation on nodes adds another layer of expressiveness, so that e.g. the term a/a actually has many different presentations in $\mathbf{Hyp}_{\mathsf{eq},\Sigma}$. Four of them are represented below

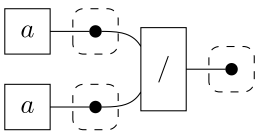


Figure (1).

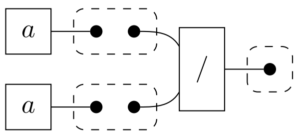


Figure (2).

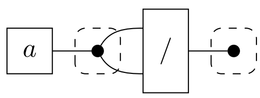


Figure (3).

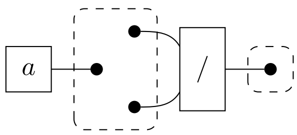


Figure (4).

The pictures above depict four left-monogamous hypergraphs (i.e. term graphs) with equivalence labelled over Σ, thus living in $\mathbf{LMH}_{\mathsf{eq},\Sigma}$. In the picture (1) each equivalence class contains only one node, and its image along the forgetful functor $\mathbf{LMH}_{\mathsf{eq},\Sigma} \to \mathbf{LMH}_\Sigma$ (the labeled extension of the functor O_{eq} in Remark 4.17) is the standard representation of a/a as a tree. The picture (2) represents a separated term graph with two occurrences of the constant a and a non-trivial equivalence on nodes. However, they do not live in $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\Sigma}$, that is, neither of them is an e-term graph: constants have no input, and nodes that are targets of edges with the same label and whose source is the empty word must belong to the same equivalence class. Consider now the pictures indicated by (3) and (4). They are both e-term graphs. Picture (3) is the representation in $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\Sigma}$ of a/a as a term graph with a shared constant; while picture (4) is a separated term graph which is the unique representation of a/a n $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\Sigma}$.

In the original paper [15] EGGs were simply defined as term graphs with an equivalence relation on nodes that is composition-closed, resulting in some ambiguity in which is the canonical EGG associated to a term. To this end, we decided to introduce two different categories, namely $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\Sigma}$ and $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\Sigma}$. In particular, e-term graphs offer a representation that directly recalls the original definition; while the sub-category of pruned, separated e-term graphs guarantees a unique representation for a term, and it seems more in touch with recent implementations [34].

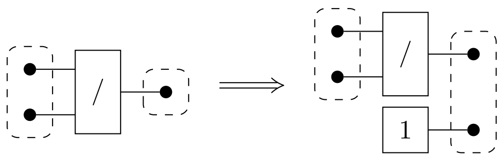
6.2. DPO rewriting on e-term graphs

According to the double-pushout (DPO) approach to rewriting in $\mathcal{M}$-adhesive categories [**?**], a rule is given by two arrows $l : K \to L$ and $r : K \to R$ and its application requires a match $m : L \to G$: the rewriting step from G to H is then given by the diagram below, whose halves are pushouts

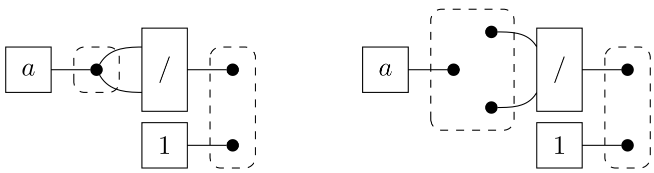
$$\begin{tikzcd} L \arrow[d, "m"'] & K \arrow[l, "l"'] \arrow[r, "r"] \arrow[d, "k"'] & R \arrow[d, "h"] \\ G & C \arrow[l, "f"] \arrow[r, "g"'] & H \end{tikzcd}$$

The situation is simpler in the EGGs approach since, as we recalled in the introduction, it chooses to just add new terms and link them to the older ones via the equivalence relation, instead of removing sub-terms [15]. So, the corresponding DPO rules are spans $L \leftarrow L \to R$ whose left-hand side is the identity.

Consider the term rewriting rule such $x/x \to 1$, from the key example in [36]. Using e-term graphs (variables are represented as nodes that do not occur among the targets of an edge), it can be modelled as the DPO rule below, concisely given by the arrow $L \to R$



The rule lives in the category $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\Sigma}$, since the intermediate and the right-hand side labelled e-term graphs are separated and pruned. As such, the same rule can be used also for DPO rewriting in $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\Sigma}$ and $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\Sigma}$, since these categories are related by inclusion. However, the rule cannot be applied to the term graphs with equivalence in the pictures (1) and (2) above: there is no match $m : L \to G$ due to the constraints on the equivalence classes. Instead, the result of the application of the rule to the e-term graphs in the pictures (3) and (4) is shown below



An EGGs rewriting system is thus a family of rules in $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\Sigma}$ of the shape $L \leftarrow L \to R$: the left-hand side is the identity, while the right-hand side is a regular mono. The format precisely captures EGGs approach: since it requires that a rule must preserve the L component, that is nodes and edges of the e-hypergraph to which L is mapped must not be deleted, the left-hand side must be the identity, while the arrow $L \to R$ must be an injection, that is a regular mono in $\mathsf{e}\text{-}\mathbf{pSep}_{\mathsf{eq},\Sigma}$ (hence, also in $\mathsf{e}\text{-}\mathbf{LMH}_{\mathsf{eq},\Sigma}$).

6.2.1. Issue 1: Left-linear rules

The properties of $\mathcal{M}$-adhesivity ensure that if the rules are spans of arrows in $\mathcal{M}$, then we can lift the standard properties of the DPO approach that hold in the category of graphs.

As noted above, the rules of an EGGs rewriting system have the shape $L \leftarrow L \to R$: the left-hand side component is the identity, hence it belongs to any possible choice of $\mathcal{M}$. However, the situation is different for the right-hand side. The arrow $L \to R$ does not belong to eLPb^+_Σ (hence, neither in eSPb^+_Σ), thus it is not currently possible to lift to the categories at hand all the properties of the $\mathcal{M}$-adhesive framework.

The asymmetry between left- and right-hand sides falls in the research about left-linear rules for $\mathcal{M}$-adhesive categories, as pursued e.g. in [3]. The solution advanced in [5] is not adequate for our purposes. Indeed, while in their double-categorical framework the rules of a rewriting system can be spans of morphisms with no restriction, the match is required to belong to $\mathcal{M}$, which is restrictive in our context. As an example, consider again the e-term graph on picture (4) above, and look again at the application of the rule above da finire e aggiungere esempio grafico

The match is regular mono, since it is an inclusion on the underlying (pruned and separated) term graphs and an isomorphism on the equivalence classes, yet it does not belong to $\mathsf{eSPb}_{\Sigma}^{+}$, due to the occurrence of the constant a. In fact, all the arrows of the diagram above are regular monos but none is in $\mathsf{eSPb}_{\Sigma}^{+}$.

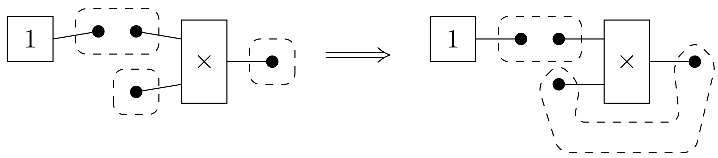
6.2.2. Issue 2: Application conditions [DA AGGIUSTARE]

The fact that the left-hand side of a rule is the identity allows for the repeated application of such rule, hence the possibility to keep on performing the same rewriting step. This is forbidden by using rules *with negative application conditions*, given as the usual span $L \leftarrow K \rightarrow R$ plus an additional arrow $n : L \rightarrow N$, such that a match $m : L \rightarrow G$ is admissible if it cannot be factorised through n [23]. For e-term graphs and rules $L \leftarrow L \rightarrow R$, it suffices to choose n as the right-hand side itself. If you consider e.g. the rule $x/x \rightarrow 1$, once 1 has been added once to the equivalence class of x/x, it is not necessary to keep on adding it. The theory of $\mathcal{M}$-adhesivity carries on for rules with negative application conditions, see [17, 18].

6.2.3. Issue 3: Cycles [DA AGGIUSTARE]

We just remarked that the DPO rules under consideration have the shape $L \leftarrow L \rightarrow R$, where the left-hand side is the identity and the right-hand side arrow is a regular mono. Despite their simplicity, though, such rules may raise some expressiveness issues.

The fact that the right-hand side is a regular mono ensures that if the e-hypergraph to whom a rule is applied is acyclic, the same property holds for the e-hypergraph obtained as the result. Note however that cycles may implicitly occur due to the presence of the equivalence on nodes for what are called collapsing rules, i.e. where the right-hand side is a single variable. Looking again at the key example of [36], consider the rule $1 \times x = x$, which can be modelled as the DPO rule below



Intuitively, the e-term graph obtained after the application of the rule seems to represent a recursive term such as $rec_x.1 \times x$, highlighting the difficulty of extracting a term representation out of an e-term graph.

7. Conclusions and further and related works

The aim of our paper was to extend the theory of $\mathcal{M}$-adhesive categories in order to include EGGs, a formalism for program optimisation. To do so, we revisited the notions of hypergraphs and term graphs with equivalence, proving that they are $\mathcal{M}$-adhesive categories, and we extended these results in order to prove the same property for e-hypergraphs and e-term graphs, i.e. where the equivalence satisfies a suitable closure constraint. Summing up, we proved that EGGs are objects of an $\mathcal{M}$-adhesive category and that optimisation steps are obtained via DPO rules (possibly with negative application conditions) whose left-hand side is in $\mathcal{M}$, and that allows for exploiting the properties of the $\mathcal{M}$-adhesive framework.

Future works. Our results open a few threads of research. The first is to check how the $\mathcal{M}$-adhesivity of EGGs can be pushed to model their rewriting via the double-pushout (DPO) approach. We have seen that the rules adopted in the literature of EGGs appears to be spans whose left-hand side is an identity and right-hand side is a regular mono (possibly with negative application conditions), and as such they fit the mould of rewriting on left-linear rewriting in $\mathcal{M}$-adhesive categories. However, it still needs to be shown how parallelism and causality, the key features for DPO rewriting on $\mathcal{M}$-adhesive categories, can be exploited in the context of implementing the EGGs updates. Moreover, extensions of the EGGs formalism could be suggested by the adhesive machinery we developed. In fact, most of the results presented here for hypergraphs can be generalised to hierarchical hypergraphs, that is, hypergraphs with a hierarchy (a partial order) among edges [22, 12]. The additional structure is useful for modelling properties such as encapsulation and sandboxing, and it seems worthwhile to check the expressiveness and applications of hierarchical EGGS.

Related works. Despite the interest they have ben raising as an efficient data structure, we are not aware of any attempt to provide an algebraic characterisation of EGGs, the only exception being [21]. We leave for future work an in-depth comparison among the two proposals, which appear to be related yet quite different. We adopted a more down-to-earth approach by equipping concretely the nodes of a term graph with an equivalence relation, thus directly corresponding to the original presentation [15], at the same time extending and generalising it [4] in the contemporary jargon of adhesive categories. The route chosen in [21] is to consider term graphs as arrows of a symmetric monoidal category, and equipping them with an enrichment over semi-lattices, the resulting formalism being reminiscent of hierarchical hypergraphs. Thus, the solution in [21] is more general than ours since it can be lifted to term graphs defined over categories other than **Set**. At the same time, we both consider the DPO approach for rewriting, even if only our solution guarantees that the resulting category is $\mathcal{M}$-adhesive, thus allowing to exploit the features of the framework as they hold for DPO graph transformation. Moreover, our construction is technically simpler, which facilitates its practical application and theoretical analysis, and aligns with the traditional methodology adopted in the graph transformation community of explicitly endowing (hyper)graphs with additional structure.

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A. Omitted proofs

This section contains the proofs which are omitted from the main body of the paper. We begin recalling a well-known fact about composition and decomposition of pullbacks [28, Lem. 1.1].

Lemma A.1. *Let $\mathbf{X}$ be a category, and consider the diagram aside, in which the right square is a pullback. Then the whole rectangle is a pullback if and only if the left square is one.*

$$\begin{tikzcd}
X \arrow[r, "f"'] \arrow[d, "a"'] & Y \arrow[r, "g"'] \arrow[d, "b"] & Z \arrow[d, "c"] \\
A \arrow[r, "h"] & B \arrow[r, "k"] & C
\end{tikzcd}$$

A.1. Proofs for Section 2

Proposition 2.6. *Let $\mathbf{X}$ be an $\mathcal{M}$-adhesive category. Then the following facts hold*

1. *every $\mathcal{M}$-pushout square is also a pullback;*
2. *every arrow in $\mathcal{M}$ is a regular mono.*

Proof.

1. Consider the cube aside in which the bottom face is an $\mathcal{M}$-pushout. By construction the top face of the cube is a pushout and the back one a pullback. The left face is a pullback because m is mono, thus the Van Kampen property yields that the front and the right faces are pullbacks too and the claim follows.

$$\begin{tikzcd}[row sep=small, column sep=small]
& A \arrow[rr, "g"] \arrow[dl, "\mathsf{id}_A"'] \arrow[dd, tail, "\mathsf{id}_A"' near end] & & B \arrow[dl, "\mathsf{id}_B"'] \arrow[dd, "\mathsf{id}_B"] \\
A \arrow[rr, "g" near start] \arrow[dd, tail, "m"'] & & B \arrow[dd, tail, "n"' near start] \\
& A \arrow[rr, "g" near end] \arrow[dl, tail, "m"] & & B \arrow[dl, tail, "n"] \\
C \arrow[rr, "f"] & & D
\end{tikzcd}$$

2. Let $m \colon X \rightarrowtail Y$ be an arrow in $\mathcal{M}$, we can then take its pushout along itself, which, by the previous point, is also a pullback. It is now immediate to see that m is the equalizer of h and k. $\square$

$$\begin{tikzcd}
X \arrow[r, tail, "m"'] \arrow[d, tail, "m"'] & Y \arrow[d, tail, "h"] \\
Y \arrow[r, tail, "k"] & Z
\end{tikzcd}$$

Lemma 2.12. *Let $f\colon X \to Y$ and $g\colon Z \to W$ be arrows admitting kernel pairs and suppose that the solid part of the four squares below is given. If the leftmost square is commutative, then there is a unique arrow $k_h\colon K_f \to K_g$ making the other three commutative.*

$$
\begin{tikzcd}
X \arrow[r, "h"] \arrow[d, "f"'] & Z \arrow[d, "g"] \\
Y \arrow[r, "t"'] & W
\end{tikzcd}
\quad
\begin{tikzcd}
K_f \arrow[r, dotted, "k_h"] \arrow[d, "\pi_f^1"'] & K_g \arrow[d, "\pi_g^1"] \\
X \arrow[r, "h"'] & Z
\end{tikzcd}
\quad
\begin{tikzcd}
K_f \arrow[r, dotted, "k_h"] \arrow[d, "\pi_f^2"'] & K_g \arrow[d, "\pi_g^2"] \\
X \arrow[r, "h"'] & Z
\end{tikzcd}
\quad
\begin{tikzcd}
K_f \arrow[r, dotted, "k_h"] \arrow[d, "{(\pi_f^1,\pi_f^2)}"'] & K_g \arrow[d, "{(\pi_g^1,\pi_g^2)}"] \\
X \times X \arrow[r, "h\times h"'] & Z \times Z
\end{tikzcd}
$$

Moreover, the following facts hold

1. *if h is a mono then k_h is a mono;*
2. *if the leftmost square is a pullback then the central two are pullbacks;*
3. *if h is mono and the leftmost square is a pullback then the rightmost is a pullback.*

Proof. We begin by computing

$$g \circ h \circ \pi_f^1 = t \circ f \circ \pi_f^1 = t \circ f \circ \pi_f^2 = g \circ h \circ \pi_f^2$$

so that existence and uniqueness of the wanted k_h follow at once from the universal property of K_g as the pullback of g along itself.

1. Let $a, b\colon T \rightrightarrows K_f$ be two arrows such that $k_h \circ a = k_h \circ b$, then we have

$$
\begin{aligned}
h \circ \pi_f^1 \circ a = \pi_g^1 \circ k_h \circ a = \pi_g^1 \circ k_h \circ b = h \circ \pi_f^1 \circ b \\
h \circ \pi_f^2 \circ a = \pi_g^2 \circ k_h \circ a = \pi_g^2 \circ k_h \circ b = h \circ \pi_f^2 \circ b
\end{aligned}
$$

   Since h is mono this entails that

$$\pi_f^1 \circ a = \pi_f^1 \circ b \qquad \pi_f^2 \circ a = \pi_f^2 \circ b$$

   and thus a must coincide with b.

2. To prove the second half of the claim, we notice that, by Lemma A.1, two rectangles below are pullbacks

$$
\begin{tikzcd}
K_f \arrow[r, "\pi_f^2"] \arrow[d, "\pi_f^1"'] & X \arrow[r, "h"] \arrow[d, "f"] & Z \arrow[d, "g"] \\
X \arrow[r, "f"'] & Y \arrow[r, "t"'] & W
\end{tikzcd}
\qquad
\begin{tikzcd}
K_f \arrow[r, "\pi_f^1"] \arrow[d, "\pi_f^2"'] & X \arrow[r, "h"] \arrow[d, "f"] & Z \arrow[d, "g"] \\
X \arrow[r, "f"'] & Y \arrow[r, "t"'] & W
\end{tikzcd}
$$

   But then the outer rectangle in the following diagrams are pullbacks too

$$
\begin{tikzcd}
K_f \arrow[r, "k_h"'] \arrow[rr, bend left, "h\circ\pi_f^2"] \arrow[d, "\pi_f^1"'] & K_g \arrow[r, "\pi_g^2"'] \arrow[d, "\pi_g^1"'] & Z \arrow[d, "g"'] \\
X \arrow[r, "h"] \arrow[rr, bend right, "t\circ f"'] & Y \arrow[r, "g"] & W
\end{tikzcd}
\qquad
\begin{tikzcd}
K_f \arrow[r, "k_h"'] \arrow[rr, bend left, "h\circ\pi_f^2"] \arrow[d, "\pi_f^1"'] & K_g \arrow[r, "\pi_g^2"'] \arrow[d, "\pi_g^1"'] & Z \arrow[d, "g"'] \\
X \arrow[r, "h"] \arrow[rr, bend right, "t\circ f"'] & Y \arrow[r, "g"] & W
\end{tikzcd}
$$

   Thus the left halves of the rectangle above are pullbacks again by Lemma A.1.

3. For the last square, let $(t_1, t_2)\colon T \to X \times X$ and $s\colon T \to K_g$ be two arrows such that

$$(\pi_g^1, \pi_g^2) \circ s = (h \times h) \circ (t_1, t_2)$$

Thus, by point 2, there exist two arrows $x, y\colon T \rightrightarrows K_f$ fitting in the diagrams below

$$
\begin{tikzcd}
T \arrow[r, dotted, "x"'] \arrow[rr, bend left, "s"] \arrow[dr, bend right, "t_1"'] & K_f \arrow[r, "k_h"'] \arrow[d, "\pi^1_f"'] & K_g \arrow[d, "\pi^1_g"] \\
& X \arrow[r, tail, "h"'] & Y
\end{tikzcd}
\qquad
\begin{tikzcd}
T \arrow[r, dotted, "y"'] \arrow[rr, bend left, "s"] \arrow[dr, bend right, "t_2"'] & K_f \arrow[r, "k_h"'] \arrow[d, "\pi^2_f"'] & K_g \arrow[d, "\pi^2_g"] \\
& X \arrow[r, tail, "h"'] & Y
\end{tikzcd}
$$

By the first point k_h is mono, thus we can deduce that $x = y$. By construction we have

$$(\pi^1_f, \pi^2_f) \circ x = (t_1, t_2) \qquad k_h \circ x = s$$

Uniqueness now follows once again from the fact that k_h is mono. $\square$

Proposition 2.13. *Let* $\mathbf{X}$ *be a strict* $\mathcal{M}$*-adhesive category with all pullbacks, and suppose that in the cube aside the top face is an* $\mathcal{M}$*-pushout and all the vertical faces are pullbacks. Then the right square is a pushout.*

$$
\begin{tikzcd}
& A' \arrow[dl, tail, "m'"'] \arrow[rr, "f'"] \arrow[dd, "a"' near start] & & B' \arrow[dl, tail, "n'"'] \arrow[dd, "b"] \\
C' \arrow[rr, crossing over, "g'" near start] \arrow[dd, "c"'] & & D' & \\
& A \arrow[dl, tail, "m"'] \arrow[rr, "f" near start] & & B \arrow[dl, tail, "n"] \\
C \arrow[rr, "g"'] & & D \arrow[from=uu, crossing over, "d" near start] &
\end{tikzcd}
\qquad
\begin{tikzcd}
K_a \arrow[r, "k_{f'}"] \arrow[d, "k_{m'}"'] & K_b \arrow[d, "k_{n'}"] \\
K_c \arrow[r, "k_{g'}"] & K_d
\end{tikzcd}
$$

Proof. By Proposition 2.6 we know that the top face of the original cube is a pullback. Thus Lemma 2.12 entails that in the cube aside the vertical faces are pullbacks. The claim now follows from strict $\mathcal{M}$-adhesivity. $\square$

$$
\begin{tikzcd}
& K_a \arrow[dl, "k_{m'}"'] \arrow[rr] \arrow[dd, "\pi^1_a"' near start] & & K_b \arrow[dl, "k_{n'}"'] \arrow[dd, "\pi^1_b"] \\
K_c \arrow[rr, crossing over, "k_{f'}" near start] \arrow[dd, "\pi^1_c"'] & & K_d & \\
& A' \arrow[dl, tail, "m'"'] \arrow[rr, "f'" near start] & & B' \arrow[dl, tail, "n'"] \\
C' \arrow[rr, "g'"] & & D' \arrow[from=uu, crossing over, "\pi^1_d" near start] &
\end{tikzcd}
$$

To prove Lemma 2.14 we need some results about pushouts and coproducts in **Set**.

Lemma A.2. *Suppose that the square aside is a pushout, with* $m\colon A \rightarrowtail B$ *an injection. Let* $\iota_m\colon E \rightarrowtail B$ *be the inclusion of the complement of the image of* m*. Then* $(D, \{n, h \circ \iota_m\})$ *is a coproduct. In particular,* $h \circ \iota_m$ *is mono.*

$$
\begin{tikzcd}
A \arrow[r, tail, "m"'] \arrow[d, "g"'] & B \arrow[d, "h"] \\
C \arrow[r, tail, "n"] & D
\end{tikzcd}
$$

Proof. Let $f\colon E \to Z$ and $k\colon C \to Z$ be two arrows with the same codomain. By definition of complement and since m is mono $(B, \{m, \iota_m\})$ is a coproduct. Let $\phi\colon B \to Z$ be the unique arrow such that

$$\phi \circ m = k \circ g \qquad \phi \circ \iota_m = f$$

By the universality of pushouts there is a unique arrow $\psi\colon D \to Z$ such that $\psi \circ n = k$ and $\psi \circ h = \phi$, thus

$$\psi \circ h \circ \iota_m = \phi \circ \iota_m = f$$

To conclude we have to show that ψ is the unique arrow $D \to Z$ such that $\psi \circ n = k$ and $\psi \circ h \circ \iota_m = f$. Let $\psi'\colon D \to Z$ be another arrow such that $\psi' \circ n = k$ and $\psi' \circ h \circ \iota_m = f$. Then we have

$$\psi' \circ h \circ m = \psi' \circ n \circ g = k \circ g = \phi \circ m = \psi \circ h \circ m$$

Since m is mono then $\psi' \circ h = \psi \circ h$, but $(D, \{n, h\})$ is a pushout and so $\psi' = \psi$. $\square$

The category of **Set** enjoys two other remarkable properties; it is *distributive* and *extensive* [10]. Distributivity amounts to the following property: for every family $\{X_i\}_i$ and an object Y, the unique morphisms ϕ and ψ fitting in the diagrams below, where j_i, k_i and h_i are coprojections, are isomorphisms

$$
\begin{tikzcd}
& Y \times X_i \arrow[dl, tail, "j_i"'] \arrow[dr, tail, "\mathsf{id}_Y \times h_i"] & \\
\sum\limits_{i \in I}(Y \times X_i) \arrow[rr, dotted, "\phi"'] & & Y \times (\sum\limits_{i \in I} X_i)
\end{tikzcd}
\qquad
\begin{tikzcd}
& X_i \times Y \arrow[dl, tail, "k_i"'] \arrow[dr, tail, "h_i \times \mathsf{id}_Y"] & \\
\sum\limits_{i \in I}(X_i \times Y) \arrow[rr, dotted, "\psi"'] & & (\sum\limits_{i \in I} X_i) \times Y
\end{tikzcd}
$$

Extensivity means that given a family of objects $\{X_i\}_{i\in I}$ and a family of commuting squares as the one on the right, where j_i is a coprojection, then all the squares are pullbacks if and only if $(Z, \{k_i\}_{i\in I})$ is a coproduct.

$$\begin{tikzcd} Z_i \arrow[r, "k_i"] \arrow[d, "f_i"'] & Z \arrow[d, "f"] \\ X_i \arrow[r, tail, "j_i"] & X \end{tikzcd}$$

Remark A.3. Notice that extensivity entails that coproducts are *disjoint*, i.e. the pullback between two coprojections is given by the initial object (with initial maps as coprojections). To see this, let $\{X, \{j_i\}_{i\in I}\}$ be the coproduct of the family $\{X_i\}_{i\in I}$ and k an element of I. Consider $(X_k, \{t_i\}_{i\in I})$, where:

$$X_i = \begin{cases} X_k & i = k \\ \emptyset & i \neq k \end{cases} \qquad t_i = \begin{cases} \mathsf{id}_{X_k} & i = k \\ ?_{X_k} & i \neq k \end{cases}$$

is a coproduct and so the squares on the right are pullbacks.

$$\begin{tikzcd} \emptyset \arrow[r, tail, "?_{X_k}"] \arrow[d, "?_{X_i}"'] & X_k \arrow[d, tail, "j_k"] \\ X_i \arrow[r, tail, "j_i"'] & X \end{tikzcd} \qquad \begin{tikzcd} X_k \arrow[r, "\mathsf{id}_X"] \arrow[d, "\mathsf{id}_X"'] & X_k \arrow[d, tail, "j_k"] \\ X_k \arrow[r, tail, "j_k"'] & X \end{tikzcd}$$

Proposition A.4. *Suppose that the square on the right is a pullback. Let $\iota\colon E \rightarrowtail Y$ be the inclusion of the complement of the image of p_1. If there exist two arrows $t_1\colon T \to E$ and $t_2\colon T \to X$ such that $g \circ \iota \circ t_1 = f \circ t_2$ then T is empty.*

$$\begin{tikzcd} P \arrow[r, "p_1"] \arrow[d, "p_2"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"] & Z \end{tikzcd}$$

Proof. By the universal property of pullbacks we get an arrow $t\colon T \to P$ such that

$$p_1 \circ t = \iota \circ t_1 \qquad p_2 \circ t = t_2$$

$$\begin{tikzcd} T \arrow[r, dotted, "s"] \arrow[rr, bend left, "t_1"] \arrow[dr, two heads, "\pi"'] & \emptyset \arrow[r, tail, "?_E"] \arrow[d, "?_I"] & E \arrow[d, tail, "\iota"] \\ & I \arrow[r, tail, "i"] & Y \end{tikzcd}$$

Let $i\colon I \rightarrowtail Y$ be the inclusion of the image of p_1, so that we have also an epi $q\colon P \twoheadrightarrow I$. By definition of complement $(Y, \{i, \iota\})$ is a coproduct, thus by Remark A.3 the inner square in the diagram on the left is a pullback and we get a dotted arrow $s\colon T \to \emptyset$ filling the diagram. But a set with an arrow towards the empty set must be empty and we conclude. □

Extensivity and distributivity, moreover, yield the following fact.

Lemma A.5. *Let $f, g\colon A \rightrightarrows B + C$ be two arrows with a coproduct as codomain. Then $(A, \{j_i\}_{i=0}^3)$ is a coproduct, where $j_i\colon A_i \rightarrowtail A$ is defined by the following pullback squares*

$$\begin{tikzcd} A_0 \arrow[r, tail, "j_0"] \arrow[d, "p_0"'] & A \arrow[d, "(f,g)"] \\ B \times B \arrow[r, tail, "i_B \times i_B"'] & (B+C) \times (B+C) \end{tikzcd} \qquad \begin{tikzcd} A_1 \arrow[r, tail, "j_1"] \arrow[d, "p_1"'] & A \arrow[d, "(f,g)"] \\ B \times C \arrow[r, tail, "i_B \times i_C"'] & (B+C) \times (B+C) \end{tikzcd}$$

$$\begin{tikzcd} A_2 \arrow[r, tail, "j_2"] \arrow[d, "p_2"'] & A \arrow[d, "(f,g)"] \\ C \times B \arrow[r, tail, "i_C \times i_B"'] & (B+C) \times (B+C) \end{tikzcd} \qquad \begin{tikzcd} A_3 \arrow[r, tail, "j_3"] \arrow[d, "p_3"'] & A \arrow[d, "(f,g)"] \\ C \times C \arrow[r, tail, "i_C \times i_C"'] & (B+C) \times (B+C) \end{tikzcd}$$

We are now ready to exploit the previous properties to prove Lemma 2.14.

Lemma 2.14. *Suppose that in* **Set** *the commuting cube in the diagram on the left is given, whose top face is a pushout, the left and bottom faces are pullbacks, and $n\colon B \rightarrowtail D$ is an injection. Then the following facts hold*

1. *the right face of the cube is a pullback;*
2. *the right square, made by the kernel pairs of the vertical arrows, is a pushout.*

$$\begin{tikzcd}[row sep=small, column sep=small] & A' \arrow[rr, "f'"] \arrow[dl, tail, "m'"'] \arrow[dd, "a" near start] & & B' \arrow[dl, tail, "n'"'] \arrow[dd, "b"] \\ C' \arrow[rr, crossing over, "g'" near start] \arrow[dd, "c"'] & & D' & \\ & A \arrow[rr, "f" near start] \arrow[dl, tail, "m"] & & B \arrow[dl, tail, "n"] \\ C \arrow[rr, "g"'] & & D \arrow[from=uu, crossing over, "d" near start] & \end{tikzcd} \qquad \begin{tikzcd} K_a \arrow[r, "k_{f'}"] \arrow[d, "k_{m'}"'] & K_b \arrow[d, "k_{n'}"] \\ K_c \arrow[r, "k_{g'}"'] & K_d \end{tikzcd}$$

Proof. We can start noticing that m, being the pullback of n, is mono too. This, in turn, entails that m', being the pullback of m, is mono, and so n' is injective too because in **Set**, as in any adhesive category, monomorphisms are stable under pushouts.

Let $\iota' \colon E' \rightarrowtail C'$ and $\iota \colon E \to C$ be the inclusions of the complement of the images of of m' and m respectively . By Lemma A.2, $(D', \{n', g' \circ \iota\})$ is a coproduct.

$$\begin{tikzcd} E' \arrow[r, tail, "\iota'"] \arrow[d, dotted, "w"'] & C' \arrow[d, "c"] \\ E \arrow[r, tail, "\iota"] & C \end{tikzcd}$$

We can also notice that $c \circ \iota'$ factors through E, as shown by the square on the right. To see this, let e be in E' and suppose that $c(\iota'(e)) = m(x)$ for some $x \in A$. Then we can apply the universal property of pullbacks to build the dotted arrow v in the diagram aside, where v_0 and v_1 are the arrows picking x and $\iota'(e)$, respectively. But then e must belong to the image of m', which is a contradiction.

$$\begin{tikzcd} 1 \arrow[r, dotted, "v"'] \arrow[rr, bend left, "v_1"] \arrow[dr, "v_0"'] & A' \arrow[r, tail, "m'"'] \arrow[d, "a"'] & C' \arrow[d, "c"] \\ & A \arrow[r, tail, "m"] & C \end{tikzcd}$$

1. Consider two arrows $t_1 \colon T \to D'$ and $t_2 \colon T \to B$ such that $d \circ t_1 = n \circ t_2$. By extensivity of **Set**, we already know that $(T, \{l_i\}_{i=0}^{1})$ is a coproduct, where $l_i \colon T_i \rightarrowtail T$ are defined by the diagram aside, whose two halves are pullbacks.

$$\begin{tikzcd} T_0 \arrow[r, tail, "l_0"'] \arrow[d, "h_0"'] & T \arrow[d, "t_1"] & T_1 \arrow[l, tail, "l_1"] \arrow[d, "h_1"] \\ E' \arrow[r, tail, "g' \circ \iota'"] & D' & B' \arrow[l, tail, "n'"'] \end{tikzcd}$$

   If we compute we get

$$g \circ c \circ \iota' \circ h_0 = d \circ g' \circ \iota' \circ h_0 = d \circ t_1 \circ l_0 = n \circ t_2 \circ l_0$$

   The bottom face of the given cube is a pullback, thus there is an arrow $u \colon T_0 \to A$ such that

$$f \circ u = n \circ t_2 \circ l_0 \qquad m \circ u = c \circ \iota' \circ h_0$$

   By Proposition A.4 we conclude that T_0 is empty. Therefore l_1 is an isomorphism and we have $n' \circ h_1 \circ l_1^{-1} = t_1$. On the other hand

$$n \circ b \circ h_1 \circ l_1^{-1} = d \circ n' \circ h_1 \circ l_1^{-1} = d \circ t_1 = n \circ t_2$$

   and we can conclude that $b \circ h_1 \circ l_1^{-1} = t_2$ because n is a mono. The claim now follows from the fact that also n' is mono.
2. By Lemma A.2 and Lemma A.5, the previous point and the third point of Lemma 2.12, we can decompose K_d as the coproduct of K_b K_1, K_2 and K_3 with coprojections given by, respectively, $k_{n'}$ and j_1, j_2 and j_3, where these objects and arrows fit in the following four pullbacks

$$\begin{tikzcd} K_b \arrow[r, tail, "k_{n'}"] \arrow[d, "(\pi_b^1, \pi_b^2)"'] & K_d \arrow[d, "(\pi_d^1, \pi_d^2)"] \\ B' \times B' \arrow[r, tail, "n' \times n'"'] & D' \times D' \end{tikzcd} \qquad \begin{tikzcd} K_1 \arrow[r, tail, "j_1"] \arrow[d, "(p_1, q_1)"'] & K_d \arrow[d, "(\pi_d^1, \pi_d^2)"] \\ B' \times E' \arrow[r, tail, "n' \times (g' \circ \iota')"'] & D' \times D' \end{tikzcd}$$

$$\begin{tikzcd} K_2 \arrow[r, tail, "j_2"] \arrow[d, "(p_2, q_2)"'] & K_d \arrow[d, "(\pi_d^1, \pi_d^2)"] \\ E' \times B' \arrow[r, tail, "(g' \circ \iota') \times n'"'] & D' \times D' \end{tikzcd} \qquad \begin{tikzcd} K_3 \arrow[r, tail, "j_3"] \arrow[d, "(p_3, q_3)"'] & K_d \arrow[d, "(\pi_d^1, \pi_d^2)"] \\ E' \times E' \arrow[r, tail, "(g' \circ \iota') \times (g' \circ \iota')"'] & D' \times D' \end{tikzcd}$$

   Let us now examine K_1 and K_2. We have

$$n \circ b \circ p_1 = d \circ n' \circ p_1 = d \circ \pi_d^1 \circ j_1 = d \circ \pi_d^2 \circ j_1 = d \circ g' \circ \iota' \circ q_1 = g \circ c \circ \iota' \circ q_1$$
$$n \circ b \circ q_2 = d \circ n' \circ p_2 = d \circ \pi_D^2 \circ j_2 = d \circ \pi_d^1 \circ j_2 = d \circ g' \circ \iota' \circ p_2 = g \circ c \circ \iota' \circ p_2$$

   Since by hypothesis the bottom face of the original cube is a pullback, we conclude that there exist arrows $k_1 \colon K_1 \to A$ and $k_2 \colon K_2 \to A$ such that

$$m \circ k_1 = c\iota' \circ q_1 \quad f \circ k_1 = b \circ p_1 \quad m \circ k_2 = c\iota' \circ q_2 \quad f \circ k_2 = b \circ q_2$$

   By Proposition A.4 we conclude that K_1 and K_2 are both empty. In particular, this implies that $(K_d, \{k_{n'}, j_3\})$ is a coproduct.

We focus now on K_3. By computing we get

$$
\begin{aligned}
&g \circ \iota \circ w \circ p_3 = g \circ c \circ \iota' \circ p_3 = d \circ g' \circ \iota' \circ p_3 = d \circ \pi_d^1 \circ j_3 \\
=&d \circ \pi_d^2 \circ j_3 = d \circ g' \circ \iota' \circ q_3 = g \circ c \circ \iota' \circ q_3 = g \circ \iota \circ w \circ q_3
\end{aligned}
$$

$$
\begin{tikzcd}
K_3 \arrow[r, "p_3"] \arrow[d, "q_3"'] \arrow[dr, dotted, "\phi_3"] & E' \arrow[dr, "\iota'"] & \\
E' \arrow[dr, "\iota'"'] & K_c \arrow[r, "\pi_c^1"] \arrow[d, "\pi_c^2"'] & C' \arrow[d, "c"] \\
& C' \arrow[r, "c"'] & C
\end{tikzcd}
$$

By Lemma A.2 $g \circ \iota$ is a monomorphism, thus $w \circ p_3 = w \circ q_3$ and therefore

$$
c \circ \iota' \circ p_3 = \iota \circ w \circ p_3 = \iota \circ w \circ q_3 = c \circ \iota' \circ q_3
$$

Thus we get the dotted $\phi_3 \colon K_3 \to K_c$ in the diagram aside. Moreover, $k_{g'} \circ \phi_3 = j_3$ as shown by the following computation

$$
\begin{aligned}
&(\pi_d^1, \pi_d^2) \circ k_{g'} \circ \phi_3 = (g' \times g') \circ (\pi_c^1, \pi_c^2) \circ \phi_3 = (g' \times g') \circ (\iota' \circ p_3, \iota' \circ q_3) \\
=&((g' \circ \iota') \times (g' \circ \iota')) \circ (p_3, q_3) = (\pi_d^1, \pi_d^2) \circ j_3
\end{aligned}
$$

By definition of complement, we also know that $(C', \{\iota', m'\})$ is a coproduct. We can then apply again Lemma A.5, decomposing K_c in four parts, given by the pullbacks below

$$
\begin{tikzcd}
H_0 \arrow[r, tail, "x_0"] \arrow[d, "(y_0,z_0)"'] & K_c \arrow[d, "(\pi_c^1,\pi_c^2)"] \\
A' \times A' \arrow[r, tail, "m' \times m'"'] & C' \times C'
\end{tikzcd}
\qquad
\begin{tikzcd}
H_1 \arrow[r, tail, "x_1"] \arrow[d, "(y_1,z_1)"'] & K_c \arrow[d, "(\pi_c^1,\pi_c^2)"] \\
A' \times E' \arrow[r, tail, "m' \times \iota'"'] & C' \times C'
\end{tikzcd}
$$

$$
\begin{tikzcd}
H_2 \arrow[r, tail, "x_2"] \arrow[d, "(y_2,z_2)"'] & K_c \arrow[d, "(\pi_c^1,\pi_c^2)"] \\
E' \times A' \arrow[r, tail, "\iota' \times m'"'] & C' \times C'
\end{tikzcd}
\qquad
\begin{tikzcd}
H_3 \arrow[r, tail, "x_3"] \arrow[d, "(y_3,z_3)"'] & K_c \arrow[d, "(\pi_c^1,\pi_c^2)"] \\
E' \times E' \arrow[r, tail, "\iota' \times \iota'"'] & C' \times C'
\end{tikzcd}
$$

Let us examine H_0. We start noticing that

$$
\begin{aligned}
(\pi_d^1, \pi_d^2) \circ k_{g'} \circ x_0 =&(g' \times g') \circ (\pi_c^1, \pi_c^2) \circ x_0 \\
=&(g' \times g') \circ (m' \times m') \circ (y_0, z_0) \\
=&((g' \circ m') \times (g' \circ m')) \circ (y_0, z_0) \\
=&((n' \circ f') \times (n' \circ f')) \circ (y_0, z_0) \\
=&(n' \times n') \circ (f' \times f') \circ (y_0, z_0)
\end{aligned}
$$

$$
\begin{tikzcd}
H_0 \arrow[rr, "(y_0,z_0)"] \arrow[d, "x_0"'] \arrow[dr, dotted, "h_0"] & & A' \times A' \arrow[d, "f' \times f'"] \\
K_c \arrow[dr, "k_{g'}"'] & K_b \arrow[r, "(\pi_b^1,\pi_b^2)"'] \arrow[d, "k_{n'}"'] & B' \times B' \arrow[d, "n' \times n'"] \\
& K_d \arrow[r, "(\pi_d^1,\pi_d^2)"] & D' \times D'
\end{tikzcd}
$$

Thus we get the dotted arrow $h_0 \colon H_0 \to K_b$ in the diagram above. Moreover, we have

$$
m \circ a \circ y_0 = c \circ m' \circ y_0 = c \circ \pi_c^1 \circ x_0 = c \circ \pi_c^2 \circ x_0 = c \circ m' \circ z_0 = m \circ a \circ z_0
$$

Since m is a mono $a \circ y_0 = a \circ z_0$ and there exists an arrow $h_0' \colon H_0 \to K_a$ such that $\pi_a^1 \circ h_0' = y_0$ and $\pi_a^2 \circ h_0' = z_0$. We now can conclude that $k_{f'} \circ h_0' = h_0$, noticing that

$$
(\pi_b^1, \pi_b^2) \circ h_0 = (f' \times f') \circ (y_0, z_0) = (f' \times f') \circ (\pi_a^1, \pi_a^2) \circ h_0' = (\pi_b^1, \pi_b^2) \circ k_{f'} \circ h_0'
$$

We can prove another property of h_0'. By computing we have

$$
(\pi_c^1, \pi_c^2) \circ k_{m'} \circ h_0' = (m' \times m') \circ (\pi_a^1, \pi_a^2) \circ h_0' = (m' \times m') \circ (y_0, z_0) = (\pi_c^1, \pi_c^2) \circ x_0
$$

so that $k_{m'} \circ h_0'$ must coincide with x_0.

As a next step, let us focus on H_1 and H_2. Two computations yield

$$\begin{aligned}(\pi_d^1,\pi_d^2)\circ k_{g'}\circ x_1 = &(g'\times g')\circ(\pi_c^1,\pi_c^2)\circ x_1\\ =&(g'\times g')\circ(m'\times\iota')\circ(y_1,z_1)\\ =&((g'\circ m')\times(g'\circ\iota'))\circ(y_1,z_1)\\ =&((n'\circ f')\times(g'\circ\iota'))\circ(y_1,z_1)\\ =&(n'\times(g'\circ\iota'))\circ(f'\circ y_1,z_1)\end{aligned}\qquad \begin{aligned}(\pi_d^1,\pi_d^2)\circ k_{g'}\circ x_2 = &(g'\times g')\circ(\pi_c^1,\pi_c^2)\circ x_2\\ =&(g'\times g')\circ(\iota'\times m')\circ(y_2,z_2)\\ =&((g'\circ\iota')\times(g'\circ m'))\circ(y_2,z_2)\\ =&((g'\circ\iota')\times(n'\circ f'))\circ(y_2,z_2)\\ =&((g'\circ\iota')\times n')\circ(y_2,f'\circ z_2)\end{aligned}$$

Hence, we have arrows $h_1\colon H_1\to K_1$ and $h_2\colon H_2\to K_2$, showing that H_1 and H_2 are empty. For H_3, let us consider the two diagrams below

$$\begin{tikzcd} H_3 \arrow[r, "x_3"] \arrow[d, dotted, "\alpha_3"] \arrow[dd, bend right, "(y_3,z_3)"'] & K_c \arrow[d, "k_{g'}"'] \\ K_3 \arrow[r, tail, "j_3"] \arrow[d, "(p_3,q_3)"] & K_d \arrow[d, "(\pi_d^1,\pi_d^2)"] \\ E'\times E' \arrow[r, tail, "(g'\circ\iota')\times(g'\circ\iota')"'] & D'\times D' \end{tikzcd}\qquad \begin{tikzcd} K_3 \arrow[dr, bend left, "\phi_3"] \arrow[d, dotted, "\beta_3"] \arrow[dd, bend right, "(p_3,q_3)"'] & \\ H_3 \arrow[r, tail, "x_3"] \arrow[d, "(y_3,z_3)"] & K_c \arrow[d, "(\pi_c^1,\pi_c^2)"] \\ E'\times E' \arrow[r, tail, "\iota'\times\iota'"'] & C'\times C' \end{tikzcd}$$

Their solid part commute. Indeed, we have

$$\begin{aligned}(\pi_d^1,\pi_d^2)\circ k_{g'}\circ x_3 = &(g'\times g')\circ(\pi_c^1,\pi_c^2)\circ x_3\\ =&(g'\times g')\circ(\iota'\times\iota')\circ(y_3\times z_3)\\ =&(g'\circ\iota')\times(g'\circ\iota')\circ(y_3,z_3)\end{aligned}\qquad \begin{aligned}(\pi_c^1,\pi_c^2)\circ\phi_3 =&(\iota'\circ p_3,\iota'\circ q_3)\\ =&(\iota'\times\iota')\circ(p_3,q_3)\end{aligned}$$

Thus we get the dotted arrows α_3 and β_3, which are mutually inverse. Indeed, on the one hand

$$j_3\circ\alpha_3\circ\beta_3 = k_{g'}\circ x_3\circ\beta_3 = k_{g'}\circ\phi_3 = j_3 \qquad (p_3,q_3)\circ\alpha_3\circ\beta_3 = (y_3,z_3)\circ\beta_3 = (p_3,q_3)$$

On the other hand, notice that

$$(\pi_c^1,\pi_c^2)\circ\phi_3\circ\alpha_3 = (\iota'\times\iota')\circ(p_3,q_3)\circ\alpha_3 = (\iota'\times\iota')\circ(y_3,z_3) = (\pi_c^1,\pi_c^2)\circ x_3$$

Therefore $\phi_3\circ\alpha_3 = x_3$ and thus $x_3\circ\beta_3\circ\alpha_3 = x_3$. Moreover

$$(y_3,z_3)\circ\beta_3\circ\alpha_3 = (p_3,q_3)\circ\alpha_3 = (y_3,z_3)$$

Summing up, we have just proved that $(K_c,\{x_0,\phi_3\})$ is a coproduct.
Now let $\gamma\colon K_b\to Z$ and $\delta\colon K_c\to Z$ be two arrows such that $\gamma\circ k_{f'} = \delta\circ k_{m'}$. We want to construct an arrow $\theta\colon K_d\to Z$ such that $\theta\circ k_{g'} = \delta$ and $\theta\circ k_{n'} = \gamma$.
We have already proved that $(K_d,\{k_n,j_3\})$ is a coproduct, thus there is a unique arrow $\theta\colon K_d\to Z$ such that $\theta\circ k_{n'} = \gamma$ and $\theta\circ j_3 = \delta\circ\phi_3$. Now, on the one hand we have

$$\theta\circ k_{g'}\circ\phi_3 = \theta\circ j_3 = \delta\circ\phi_3$$

On the other hand, we can conclude that $\theta\circ k_{g'} = \delta$ as wanted, since

$$\theta\circ k_{g'}\circ x_0 = \theta\circ k_{n'}\circ h_0 = \gamma\circ h_0 = \gamma\circ k_{f'}\circ h_0' = \delta\circ k_{m'}\circ h_0' = \delta\circ x_0$$

We are left with uniqueness. Let θ' be another arrow $K_d\to Z$ such that $\theta'\circ k_{n'} = \gamma$ and $\theta'\circ k_{g'} = \delta$. If we compute we get

$$\theta'\circ j_3 = \theta'\circ k_{g'}\circ\phi_3 = \delta\circ\phi_3 = \theta\circ j_3$$

We already know that $\theta'\circ k_{n'} = \theta\circ k_{n'}$, so the previous identity entails that $\theta = \theta'$. $\square$

Lemma 3.17. *There is an equivalence of categories $\mathbf{Set}^{\mathbf{H}} \to \mathbf{Hyp}$ sending a functor $F \colon \mathbf{H} \to \mathbf{Set}$ to $\mathcal{G}_F := (E_F, F(\bullet), s_F, t_F)$, where $(E_F, \{\iota^F_{k,l}\})$ is a coproduct for $\{F(k,l)\}_{k,l \in \mathbb{N}}$ and $s_F, t_F \colon E_F \rightrightarrows V_F^\star$ are the unique arrows fitting in the diagrams aside, where $(F(\bullet), \{v^F_n\}_{n\in\mathbb{N}})$ is the coproduct coming from the first point of Proposition 3.1 and $s^F_{k,l}, t^F_{k,l} \colon F(k,l) \to F(\bullet)^k$, $t^F_{k,l} \colon F(k,l) \to F(\bullet)^l$ are the arrows induced by, respectively, $\{F(f_i)\}_{i=0}^{k-1}$ and $\{F(f_i)\}_{i=k}^{k+l-1}$.*

$$\begin{tikzcd} F(k,l) \arrow[r, "s^F_{k,l}"'] \arrow[d, tail, "\iota^F_{k,l}"'] & F(\bullet)^k \arrow[d, tail, "v^F_k"] \\ E_F \arrow[r, dotted, "s_F"] & F(\bullet)^\star \end{tikzcd}$$

$$\begin{tikzcd} F(k,l) \arrow[r, "t^F_{k,l}"'] \arrow[d, tail, "\iota^F_{k,l}"'] & F(\bullet)^l \arrow[d, tail, "v^F_l"] \\ E_F \arrow[r, dotted, "t_F"] & F(\bullet)^\star \end{tikzcd}$$

Proof. Let $\eta \colon F \to G$ be a natural transformation between $F, G \colon \mathbf{H} \rightrightarrows \mathbf{Set}$. By the universal property of coproducts there is an arrow $\hat{\eta} \colon E_F \to E_G$ fitting in the diagram aside.

$$\begin{tikzcd} F(k,l) \arrow[r, "\eta_{(k,l)}"] \arrow[d, tail, "\iota^F_{k,l}"'] & G(k,l) \arrow[d, tail, "\iota^G_{k,l}"] \\ E_F \arrow[r, dotted, "\hat{\eta}"'] & E_G \end{tikzcd}$$

For every $i, n \in \mathbb{N}$ with $i < n$, let $\pi^G_{n,i} \colon G(\bullet)^n \to G(\bullet)$ be the i^{th} projections, then

$$\pi^G_{n,i} \circ s^G_{k,l} \circ \eta_{(k,l)} = G(f_i) \circ \eta_{(k,l)} = \eta_\bullet \circ F(f_i) = \eta_\bullet \circ \pi^G_{n,i} \circ s^F_{k,l} = \pi^G_{n,i} \circ \eta^k_\bullet \circ s^F_{k,l}$$
$$\pi^G_{n,i} \circ t^G_{k,l} \circ \eta_{(k,l)} = G(f_{i+k}) \circ \eta_{(k,l)} = \eta_\bullet \circ F(f_{i+k}) = \eta_\bullet \circ \pi^G_{n,i} \circ t^F_{k,l} = \pi^G_{n,i} \circ \eta^k_\bullet \circ t^F_{k,l}$$

therefore $s^G_{k,l} \circ \eta_{(k,l)} = \eta^k_\bullet \circ s^F_{k,l}$ and $t^G_{k,l} \circ \eta_{(k,l)} = \eta^k_\bullet \circ t^F_{k,l}$. We can use these identities to show that $(\hat{\eta}, \eta_\bullet)$ is a morphism of **Hyp**

$$s_G \circ \hat{\eta} \circ \iota^F_{k,l} = s_G \circ \iota^G_{k,l} \circ \eta_{(k,l)} = v^F_k \circ s^G_{k,l} \circ \eta_{(k,l)} = v^F_k \circ \eta^k_\bullet \circ s^F_{k,l} = \eta^\star_\bullet \circ v^F_k \circ s^F_{k,l} = \eta^\star_\bullet \circ s^F \circ \iota^F_{k,l}$$
$$t_G \circ \hat{\eta} \circ \iota^F_{k,l} = t_G \circ \iota^G_{k,l} \circ \eta_{(k,l)} = v^F_k \circ t^G_{k,l} \circ \eta_{(k,l)} = v^F_k \circ \eta^k_\bullet \circ t^F_{k,l} = \eta^\star_\bullet \circ v^F_k \circ t^F_{k,l} = \eta^\star_\bullet \circ t^F \circ \iota^F_{k,l}$$

Thus for every $\eta \colon F \to G$ we get an arrow $\mathcal{G}_\eta \colon \mathcal{G}_F \to \mathcal{G}_G$. By definition $\mathcal{G}_{\mathsf{id}_F}$ is $\mathsf{id}_{\mathcal{G}_F}$. Moreover, if ψ is another natural transformation $G \to H$ we have

$$\widehat{\psi \circ \eta} \circ \iota^F_{k,l} = \iota^H_{k,l} \circ \psi_{(k,l)} \circ \eta_{(k,l)} = \hat{\psi} \circ \iota^G_{k,l} \circ \eta_{(k,l)} = \hat{\psi} \circ \hat{\eta} \circ \iota^F_{k,l}$$

and so $\widehat{\psi \circ \eta} = \hat{\psi} \circ \hat{\eta}$.

Summing up we have just defined a functor $\mathcal{G}_{-} \colon \mathbf{Set}^{\mathbf{H}} \to \mathbf{Hyp}$. We have now to show that such functor is full, faithful and essentially surjective.

- For faithfulness, suppose that $\eta, \psi \colon F \rightrightarrows G$ are arrows such that $\mathcal{G}_\eta = \mathcal{G}_\psi$. Then, by definition $\eta_\bullet = \psi_\bullet$ and, for every $(k,l) \in \mathbf{H}$
$$\iota^G_{k,l} \circ \eta_{(k,l)} = \hat{\eta} \circ \iota^F_{k,l} = \hat{\psi} \circ \iota^F_{k,l} = \iota^G_{k,l} \circ \psi_{(k,l)}$$
Since $\iota^G_{k,l}$ is mono we conclude that $\eta_{(k,l)} = \psi_{(k,l)}$ and so $\eta = \psi$.

- To prove that $\mathcal{G}$ is full, let $(f,g) \colon \mathcal{G}_F \to \mathcal{G}_G$ be a morphism of hypergraphs and define $f_{k,l}$ to be $f \circ \iota^F_{k,l}$. Now, notice that, by extensivity, the diagram on the right is a pullback.
$$\begin{tikzcd} G(k,l) \arrow[r, tail, "\iota^G_{k,l}"] \arrow[d, "!_{G(k,l)}"'] & E_G \arrow[d, "(\mathsf{lg}_{G(\bullet)} \circ s_G, \mathsf{lg}_{G(\bullet)} \circ t_G)"'] \\ 1 \arrow[r, "(\delta_k, \delta_l)"'] & \mathbb{N} \times \mathbb{N} \end{tikzcd}$$
Now, by Remark 3.4 we have
$$\mathsf{lg}_{G(\bullet)} \circ s_G \circ f_{k,l} = \mathsf{lg}_{G(\bullet)} \circ s_G \circ f \circ \iota^F_{k,l} = \mathsf{lg}_{G(\bullet)} \circ g^\star \circ s_F \circ \iota^F_{k,l}$$
$$= \mathsf{lg}_{F(\bullet)} \circ s_F \circ \iota^F_{k,l} = \delta_k \circ !_{F(k,l)}$$
$$\mathsf{lg}_{G(\bullet)} \circ t_G \circ f_{k,l} = \mathsf{lg}_{G(\bullet)} \circ t_G \circ f \circ \iota^F_{k,l} = \mathsf{lg}_{G(\bullet)} \circ g^\star \circ t_F \circ \iota^F_{k,l} = \mathsf{lg}_{F(\bullet)} \circ t_F \circ \iota^F_{k,l} = \delta_l \circ !_{F(k,l)}$$

Thus there exists an arrow $\eta_{k,l} \colon F(k,l) \to H(k,l)$ fitting in the diagram on the right. Let $\eta_\bullet \colon F(\bullet) \to H(\bullet)$ be simply the arrow g, then the collection of all the $\eta_{k,l}$ and of $\eta_\bullet$ defines a natural transformation $\eta \colon F \to H$. To see this, we can first notice that for every object $(k,l) \in \mathbf{H}$ we have

$$\begin{tikzcd}
F(k,l) \arrow[r, tail, "\iota^F_{k,l}"'] \arrow[dr, dotted, "\eta_{(k,l)}"'] \arrow[ddr, bend right, "!_{F(k,l)}"'] & E_F \arrow[dr, "f"] & \\
& G(k,l) \arrow[r, tail, "\iota^G_{k,l}"] \arrow[d, "!_{G(k,l)}"'] & E_G \arrow[d, "(\mathsf{lg}_{G(\bullet)} \circ s_G, \mathsf{lg}_{G(\bullet)} \circ t_G)"'] \\
& 1 \arrow[r, "(\delta_k, \delta_l)"'] & \mathbb{N} \times \mathbb{N}
\end{tikzcd}$$

$$v^G_k \circ g^k \circ s^F_{k,l} = g^\star \circ v^F_k \circ s^F_{k,l} = g^\star \circ s_F \circ \iota^F_{k,l} = s_G \circ f \circ \iota^F_{k,l} = s_G \circ \iota^G_{k,l} \circ \eta_{(k,l)} = v^G_k \circ s^G_{k,l} \circ \eta_{(k,l)}$$
$$v^G_l \circ g^l \circ t^F_{k,l} = g^\star \circ v^F_l \circ t^F_{k,l} = g^\star \circ t_F \circ \iota^F_{k,l} = t_G \circ f \circ \iota^F_{k,l} = t_G \circ \iota^G_{k,l} \circ \eta_{(k,l)} = v^G_l \circ t^G_{k,l} \circ \eta_{(k,l)}$$

so that $g^k \circ s^F_{k,l} = s^G_{k,l} \circ \eta_{(k,l)}$ and $g^l \circ t^F_{k,l} = t^G_{k,l} \circ \eta_{(k,l)}$.

Now, let $f_i \colon (k,l) \to \bullet$ be an arrow in $\mathbf{H}$. We have two cases

– if $i < k$ then

$$\eta_\bullet \circ F(f_i) = g \circ F(f_i) = g \circ \pi^F_{k,i} \circ s^F_{k,l} = \pi^G_{k,i} \circ g^k \circ s^F_{k,l} = \pi^G_{k,i} \circ s^G_{k,l} \circ \eta_{(k,l)} = G(f_i) \circ \eta_{(k,l)}$$

– if $k \leq i < k + l - 1$ then

$$\eta_\bullet \circ F(f_i) = g \circ F(f_i) = g \circ \pi^F_{l,i-k} \circ t^F_{k,l} = \pi^G_{l,i-k} \circ g^l \circ t^F_{k,l} = \pi^G_{k,i-k} \circ t^G_{k,l} \circ \eta_{(k,l)} = G(f_i) \circ \eta_{(k,l)}$$

By construction $f_{k,l} = \iota^G_{k,l} \circ \eta_{(k,l)}$ and so $(\hat{\eta}, \eta_\bullet) = (f, g)$ as wanted.

- We are left with essential surjectivity of $\mathcal{G}_{(-)}$. Given a hypergraph $\mathcal{G} = (E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G})$ we can define

$$F_\mathcal{G}(k,l) := \{e \in E_\mathcal{G} \mid \mathsf{lg}_{V_\mathcal{G}}(s_\mathcal{G}(e)) = k \text{ and } \mathsf{lg}_{V_\mathcal{G}}(t_\mathcal{G}(e)) = l\} \qquad F_\mathcal{G}(\bullet) := V_\mathcal{G}$$

while for every arrow $f_i \colon (k,l) \to \bullet$ we put

$$F_\mathcal{G}(f_i) \colon F_\mathcal{G}(k,l) \to F_\mathcal{G}(\bullet) \qquad x \mapsto \begin{cases} s_\mathcal{G}(x)_i & i < k \\ t_\mathcal{G}(x)_{i-k} & i \leq k < k + l - 1 \end{cases}$$

$F_\mathcal{G}$ so defined is a functor $\mathbf{H} \to \mathbf{Set}$. Let $\iota^\mathcal{G}_{k,l} \colon F_\mathcal{G}(k,l) \rightarrowtail E_\mathcal{G}$ be the inclusion, by definition the diagram on the right is a pullback and thus, by extensivity, $(E_\mathcal{G}, \{\iota^\mathcal{G}_{k,l}\}_{k,l \in \mathbb{N}})$ is a coproduct and so we have an isomorphism $\phi \colon E_\mathcal{G} \to E_{\mathcal{F}_\mathcal{G}}$ such that, for every $k, l \in \mathbb{N}$, $\phi \circ \iota^\mathcal{G}_{k,l} = \iota^{F_\mathcal{G}}_{k,l}$.

$$\begin{tikzcd}
F_\mathcal{G}(k,l) \arrow[r, tail, "\iota^\mathcal{G}_{k,l}"] \arrow[d, "!_{G(k,l)}"'] & E_\mathcal{G} \arrow[d, "(\mathsf{lg}_{G(\bullet)} \circ s_G, \mathsf{lg}_{G(\bullet)} \circ t_G)"'] \\
1 \arrow[r, tail, "(\delta_k, \delta_l)"] & \mathbb{N} \times \mathbb{N}
\end{tikzcd}$$

Moreover, by definition of $F_\mathcal{G}(k,l)$, the solid parts of the diagrams aside commute and so Remark 3.5 gives us the dotted arrows $s^\mathcal{G}_{k,l} \colon F_\mathcal{G}(k,l) \to V^k_\mathcal{G}$.

$$\begin{tikzcd}
F_\mathcal{G}(k,l) \arrow[r, tail, "\iota^\mathcal{G}_{k,l}"'] \arrow[d, dotted, "s^\mathcal{G}_{k,l}"'] \arrow[dd, bend left, "!_{F_\mathcal{G}(k,l)}"] & E_\mathcal{G} \arrow[d, "s_\mathcal{G}"] \\
V^k_\mathcal{G} \arrow[r, tail, "v^{F_\mathcal{G}}_k"] \arrow[d, "!_{V^k_\mathcal{G}}"'] & V^\star_\mathcal{G} \arrow[d, "\mathsf{lg}_{V_\mathcal{G}}"] \\
1 \arrow[r, tail, "\delta_k"] & \mathbb{N}
\end{tikzcd}
\qquad
\begin{tikzcd}
F_\mathcal{G}(k,l) \arrow[r, tail, "\iota^\mathcal{G}_{k,l}"'] \arrow[d, dotted, "t^\mathcal{G}_{k,l}"'] \arrow[dd, bend left, "!_{F_\mathcal{G}(k,l)}"] & E_\mathcal{G} \arrow[d, "t_\mathcal{G}"'] \\
V^l_\mathcal{G} \arrow[r, tail, "v^{F_\mathcal{G}}_l"] \arrow[d, "!_{V^l_\mathcal{G}}"'] & V^\star_\mathcal{G} \arrow[d, "\mathsf{lg}_{V_\mathcal{G}}"'] \\
1 \arrow[r, tail, "\delta_l"] & \mathbb{N}
\end{tikzcd}$$

Now for every $i < k$ and $j < l$ we have

$$\pi^{F_\mathcal{G}}_{k,i}(s^\mathcal{G}_{k,l}(e)) = (v^{F_\mathcal{G}}_k(s^\mathcal{G}_{k,l}(e)))_i = F_\mathcal{G}(f_i)(e) = \pi^{F_\mathcal{G}}_{k,i}(s^{F_\mathcal{G}}_{k,l}(e))$$
$$\pi^{F_\mathcal{G}}_{l,j}(t^\mathcal{G}_{k,l}(e)) = (v^{F_\mathcal{G}}_l(t^\mathcal{G}_{k,l}(e)))_j = F_\mathcal{G}(f_{j+k})(e) = \pi^{F_\mathcal{G}}_{l,j}(t^{F_\mathcal{G}}_{k,l}(e))$$

but this means that $s^\mathcal{G}_{k,l} = s^{F_\mathcal{G}}_{k,l}$ and $t^\mathcal{G}_{k,l} = t^{F_\mathcal{G}}_{k,l}$ and so

$$s_{F_\mathcal{G}} \circ \phi \circ \iota^\mathcal{G}_{k,l} = s_{F^\mathcal{G}} \circ \iota^{F_\mathcal{G}}_{k,l} = v^{F_\mathcal{G}}_k \circ s^{F_\mathcal{G}}_{k,l} = v^{F_\mathcal{G}}_k \circ s^\mathcal{G}_{k,l} = s_\mathcal{G} \circ \iota^\mathcal{G}_{k,l}$$
$$t_{F_\mathcal{G}} \circ \phi \circ \iota^\mathcal{G}_{k,l} = t_{F^\mathcal{G}} \circ \iota^{F_\mathcal{G}}_{k,l} = v^{F_\mathcal{G}}_l \circ t^{F_\mathcal{G}}_{k,l} = v^{F_\mathcal{G}}_l \circ t^\mathcal{G}_{k,l} = t_\mathcal{G} \circ \iota^\mathcal{G}_{k,l}$$

Therefore $(\phi, \mathsf{id}_{V_\mathcal{G}})$ is an isomorphism of hypergraphs $\mathcal{G} \to \mathcal{G}_{F_\mathcal{G}}$ and the thesis follows. $\square$

A.2.1. Left-monogamous hypergraphs - proofs

Proposition 3.23. *Given a hypergraph $\mathcal{H}$, the following facts are equivalent*

1. *$\mathcal{H}$ is left-monogamous;*
2. *for every $e \in E_{\mathcal{H}}$, $t_{\mathcal{H}}(e)$ has no repeated letter and $t_{\mathcal{H}}(e)$ is disjoint from $t_{\mathcal{H}}(e')$ for every other hyperedge e'.*

Proof. $(1 \Rightarrow 2)$ If $t_{\mathcal{H}}(e)$ has a repeated letter v, then $\mathsf{out}_{\mathcal{H}}(v,e)$ is at least 2, thus $\mathcal{H}$ is not left-monogamous. By definition, if v appears in $t_{\mathcal{H}}(e)$ and $t_{\mathcal{H}}(e')$ for $e \neq e'$, then $\mathcal{H}$ is not in **LMH**,

$(2 \Rightarrow 1)$ Let v be an element of $V_{\mathcal{H}}$. Since $t_{\mathcal{H}}(e)$ has no repeated letter then $\mathsf{out}_{\mathcal{H}}(v,e) \leq 1$ for every hyperedge e. On the other hand, since $t_{\mathcal{H}}(e)$ is disjoint from $t_{\mathcal{H}}(e')$ for every e different from e, we also get that there is at most one hyperedge such that $\mathsf{out}_{\mathcal{H}}(v,e) = 1$. □

Proposition 3.26. **LMH** *is closed in* **Hyp** *under subobject: if $(f,g)\colon \mathcal{G} \to \mathcal{H}$ is a mono in* **Hyp** *and $\mathcal{H}$ is left-monogamous, then $\mathcal{G}$ is left-monogamous too.*

Proof. By Corollary 3.8 we know that $f\colon E_{\mathcal{G}} \to E_{\mathcal{H}}$ and $g\colon V_{\mathcal{G}} \to V_{\mathcal{H}}$ are injective. Now, given $v \in V_{\mathcal{G}}$ and $w \in V_{\mathcal{H}}$, we can define

$$E^{\mathcal{G}}_v = \{e \in E_{\mathcal{G}} \mid \mathsf{out}_{\mathcal{G}}(v,e) \neq 0\} \qquad E^{\mathcal{H}}_w = \{e \in E_{\mathcal{H}} \mid \mathsf{out}_{\mathcal{G}}(w,e) \neq 0\}$$

Now, if $e \in E^{\mathcal{G}}_v$ then $g(v)$ is a letter of $t_{\mathcal{H}}(f(e))$ and so $f(e)$ belongs to $E^{\mathcal{H}}_{g(v)}$. Thus there exists an arrow $f_v\colon E^{\mathcal{G}}_v \to E^{\mathcal{H}}_{g(v)}$ as in the diagram aside. Since f is injective then f_v is injective too and we know, by left-monogamy of $\mathcal{H}$, that $E^{\mathcal{H}}_{g(v)}$ is finite. Therefore also $E^{\mathcal{G}}_v$ is finite.

$$\begin{tikzcd} E^{\mathcal{G}}_v \arrow[r, dotted, "f_v"] \arrow[d, tail, "\iota^{\mathcal{G}}_v"'] & E^{\mathcal{H}}_{g(v)} \arrow[d, tail, "\iota^{\mathcal{H}}_v"] \\ E_{\mathcal{G}} \arrow[r, tail, "f"] & E_{\mathcal{H}} \end{tikzcd}$$

Given an hyperedge e, if $t_{\mathcal{G}}(e)$ has a repeated letter, then there are two different indexes $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{L}(e)})$, $1 \leq k \leq \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{L}(e')})$ such that $t_{\mathcal{G}}(e)_i = t_{\mathcal{G}}(e)_k$ and so

$$t_{\mathcal{H}}(f(e))_k = g^{\star}(t_{\mathcal{G}}(e))_k = g(t_{\mathcal{G}}(e)_k) = g(t_{\mathcal{G}}(e)_i) = g^{\star}(t_{\mathcal{G}}(e))_i = t_{\mathcal{H}}(f(e))_i$$

But $\mathcal{H}$ is left-monogamous and thus, by Proposition 3.23, $f(e)$ cannot have a repeated letter.

Moreover, let e and e' be two non-disjoint hyperedges, i.e. there are indexes $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{L}(e)})$, $1 \leq k \leq \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{L}(e')})$ such that $t_{\mathcal{G}}(e)_i = t_{\mathcal{G}}(e')_k$. Then

$$t_{\mathcal{H}}(f(e'))_k = g^{\star}(t_{\mathcal{G}}(e'))_k = g(t_{\mathcal{G}}(e')_k) = g(t_{\mathcal{G}}(e)_i) = g^{\star}(t_{\mathcal{G}}(e))_i = t_{\mathcal{H}}(f(e))_i$$

Since $\mathcal{H}$ is left-monogamous then $f(e) = f(e')$ and so, by injectivity of f, $e = e'$. Thus every hyperedge e is disjoint from any other $e' \in E_{\mathcal{G}}$. We then conclude appealing to Proposition 3.23. □

Lemma 3.28. **LMH** *is closed in* **Hyp** *under binary products.*

Proof. Consider two hypergraphs $\mathcal{G}_0$ and $\mathcal{G}_1$ and their product $(\mathcal{P}, \{(\pi^E_{\mathcal{G}_i}, \pi^V_{\mathcal{G}_i})\}_{i=0}^1)$. By Corollary 3.19 we know that $(V_{\mathcal{P}}, \{\pi^V_{\mathcal{G}_i}\}_{i=0}^1))$ is a product and the diagrams below on the right commute for $i \in \{0,1\}$.

Let e be an element of $E_{\mathcal{P}}$, we know that there exists h in $E_{k,l}$ and $k,l \in \mathbb{N}$ such that $\iota_{k,l}(h) = e$.

Suppose that there exists $1 \leq n,m \leq l$, with $n \neq m$ such that

$$t_{\mathcal{P}}(e)_n = t_{\mathcal{P}}(e)_m$$

then for every $i \in \{0,1\}$ we have

$$\begin{tikzcd} E^i_{k,l} \arrow[r, tail, "\iota^i_{k,l}"] & E_{\mathcal{G}_i} \arrow[r, "t_{\mathcal{G}_i}"] & V^{\star}_{\mathcal{G}_i} \\ E_{k,l} \arrow[u, "\pi^i_{k,l}"'] \arrow[d, tail, "\iota_{k,l}"] & & \\ E_{\mathcal{P}} \arrow[rr, "t_{\mathcal{P}}"'] & & V^{\star}_{\mathcal{P}} \arrow[uu, "(\pi^V_{\mathcal{G}_i})^{\star}"'] \end{tikzcd} \qquad \begin{tikzcd} E_{k,l} \arrow[r, "\pi^i_{k,l}"] \arrow[d, "\iota_{k,l}"'] & E^i_{k,l} \arrow[d, tail, "\iota^i_{k,l}"] \\ E_{\mathcal{P}} \arrow[r, "\pi^E_{\mathcal{G}_i}"'] & E_{\mathcal{G}_i} \end{tikzcd}$$

$$\begin{aligned} & t_{\mathcal{G}_i}(\iota^i_{k,l}(\pi^i_{k,l}(h)))_n = ((\pi^V_{\mathcal{G}_i})^{\star}(t_{\mathcal{P}}(\iota_{k,l}(h))))_n = \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(\iota_{k,l}(h))_n) = \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(e)_n) \\ =& \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(e)_m) = \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(\iota_{k,l}(h))_m) = ((\pi^V_{\mathcal{G}_i})^{\star}(t_{\mathcal{P}}(\iota_{k,l}(h))))_m = t_{\mathcal{G}_i}(\iota^i_{k,l}(\pi^i_{k,l}(h)))_m \end{aligned}$$

But this contradicts the the left-monogamy of $\mathcal{G}_0$ and $\mathcal{G}_1$.

Now, let now e and e' be two non-disjoint hyperedges. We know that $e = \iota_{k,l}(h)$ and $e' = \iota_{k',l'}(h')$ for some $h \in E_{k,l}$ and $h' \in E_{k',l'}$ and that there exist $1 \leq n \leq l$ and $1 \leq m \leq l'$ such that $t_{\mathcal{P}}(e)_n = t_{\mathcal{P}}(e')_m$. Computing we get

$$\begin{aligned} t_{\mathcal{G}_i}(\iota^i_{k,l}(\pi^i_{k,l}(h)))_n &= ((\pi^V_{\mathcal{G}_i})^\star(t_{\mathcal{P}}(\iota_{k,l}(h))))_n = \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(\iota_{k,l}(h))_n) = \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(e)_n) \\ =&\pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(e')_m) = \pi^V_{\mathcal{G}_i}(t_{\mathcal{P}}(\iota_{k',l'}(h'))_m) = ((\pi^V_{\mathcal{G}_i})^\star(t_{\mathcal{P}}(\iota_{k',l'}(h'))))_m = t_{\mathcal{G}_i}(\iota^i_{k',l'}(\pi^i_{k',l'}(h')))_m \end{aligned}$$

By left-monogamy this entails that $\iota^i_{k,l}(\pi^i_{k,l}(h)) = \iota^i_{k',l'}(\pi^i_{k',l'}(h'))$. Since coproducts in **Set** are disjoint then $k = k'$ and $l = l'$ and since $\iota^i_{k,l}$ is injective we get

$$\pi^0_{k,l}(h) = \pi^0_{k,l}(h') \qquad \pi^1_{k,l}(h) = \pi^1_{k,l}(h')$$

Thus $h = h'$ and $e = e'$ too. We conclude by Proposition 3.23. $\square$

Lemma 3.31. *Let $(f, g)\colon \mathcal{G} \to \mathcal{H}$ be a morphism of hypergraphs with $\mathcal{H}$ left-monogamous, then the following facts hold*

1. *if g sends input nodes to input nodes, then (f, g) is downward-closed;*
2. *if g is mono and (f, g) is downward-closed, then g sends input nodes to input nodes.*

Proof.
1. Suppose that $v \in V_{\mathcal{G}}$ is a node such that $g(v)$ is a letter of $t_{\mathcal{H}}(h)$ for some hyperedge h, hence $g(v)$ is not an input node and so v does not belong to $\mathsf{IN}(\mathcal{G})$. Therefore there is $k \in E_{\mathcal{G}}$ such that v occurs in $t_{\mathcal{G}}(k)$. But then $\mathsf{out}_{\mathcal{H}}(g(v), f(k)) \neq 0$ and so, by left-monogamy, $f(k)$ must coincide with h as wanted.
2. Let v be an input node and suppose that there is $h \in E_{\mathcal{H}}$ such that $g(v)$ appears in $t_{\mathcal{H}}(h)$. Thus there is $k \in E_{\mathcal{G}}$ such that $f(k) = h$. Now, let $1 \leq n \leq \mathsf{lg}_{V_{\mathcal{H}}}(h)$ be such that $t_{\mathcal{H}}(h)_n = g(v)$, then, by Remark 3.4, $\mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{G}}(k)) = \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{H}}(h))$ and

$$g(t_{\mathcal{G}}(k)_n) = g^\star(t_{\mathcal{G}}(k)_n) = (g^\star(t_{\mathcal{G}}(k)))_n = t_{\mathcal{H}}(f(k))_n = t_{\mathcal{H}}(h)_n = g(v)$$

By injectivity of g we have $t_{\mathcal{G}}(k)_n = g(v)$, contradicting the assumption that v is an input node. $\square$

Proposition 3.32. *Let $(f_1, g_1)\colon \mathcal{G} \to \mathcal{H}$ and $(f_2, g_2)\colon \mathcal{G} \to \mathcal{K}$ be two morphisms in* **LMH**. *If (f_1, g_1) is mono and downward-closed, then the pushout $\mathcal{P}$ of (f_1, g_1) along (f_2, g_2) in* **Hyp** *is left-monogamous.*

Proof. By Proposition 3.7 and Lemma B.3, we know that there pushout squares in **Set** as the ones aside.

$$\begin{tikzcd} E_{\mathcal{G}} \arrow[r, "f_2"] \arrow[d, tail, "f_1"'] & E_{\mathcal{K}} \arrow[d, tail, "k_E"] \\ E_{\mathcal{H}} \arrow[r, "h_E"'] & E_{\mathcal{P}} \end{tikzcd} \qquad \begin{tikzcd} V_{\mathcal{G}} \arrow[r, "g_2"] \arrow[d, tail, "g_1"'] & V_{\mathcal{K}} \arrow[d, tail, "k_V"] \\ V_{\mathcal{H}} \arrow[r, "h_V"'] & V_{\mathcal{P}} \end{tikzcd}$$

By Corollary 3.8, f_1 and g_1 are monos, thus also k_V and k_E are injective as they are the pushout of injective arrows. By Remark 3.2, moreover, we know that $k_V^\star$ is injective too.

Moreover, the first of the diagrams above fits in other two diagrams as the ones aside. Let $\iota_E\colon A \rightarrowtail E_{\mathcal{H}}$ and $\iota_V\colon B \rightarrowtail V_{\mathcal{H}}$ be the inclusions of the complements of the images of, respectively f_1 and g_1. By Lemma A.2, we know that $(E_{\mathcal{P}}, \{k_E, h_E \circ \iota_E\})$ and $(V_{\mathcal{P}}, \{k_V, h_V \circ \iota_V\})$ are coproducts.

$$\begin{tikzcd} V^\star_{\mathcal{G}} \arrow[rrr, "g_2^\star"] \arrow[ddd, tail, "g_1^\star"'] & & & V^\star_{\mathcal{K}} \arrow[ddd, tail, "k_V^\star"] \\ & E_{\mathcal{G}} \arrow[ul, "s_{\mathcal{G}}"] \arrow[r, "f_2"] \arrow[d, tail, "f_1"'] & E_{\mathcal{K}} \arrow[ur, "s_{\mathcal{K}}"'] \arrow[d, tail, "k_E"] & \\ & E_{\mathcal{H}} \arrow[dl, "s_{\mathcal{H}}"'] \arrow[r, "h_E"'] & E_{\mathcal{P}} \arrow[dr, "s_{\mathcal{P}}"] & \\ V^\star_{\mathcal{H}} \arrow[rrr, "h_V^\star"] & & & V^\star_{\mathcal{P}} \end{tikzcd} \qquad \begin{tikzcd} V^\star_{\mathcal{G}} \arrow[rrr, "g_2^\star"] \arrow[ddd, tail, "g_1^\star"'] & & & V^\star_{\mathcal{K}} \arrow[ddd, tail, "k_V^\star"] \\ & E_{\mathcal{G}} \arrow[ul, "t_{\mathcal{G}}"] \arrow[r, "f_2"] \arrow[d, tail, "f_1"'] & E_{\mathcal{K}} \arrow[ur, "t_{\mathcal{K}}"'] \arrow[d, tail, "k_E"] & \\ & E_{\mathcal{H}} \arrow[dl, "t_{\mathcal{H}}"'] \arrow[r, "h_E"'] & E_{\mathcal{P}} \arrow[dr, "t_{\mathcal{P}}"] & \\ V^\star_{\mathcal{H}} \arrow[rrr, "h_V^\star"] & & & V^\star_{\mathcal{P}} \end{tikzcd}$$

We want to use Proposition 3.23 to prove that $\mathcal{P}$ is left-monogamous. To do so, we need to show that the target of every element of $E_{\mathcal{P}}$ has no repeated letter and that every two different hyperedges of $\mathcal{P}$ have disjoint targets.

Suppose that there exists $e \in E_{\mathcal{P}}$ such that $\mathsf{out}_{\mathcal{P}}(v, e) > 1$ for some $v \in V_{\mathcal{P}}$. So that there are two different indexes $1 \leq n, m \leq \mathsf{lg}_{V_{\mathcal{P}}}(e)$ such that $t_{\mathcal{P}}(e)_n = t_{\mathcal{P}}(e)_m$. We have two cases.

- $e = k_E(h)$ for some $h \in E_{\mathcal{K}}$. Then we have

$$\begin{aligned} k_V(t_{\mathcal{K}}(h)_n) &= (k_V^\star(t_{\mathcal{K}}(h)))_n = t_{\mathcal{P}}(k_E(h))_n = t_{\mathcal{P}}(e)_n \\ =& t_{\mathcal{P}}(e)_m = t_{\mathcal{P}}(k_E(h))_m = (k_V^\star(t_{\mathcal{K}}(h)))_m = k_V(t_{\mathcal{K}}(h)_m) \end{aligned}$$

  Since k_V is injective, then $t_{\mathcal{K}}(h)_n = t_{\mathcal{K}}(h)_m$, which contradicts the left monogamy of $\mathcal{K}$.

- $e = h_E(\iota_E(h))$ for some h in the complement A of the image of f_1. Now, suppose that there exists $u \in V_{\mathcal{G}}$ such that $t_{\mathcal{H}}(h)_n = g_1(u)$ or $t_{\mathcal{H}}(h)_m = g_1(u)$. Since (f_1, g_1) is downward-closed in both cases we get $h_1 \in E_{\mathcal{G}}$ such that $f_1(h_1) = h$, which is absurd.

  Thus there exist $b_1, b_2 \in B$ such that

$$t_{\mathcal{H}}(\iota_E(h))_n = \iota_V(b_1) \qquad t_{\mathcal{H}}(\iota_E(h))_m = \iota_V(b_2)$$

  But then we have

$$\begin{aligned} h_V(\iota_V(b_1)) &= h_V(t_{\mathcal{H}}(\iota_E(h))_n) = (h_V^\star(t_{\mathcal{H}}(\iota_E(h))))_n = t_{\mathcal{P}}(h_E(\iota_E(h)))_n = t_{\mathcal{P}}(e)_n \\ =& t_{\mathcal{P}}(e)_m = t_{\mathcal{P}}(h_E(\iota_E(h)))_m = (h_V^\star(t_{\mathcal{H}}(\iota_E(h))))_m = h_V(t_{\mathcal{H}}(\iota_E(h))_m) = h_V(\iota_V(b_2)) \end{aligned}$$

  By Lemma A.2 $h_V \circ \iota_V$ is injective and so $b_1 = b_2$. Hence $t_{\mathcal{H}}(\iota_E(h))_n = t_{\mathcal{H}}(\iota_E(h))_m$, which is absurd since $\mathcal{H}$ is left-monogamous.

Suppose that there are $e_1, e_2 \in E_{\mathcal{P}}$ such that $t_{\mathcal{P}}(e_1)_n = t_{\mathcal{P}}(e_2)_m$ for some indexes $1 \leq n \leq \mathsf{lg}_{V_{\mathcal{P}}}(e_1)$ and $1 \leq m \leq \mathsf{lg}_{V_{\mathcal{P}}}(e_2)$. We have to show that $e_1 = e_2$. We can again use Lemma A.2 to split the cases.

- $e_1 = k_E(k_1)$ and $e_2 = k_E(k_2)$ for some k_1 and $k_2 \in E_{\mathcal{K}}$. Then we have

$$\begin{aligned} k_V(t_{\mathcal{K}}(k_1)_n) &= (k_V^\star(t_{\mathcal{K}}(k_1)))_n = t_{\mathcal{P}}(k_E(k_1))_n = t_{\mathcal{P}}(h_1)_n \\ =& t_{\mathcal{P}}(h_2)_m = t_{\mathcal{P}}(k_E(k_2))_m = (k_V^\star(t_{\mathcal{K}}(k_2)))_m = k_V(t_{\mathcal{K}}(k_2)_m) \end{aligned}$$

  Since k_V is injective, $t_{\mathcal{K}}(k_1)_n = t_{\mathcal{K}}(k_1)_m$. And $\mathcal{K}$ is left-monogamous, so we conclude that $k_1 = k_2$ and so also e_1 and e_2 are equal.

- $e_1 = k_E(k)$ and $e_2 = h_E(\iota_E(h))$ for some $k \in E_{\mathcal{K}}$ and $h \in A$. then we have

$$\begin{aligned} k_V(t_{\mathcal{K}}(k)_n) &= (k_V^\star(t_{\mathcal{K}}(k)))_n = t_{\mathcal{P}}(k_E(k))_n = t_{\mathcal{P}}(e_1)_n \\ =& t_{\mathcal{P}}(e_2)_m = t_{\mathcal{P}}(h_E(\iota_E(h)))_m = (h_V^\star(t_{\mathcal{H}}(\iota_E(h))))_m = h_V(t_{\mathcal{H}}(\iota_E(h))_m) \end{aligned}$$

  By the first point of Proposition 2.6, there exists $w \in V_{\mathcal{G}}$ such that

$$t_{\mathcal{K}}(k)_n = g_2(w) \qquad t_{\mathcal{H}}(\iota_E(h))_m = g_1(w)$$

  Since (f_1, g_1) is downward-closed, the second equation above entails that there exists $a \in E_{\mathcal{G}}$ such that $\iota_E(h) = f_1(a)$. But this is absurd since coproducts in **Set** are disjoint.

- $e_1 = h_E(\iota_E(h))$ and $e_2 = k_E(k)$ for some $k \in E_{\mathcal{K}}$ and $h \in A$. This is done as in the previous case swapping the role of e_1 and e_2.

- $e_1 = h_E(\iota_E(h_1))$ and $e_2 = h_E(\iota_E(h_1))$ for some h_1 and $h_2 \in A$. Suppose that there exist $x_1, x_2 \in V_{\mathcal{G}}$ such that

$$t_{\mathcal{H}}(\iota_E(h_1))_n = g_1(x_1) \qquad t_{\mathcal{H}}(\iota_E(h_1))_m = g_1(x_2)$$

  Being (f_1, g_1) downward-closed, these identities entails the existence of $a_1, a_2 \in E_{\mathcal{G}}$ such that

$$\iota_E(h_1) = f_1(a_1) \qquad \iota_E(h_2 = f_1(a_2))$$

which is absurd. Thus there there exist $b_1, b_2 \in B$ such that

$$t_{\mathcal{H}}(\iota_E(h_1))_n = \iota_V(b_1) \qquad t_{\mathcal{H}}(\iota_E(h_2))_m = \iota_V(b_2)$$

Moreover

$$\begin{aligned} h_V(\iota_V(b_1)) = h_V(t_{\mathcal{H}}(\iota_E(h_1))_n) = (h_V^\star(t_{\mathcal{H}}(\iota_E(h_1))))_n = t_{\mathcal{P}}(h_E(\iota_E(h_1)))_n = t_{\mathcal{P}}(e_1)_n \\ = t_{\mathcal{P}}(e_2)_m = t_{\mathcal{P}}(h_E(\iota_E(h_2)))_m = (h_V^\star(t_{\mathcal{H}}(\iota_E(h_2))))_m = h_V(t_{\mathcal{H}}(\iota_E(h_2))_m) = h_V(\iota_V(b_2)) \end{aligned}$$

But $h_V \circ \iota_V \colon B \rightarrowtail V_{\mathcal{P}}$ is injective and so b_1 and b_2 are equal. Hence $t_{\mathcal{H}}(\iota_E(h_1))_n = t_{\mathcal{H}}(\iota_E(h_2))_m$. Since $\mathcal{H}$ is in **LMH** we get that $\iota_E(h_1) = \iota_E(h_2)$ and so e_1 is equal to e_2. $\square$

Lemma 3.34. *Let $(f, g) \colon \mathcal{G} \to \mathcal{H}$ be s monomorphism in* **LMH***. Then the following facts are equivalent*

1. *(f, g) is a regular mono;*
2. *(f, g) sends input nodes to input nodes;*
3. *(f, g) is downward-closed.*

Proof. $(1 \Rightarrow 2)$ Let (h_1, k_1) and $(h_2, k_2) : \mathcal{H} \rightrightarrows \mathcal{K}$ be two arrows equalized by (f, g) and define

$$E_{h_1,h_2} := \{e \in E_{\mathcal{H}} \mid h_1(e) = h_2(e)\} \qquad E_{k_1,k_2} := \{v \in V_{\mathcal{H}} \mid k_1(v) = k_2(v)\}$$

$$\begin{tikzcd} E_{\mathcal{G}} \arrow[r, dotted, "\phi"'] \arrow[rr, bend left, tail, "f"] \arrow[d, "s_{\mathcal{G}}"'] & E_{h_1,h_2} \arrow[r, tail, "\iota_E"'] \arrow[d, dotted, "s_{\mathcal{E}}"'] & E_{\mathcal{H}} \arrow[d, "s_{\mathcal{H}}"'] \\ V_{\mathcal{G}}^\star \arrow[r, dotted, "\psi^\star"] \arrow[rr, bend right, tail, "g^\star"'] & V_{k_1,k_2}^\star \arrow[r, tail, "\iota_V^\star"] & V_{\mathcal{H}}^\star \end{tikzcd}$$

$$\begin{tikzcd} E_{\mathcal{G}} \arrow[r, dotted, "\phi"'] \arrow[rr, bend left, tail, "f"] \arrow[d, "t_{\mathcal{G}}"'] & E_{h_1,h_2} \arrow[r, tail, "\iota_E"'] \arrow[d, dotted, "t_{\mathcal{E}}"'] & E_{\mathcal{H}} \arrow[d, "t_{\mathcal{H}}"'] \\ V_{\mathcal{G}}^\star \arrow[r, dotted, "\psi^\star"] \arrow[rr, bend right, tail, "g^\star"'] & V_{k_1,k_2}^\star \arrow[r, tail, "\iota_V^\star"] & V_{\mathcal{H}}^\star \end{tikzcd}$$

Let $\iota_E \colon E_{h_1,h_2} \rightarrowtail E_{\mathcal{H}}$ and $\iota_V \colon V_{k_1,k_2} \rightarrowtail V_{\mathcal{H}}$ be the corresponding inclusions. Let $s_{\mathcal{E}}, t_{\mathcal{E}} \colon E_{h_1,k_1} \rightrightarrows V^\star_{k_1,k_2}$ be the unique arrows fitting in the diagram aside and define $\mathcal{E}$ as $(E_{h_1,h_2}, V_{k_1,k_2}, s_{\mathcal{E}}, t_{\mathcal{E}})$. We know by Propositions 3.1, 3.7 and Corollary B.4, that $(\mathcal{E}, (\iota_E, \iota_V))$, is an equalizer for (h_1, k_1) and (h_2, k_2). Thus there is an isomorphism $(\phi, \psi) \colon \mathcal{G} \to \mathcal{E}$ whose components fit in the diagram aside and such that

$$\iota_V \circ \psi = g \qquad \iota_E \circ \phi = f$$

Now, let $v \in V_{\mathcal{G}}$ be an input node and suppose that there exists $h \in E_{\mathcal{H}}$ such that $\mathsf{out}_{\mathcal{H}}(g(v), h) > 0$. For such an hyperedge. For such an hyperedge we have

$$\mathsf{out}_{\mathcal{K}}(k_1(g(v)), h_1(h)) > 0 \qquad \mathsf{out}_{\mathcal{K}}(k_2(g(v)), h_2(h)) > 0$$

but we also have that

$$k_1(g(v)) = k_1(\iota_V(\psi(v)) = k_2(\iota_V(\psi(v))) = k_2(g(v))$$

and so, by left-monogamy of $\mathcal{K}$, h must belong to E_{h_1,h_2}. Thus there exists $e \in E_{\mathcal{G}}$ such that $\iota_E(\phi(e)) = h$ and such that $\psi(e)$ is a letter of $t_{\mathcal{E}}(\phi(e))$. But then

$$\mathsf{out}_{\mathcal{G}}(v, e) = \mathsf{out}_{\mathcal{E}}(\psi(v), \phi(e)) > 0$$

contradicting the fact that v is an input node.

$(2 \Rightarrow 3)$ This follows at once from Lemma 3.31.

$(3 \Rightarrow 1)$ Let us consider the pushout of (f, g) along itself in **Hyp** as depicted by the first square below. By Proposition 3.7 and Lemma B.3 we know that the other two squares are pushouts too

$$\begin{tikzcd} \mathcal{G} \arrow[r, "{(f,g)}"] \arrow[d, "{(f,g)}"'] & \mathcal{H} \arrow[d, "{(p_1,p_2)}"] \\ \mathcal{H} \arrow[r, "{(q_1,q_2)}"'] & \mathcal{P} \end{tikzcd} \qquad \begin{tikzcd} E_{\mathcal{G}} \arrow[r, tail, "f"] \arrow[d, tail, "f"'] & E_{\mathcal{H}} \arrow[d, "p_1"] \\ E_{\mathcal{H}} \arrow[r, "q_1"'] & E_{\mathcal{P}} \end{tikzcd} \qquad \begin{tikzcd} V_{\mathcal{G}} \arrow[r, tail, "g"] \arrow[d, tail, "g"'] & V_{\mathcal{H}} \arrow[d, "p_2"] \\ V_{\mathcal{H}} \arrow[r, "q_2"'] & V_{\mathcal{P}} \end{tikzcd}$$

By Corollary 3.8 we also know that f and g are monos in **Set** and so, by the dual of Proposition 2.11 $(E_{\mathcal{G}}, f)$ and $(V_{\mathcal{G}}, g)$ are equalizers for, respectively, $p_1, q_1 \colon E_{\mathcal{H}} \rightrightarrows E_{\mathcal{P}}$ and $p_2, q_2 \colon V_{\mathcal{H}} \rightrightarrows V_{\mathcal{P}}$.

By Proposition 3.7 and Corollary B.4 this entails that $(\mathcal{G}, (f, g))$ is the equalizer in **Hyp** of (p_1, p_2) and (q_1, q_2). By Proposition 3.32, $\mathcal{P}$ is left-monogamous, thus by Corollary 3.27 (f, g) is regular mono. $\square$

A.2.2. Separated hypergraphs - proofs

Proposition 3.45. *There exists a functor* $\mathsf{sp}\colon \mathbf{Hyp} \to \mathbf{Sep}$ *sending* $\mathcal{G}$ *to* $\mathsf{sp}(\mathcal{G})$.

Proof. Let $(h_1, h_2)\colon \mathcal{G} \to \mathcal{H}$ be a morphism of hypergraphs. For every hyperedge $e \in E_{\mathcal{G}}$, the universal property of coproducts tells us that there are arrows $f_e\colon V_e \to V_{h_1(e)}$ fitting in the middle of the diagram below. These, in turn, yield the dotted $\hat{h}_1\colon V_{\mathsf{sp}(\mathcal{G})} \to V_{\mathsf{sp}(\mathcal{G})}$ on the bottom.

$$\begin{tikzcd} 1 \arrow[r, "\mathsf{id}_1"] \arrow[d, tail, "\iota_i^e"] & 1 \arrow[d, tail, "\iota_i^{h_1(e)}"'] \\ V_e \arrow[r, dotted, "f_e"] \arrow[d, tail, "\iota_e"] & V_{h_1(e)} \arrow[d, tail, "\iota_{h_1(e)}"'] \\ V_{\mathsf{sp}(\mathcal{G})} \arrow[r, dotted, "\hat{h}_1"] & V_{\mathsf{sp}(\mathcal{H})} \end{tikzcd}$$

Given $e \in E_{\mathcal{G}}$, if $n_e = \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ and $m_e = \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{G}}(e))$, we have

$$\hat{h}_1^{\star}(s_{\mathsf{sp}(\mathcal{G})}(e)) = \hat{h}_1^{\star}(\prod_{i=1}^{n_e} \iota_e(\iota_i^e(0))) = \prod_{i=1}^{n_e} \hat{h}_1(\iota_e(\iota_i^e(0))) = \prod_{i=1}^{n_e} \iota_{h_1(e)}(\iota_i^{h_1(e)}(0)) = s_{\mathsf{sp}(\mathcal{G})}(h_1(e))$$

$$\hat{h}_1^{\star}(t_{\mathsf{sp}(\mathcal{G})}(e)) = \hat{h}_1^{\star}(\prod_{i=n_e+1}^{m_e} \iota_e(\iota_i^e(0))) = \prod_{i=n_e+1}^{m_e} \hat{h}_1(\iota_e(\iota_i^e(0))) = \prod_{i=n_e+1}^{m_e} \iota_{h_1(e)}(\iota_i^{h_1(e)}(0)) = t_{\mathsf{sp}(\mathcal{G})}(h_1(e))$$

so that $(\mathsf{id}_{E_{\mathcal{G}}}, \hat{h}_1)$ is a morphism of hypergraphs.

If h_1 is the identity, then clearly $\hat{h}_1 = \mathsf{id}_{V_{\mathsf{sp}(\mathcal{G})}}$. If (k_1, k_2) is another morphism $\mathcal{H} \to \mathcal{K}$, then we have

$$\hat{k}_1 \circ \hat{h}_1 \circ \iota_e \circ \iota_i^e = \hat{k}_1 \circ \iota_{h_1(e)} \circ \iota_i^{h_1(e)} = \iota_{k_1(h_1(e))} \circ \iota_i^{k_1(h_1(e))}$$

which entails the preservation of composition. □

Proposition 3.50. *The following properties hold*

1. *the inclusion functor* $P\colon \mathbf{pHyp} \to \mathbf{Hyp}$ *has a right adjoint* pr*;*
2. *the inclusion functor* $S_p\colon \mathbf{pSep} \to \mathbf{pHyp}$ *has a right adjoint* sp_P.

Proof. 1. Given a hypergraph $\mathcal{G}$, let $\iota_{\mathsf{pr}(\mathcal{G})}\colon V_{\mathsf{pr}(\mathcal{G})} \rightarrowtail V_{\mathcal{G}}$ be the inclusion of non isolated nodes. By definition we know that there exists $s_{\mathsf{pr}(\mathcal{G})}, t_{\mathsf{pr}(\mathcal{G})}\colon E_{\mathcal{G}} \rightrightarrows V_{\mathsf{pr}(\mathcal{G})}$ such that

$$s_{\mathcal{G}} = \iota_{\mathsf{pr}(\mathcal{G})} \circ s_{\mathsf{pr}(\mathcal{G})} \qquad t_{\mathcal{G}} = \iota_{\mathsf{pr}(\mathcal{G})} \circ t_{\mathsf{pr}(\mathcal{G})}$$

Thus we can define $\mathsf{pr}(\mathcal{G})$ as $(E_{\mathcal{G}}, V_{\mathsf{pr}(\mathcal{G})}, s_{\mathsf{pr}(\mathcal{G})}, t_{\mathsf{pr}(\mathcal{G})})$. Clearly $(\mathsf{id}_{E_{\mathcal{G}}}, \iota_{\mathsf{pr}(\mathcal{G})})$ is a morphism $P(\mathsf{pr}(\mathcal{G})) \to \mathcal{G}$. Let $(f, g)\colon P(\mathcal{H}) \to \mathcal{G}$ be a morphism with pruned domain. For every $v \in V_{\mathcal{H}}$ we have two cases:

- v occurs in $s_{\mathcal{G}}(e)$ for some $e \in E_{\mathcal{G}}$. Let $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ be such that $v = s_{\mathcal{G}}(e)_i$, then we have

$$g(v) = g(s_{\mathcal{G}}(e)_i) = (g^{\star}(s_{\mathcal{G}}(e)))_i = s_{\mathcal{H}}(f(e))_i$$

- v occurs in $t_{\mathcal{H}}(e)$ for some $e \in E_{\mathcal{H}}$. Let $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{H}}}(t_{\mathcal{G}}(e))$ be such that $v = t_{\mathcal{G}}(e)_i$, then we have

$$g(v) = g(t_{\mathcal{G}}(e)_i) = (g^{\star}(t_{\mathcal{G}}(e)))_i = t_{\mathcal{H}}(f(e))_i$$

Thus $g(v)$ belongs to $V_{\mathsf{pr}(\mathcal{G})}$ for every node v of $\mathcal{H}$ and so there exists $\hat{g}\colon V_{\mathcal{H}} \to V_{\mathsf{pr}(\mathcal{G})}$ such that $g = \iota_{\mathsf{pr}(\mathcal{G})} \circ \hat{g}$.
Computing we get

$$\iota^{\star}_{\mathsf{pr}(\mathcal{G})} \circ s_{\mathsf{pr}(\mathcal{G})} \circ f = s_{\mathcal{G}} \circ f = g^{\star} \circ s_{\mathcal{H}} = \iota^{\star}_{\mathsf{pr}(\mathcal{G})} \circ \hat{g}^{\star} \circ s_{\mathcal{H}}$$
$$\iota^{\star}_{\mathsf{pr}(\mathcal{G})} \circ t_{\mathsf{pr}(\mathcal{G})} \circ f = t_{\mathcal{G}} \circ f = g^{\star} \circ t_{\mathcal{H}} = \iota^{\star}_{\mathsf{pr}(\mathcal{G})} \circ \hat{g}^{\star} \circ t_{\mathcal{H}}$$

Since $(-)^{\star}$ sends monos to mono it follows that $(f, \hat{g})$ is a morphsim $\mathcal{H} \to \mathsf{pr}(\mathcal{G})$ which, by construction, is such that $(\mathsf{id}_{E_{\mathcal{G}}}, \iota_{\mathsf{pr}(\mathcal{G})}) \circ (f, \hat{g}) = (f, g)$. Uniqueness follows at once since $\iota_{\mathsf{pr}(\mathcal{G})}$ is mono.

2. Let $\mathcal{G}$ be a hypergraph. For every $e \in E_{\mathcal{G}}$, let $n_e = \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ and $m_e = \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{G}}(e))$. Given $1 \leq i \leq n_e + m_e$, we can define $\delta_i^e \colon \rightarrowtail V_{\mathcal{G}}$ as

$$\delta_i^e(0) = \begin{cases} s_{\mathcal{G}}(e)_i & i \leq n_e \\ t_{\mathcal{G}}(e)_{i-n_e} & n_e < i \end{cases}$$

In turn, this allows us to define $\epsilon_G \colon V_{\mathsf{sp}(\mathcal{G})} \to V_{\mathcal{G}}$ as the unique one fitting in the diagram below.

Now, for every hyperedge $e \in E_{\mathcal{G}}$ we have

$$\begin{tikzcd} 1 \arrow[rr, "\mathsf{id}_1"] \arrow[d, tail, "\iota_i^e"] & & 1 \arrow[dd, tail, "\delta_i^e"] \\ V_e \arrow[d, tail, "\iota_e"] & & \\ V_{\mathsf{sp}(\mathcal{G})} \arrow[rr, dotted, "\epsilon_{\mathcal{G}}"] & & V_{\mathcal{G}} \end{tikzcd}$$

$$\epsilon_{\mathcal{G}}^{\star}(s_{\mathsf{sp}(\mathcal{G})}(e)) = \epsilon_{\mathcal{G}}^{\star}(\prod_{i=1}^{n_e} \iota_e(\iota_i^e(0))) = \prod_{i=1}^{n_e} \epsilon_{\mathcal{G}}(\iota_e(\iota_i^e(0))) = \prod_{i=1}^{n_e} \delta_i^e(0) = \prod_{i=1}^{n_e} s_{\mathcal{G}}(e)_i = s_{\mathcal{G}}(e)$$

$$\epsilon_{\mathcal{G}}^{\star}(t_{\mathsf{sp}(\mathcal{G})}(e)) = \epsilon_{\mathcal{G}}^{\star}(\prod_{i=n_e+1}^{n_e+m_e} \iota_e(\iota_i^e(0))) = \prod_{i=n_e+1}^{n_e+m_e} \epsilon_{\mathcal{G}}(\iota_e(\iota_i^e(0))) = \prod_{n_e+i=1}^{n_e+m_e} \delta_i^e(0) = \prod_{n_e+i=1}^{n_e+m_e} t_{\mathcal{G}}(e)_i = t_{\mathcal{G}}(e)$$

Hence $(\mathsf{id}_{E_{\mathcal{G}}}, \epsilon_G)$ is a morphism $S_p(\mathsf{sp}(\mathcal{G})) \to \mathcal{G}$. Now, let $(f, g) \colon P(\mathcal{H}) \to \mathcal{G}$ be a morphism of pruned hypergraphs with $\mathcal{H}$ separated. Given a node v in $V_{\mathcal{H}}$ we know that there exists exactly one e_v such that v occurs in $s_{\mathcal{H}}(e_v)$ or in $t_{\mathcal{H}}(e_v)$. Moreover, v occurs exactly once and not in both. Let i_v be the index at which it appears. We define

$$\hat{g}(v) := \begin{cases} \iota_{f(e_v)}(\iota_{i_v}^{f(e_v)}(0)) & v \text{ occurs in } s_{\mathcal{H}}(e) \\ \iota_{f(e_v)}(\iota_{n_e+i_v}^{f(e_v)}(0)) & v \text{ occurs in } t_{\mathcal{H}}(e) \end{cases}$$

Computing we get

$$\hat{g}^{\star}(s_{\mathcal{H}}(e)) = \hat{g}^{\star}(\prod_{i=1}^{n_e} s_{\mathcal{H}}(e)_i) = \prod_{i=1}^{n_e}(\hat{g}(s_{\mathcal{H}}(e)_i)) = \prod_{i=1}^{n_e} \iota_{f(e)}(\iota_i^{f(e)}(0)) = s_{\mathsf{sp}(\mathcal{G})}(f(e))$$

$$\hat{g}^{\star}(t_{\mathcal{H}}(e)) = \hat{g}^{\star}(\prod_{i=1}^{m_e} t_{\mathcal{H}}(e)_i) = \prod_{i=1}^{m_e}(\hat{g}(t_{\mathcal{H}}(e)_i)) = \prod_{i=1}^{m_e} \iota_{f(e)}(\iota_{n_e+i}^{f(e)}(0))$$

$$= \prod_{i=n_e+1}^{n_e+m_e} \iota_{f(e)}(\iota_i^{f(e)}(0)) = t_{\mathsf{sp}(\mathcal{G})}(f(e))$$

proving that $(f, \hat{g}) \colon \mathcal{H} \to \mathsf{sp}(\mathcal{G})$ is a morphism of **pSep**. Notice that

$$\epsilon_{\mathcal{G}}(\hat{g}(v)) = \begin{cases} \epsilon_{\mathcal{G}}(\iota_{f(e_v)}(\iota_{i_v}^{f(e_v)}(0))) & v \text{ occurs in } s_{\mathcal{H}}(e_v) \\ \epsilon_{\mathcal{G}}(\iota_{f(e_v)}(\iota_{n_e+i_v}^{f(e_v)}(0))) & v \text{ occurs in } t_{\mathcal{H}}(e_v) \end{cases} = \begin{cases} \delta_{i_v}^{f(e_v)}(0) & v \text{ occurs in } s_{\mathcal{G}}(e) \\ \delta_{i_v}^{f(e_v)}(0) & v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases}$$

$$= \begin{cases} s_{\mathcal{G}}(f(e_v))_{i_v} & v \text{ occurs in } s_{\mathcal{G}}(e) \\ t_{\mathcal{G}}(f(e_v))_{i_v} & v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases} = \begin{cases} (g^{\star}(s_{\mathcal{H}}(e_v)))_{i_v} & v \text{ occurs in } s_{\mathcal{G}}(e) \\ (g^{\star}(t_{\mathcal{H}}(e_v)))_{i_v} & v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases}$$

$$= \begin{cases} g(s_{\mathcal{H}}(e_v)_{i_v}) & v \text{ occurs in } s_{\mathcal{G}}(e) \\ g(t_{\mathcal{H}}(e_v)_{i_v}) & v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases} = g(v)$$

and so $(\mathsf{id}_{E_{\mathcal{G}}}, \epsilon_G) \circ S_p(f, \hat{g}) = (f, g)$.

Now let (h, k) be another morphism $\mathcal{H} \to \mathsf{sp}(\mathcal{G})$ such that $(\mathsf{id}_{E_{\mathcal{G}}}, \epsilon_G) \circ S_p(h, k) = (f, g)$. This entails immediately that $h = f$ and so uniqueness follows from Remark 3.49. $\square$

Let $\mathcal{G}$ be a hypergraph. For every $e \in E_{\mathcal{G}}$, let $n_e = \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ and $m_e = \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{G}}(e))$. Given $1 \leq i \leq n_e + m_e$, we can define $\delta_i^e \colon \rightarrowtail V_{\mathcal{G}}$ as

$$\delta_i^e(0) = \begin{cases} s_{\mathcal{G}}(e)_i & \text{if } i \leq n_e \\ t_{\mathcal{G}}(e)_{i-n_e} & \text{otherwise} \end{cases}$$

In turn, this allows us to define $\epsilon_G\colon V_{\mathsf{sp}(\mathcal{G})} \to V_{\mathcal{G}}$ as the unique one fitting in the diagram below.

Now, for every hyperedge $e \in E_{\mathcal{G}}$ we have

$$
\begin{tikzcd}
1 \arrow[r, "\mathsf{id}_1"] \arrow[d, "\iota_i^e"'] & 1 \arrow[dd, "\delta_i^e"] \\
V_e \arrow[d, "\iota_e"'] & \\
V_{\mathsf{sp}(\mathcal{G})} \arrow[r, dotted, "\epsilon_{\mathcal{G}}"] & V_{\mathcal{G}}
\end{tikzcd}
$$

$$\epsilon_{\mathcal{G}}^{\star}(s_{\mathsf{sp}(\mathcal{G})}(e)) = \epsilon_{\mathcal{G}}^{\star}(\prod_{i=1}^{n_e} \iota_e(\iota_i^e(0))) = \prod_{i=1}^{n_e} \epsilon_{\mathcal{G}}(\iota_e(\iota_i^e(0))) = \prod_{i=1}^{n_e} \delta_i^e(0) = \prod_{i=1}^{n_e} s_{\mathcal{G}}(e)_i = s_{\mathcal{G}}(e)$$

$$\epsilon_{\mathcal{G}}^{\star}(t_{\mathsf{sp}(\mathcal{G})}(e)) = \epsilon_{\mathcal{G}}^{\star}(\prod_{i=n_e+1}^{n_e+m_e} \iota_e(\iota_i^e(0))) = \prod_{i=n_e+1}^{n_e+m_e} \epsilon_{\mathcal{G}}(\iota_e(\iota_i^e(0))) = \prod_{n_e+i=1}^{n_e+m_e} \delta_i^e(0) = \prod_{n_e+i=1}^{n_e+m_e} t_{\mathcal{G}}(e)_i = t_{\mathcal{G}}(e)$$

Hence $(\mathsf{id}_{E_{\mathcal{G}}}, \epsilon_G)$ is a morphism $P(\mathsf{sp}(\mathcal{G})) \to \mathcal{G}$. Now, let $(f, g)\colon P(\mathcal{H}) \to \mathcal{G}$ be a morphism of pruned hypergraphs with $\mathcal{H}$ separated. Given a node v in $V_{\mathcal{H}}$ we know that there exists exactly one e_v such that v occurs in $s_{\mathcal{H}}(e_v)$ or in $t_{\mathcal{H}}(e_v)$. Moreover, v occurs exactly once and not in both. Let i_v be the index at which it appears. We define

$$\hat{g}(v) := \begin{cases} \iota_{f(e_v)}(\iota_{i_v}^{f(e_v)}(0)) & \text{if } v \text{ occurs in } s_{\mathcal{H}}(e) \\ \iota_{f(e_v)}(\iota_{n_e+i_v}^{f(e_v)}(0)) & \text{if } v \text{ occurs in } t_{\mathcal{H}}(e) \end{cases}$$

Computing we get

$$\hat{g}^{\star}(s_{\mathcal{H}}(e)) = \hat{g}^{\star}(\prod_{i=1}^{n_e} s_{\mathcal{H}}(e)_i) = \prod_{i=1}^{n_e}(\hat{g}(s_{\mathcal{H}}(e)_i)) = \prod_{i=1}^{n_e} \iota_{f(e)}(\iota_i^{f(e)}(0)) = s_{\mathsf{sp}(\mathcal{G})}(f(e))$$

$$\hat{g}^{\star}(t_{\mathcal{H}}(e)) = \hat{g}^{\star}(\prod_{i=1}^{m_e} t_{\mathcal{H}}(e)_i) = \prod_{i=1}^{m_e}(\hat{g}(t_{\mathcal{H}}(e)_i)) = \prod_{i=1}^{m_e} \iota_{f(e)}(\iota_{n_e+i}^{f(e)}(0)) = \prod_{i=n_e+1}^{n_e+m_e} \iota_{f(e)}(\iota_i^{f(e)}(0)) = t_{\mathsf{sp}(\mathcal{G})}(f(e))$$

proving that $(f, \hat{g})\colon \mathcal{H} \to \mathsf{sp}(\mathcal{G})$ is a morphism of **pSep**. Notice that

$$\epsilon_{\mathcal{G}}(\hat{g}(v)) = \begin{cases} \epsilon_{\mathcal{G}}(\iota_{f(e_v)}(\iota_{i_v}^{f(e_v)}(0))) & \text{if } v \text{ occurs in } s_{\mathcal{H}}(e_v) \\ \epsilon_{\mathcal{G}}(\iota_{f(e_v)}(\iota_{n_e+i_v}^{f(e_v)}(0))) & \text{if } v \text{ occurs in } t_{\mathcal{H}}(e_v) \end{cases} = \begin{cases} \delta_{i_v}^{f(e_v)}(0) & \text{if } v \text{ occurs in } s_{\mathcal{G}}(e) \\ \delta_{i_v}^{f(e_v)}(0) & \text{if } v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases}$$

$$= \begin{cases} s_{\mathcal{G}}(f(e_v))_{i_v} & \text{if } v \text{ occurs in } s_{\mathcal{G}}(e) \\ t_{\mathcal{G}}(f(e_v))_{i_v} & \text{if } v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases} = \begin{cases} (g^{\star}(s_{\mathcal{H}}(e_v)))_{i_v} & \text{if } v \text{ occurs in } s_{\mathcal{G}}(e) \\ (g^{\star}(t_{\mathcal{H}}(e_v)))_{i_v} & \text{if } v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases}$$

$$= \begin{cases} g(s_{\mathcal{H}}(e_v)_{i_v}) & \text{if } v \text{ occurs in } s_{\mathcal{G}}(e) \\ g(t_{\mathcal{H}}(e_v)_{i_v}) & \text{if } v \text{ occurs in } t_{\mathcal{G}}(e) \end{cases} = g(v)$$

and so $(\mathsf{id}_{E_{\mathcal{G}}}, \epsilon_G) \circ P(f, \hat{g}) = (f, g)$.

Now let (h, k) be another morphism $\mathcal{H} \to \mathsf{sp}(\mathcal{G})$ such that $(\mathsf{id}_{E_{\mathcal{G}}}, \epsilon_G) \circ P(h, k) = (f, g)$. This entails immediately that $h = f$ and so uniqueness follows from Remark 3.49.

Lemma 3.51. **Sep** *is closed in* **Hyp** *under connected limits.*

Proof. Let $F\colon \mathbf{D} \to \mathbf{Sep}$ be a diagram with a connected category as domain and let $(\mathcal{L}, \{(p_E^D, p_V^D)\}_{D \in \mathbf{D}})$ be a limit for $S \circ F$. We have to show that $\mathcal{L}$ is separated.

By Corollary B.4 and Propositions 3.1 and 3.7 we know that $(V_{\mathcal{L}}, \{p_{V,D}\}_{D \in \mathbf{D}})$ and $(E_{\mathcal{L}}, \{p_{E,D}\}_{D \in \mathbf{D}})$ are limiting cones for, respectively, $U_{\mathbf{Hyp}} \circ S \circ F$ and $V_{\mathbf{Hyp}} \circ S \circ F$, where $U_{\mathbf{Hyp}}, V_{\mathbf{Hyp}}$. We proceed by cases.

- Suppose that there exists a node $v \in V_{\mathcal{L}}$ appearing two times in $s_{\mathcal{L}}(e)$ for some hyperedge $e \in E_{\mathcal{L}}$. Thus there are indexes $1 \le i, k \le \mathsf{lg}_{V_{\mathcal{L}}}(s_{\mathcal{L}(e)})$ with $i \neq k$ such that

$$(s_{\mathcal{L}}(e))_i = v \qquad (s_{\mathcal{L}}(e))_k = v$$

  Since $\mathbf{D}$ is connected then it is non empty. Let D be an object of $\mathbf{D}$. Then we have

$$\begin{aligned}(s_{S(F(D))}(p_E^D(e)))_i &= (p_{V,D}^{\star}(s_{\mathcal{L}}(e)))_i = p_{V,D}^{\star}((s_{\mathcal{L}}(e))_i) \\ &= p_{V,D}(v) = p_{V,D}((s_{\mathcal{L}}(e))_k) = (p_{V,D}^{\star}(s_{\mathcal{L}}(e)))_k = (s_{S(F(D))}(p_{E,D}(e)))_k\end{aligned}$$

contradicting the fact that $S(F(D))$ is separated.

Similarly, suppose that there is $v \in V_{\mathcal{H}}$ such that

$$(t_{\mathcal{L}}(e))_i = v \qquad (t_{\mathcal{L}}(e))_k = v$$

for some $e \in E_{\mathcal{H}}$ and $1 \leq i, k \leq \mathsf{lg}_{V_{\mathcal{L}}}(s_{\mathcal{L}(e)})$ with $i \neq k$. Then for any $D \in \mathbf{D}$

$$\begin{aligned}(t_{S(F(D))}(p_E^D(e)))_i &= (p^\star_{V,D}(t_{\mathcal{L}}(e)))_i = p^\star_{V,D}((t_{\mathcal{L}}(e))_i)\\ =&p_{V,D}(v) = p_{V,D}((t_{\mathcal{L}}(e))_k) = (p^\star_{V,D}(t_{\mathcal{L}}(e)))_k = (t_{S(F(D))}(p_{E,D}(e)))_k\end{aligned}$$

which yields again a contradiction.

- Suppose that there exist two hyperedges $e, e' \in E_{\mathcal{L}}$ and a node $v \in V_{\mathcal{L}}$ such that

  $$(s_{\mathcal{L}}(e))_i = v \qquad (t_{\mathcal{L}}(e'))_k = v$$

  for indexes $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{L}}}(s_{\mathcal{L}(e)})$ and $1 \leq k \leq \mathsf{lg}_{V_{\mathcal{L}}}(t_{\mathcal{L}(e)})$. We have already noticed that $\mathbf{D}$ is non empty. So, let $D \in \mathbf{D}$. Then we have

  $$\begin{aligned}(s_{S(F(D))}(p_E^D(e)))_i &= (p^\star_{V,D}(s_{\mathcal{L}}(e)))_i = p^\star_{V,D}((s_{\mathcal{L}}(e))_i)\\ =&p_{V,D}(v) = p_{V,D}((t_{\mathcal{L}}(e'))_k) = (p^\star_{V,D}(t_{\mathcal{L}}(e')))_k = (t_{S(F(D))}(p_{E,D}(e')))_k\end{aligned}$$

  But this is a contradiction since, by hypothesis, $S(F(D))$ is separated.

- Let now e, e' be two distinct hyperedges of $\mathcal{L}$ such that there exists $v \in V_{\mathcal{L}}$ with the property that

  $$(s_{\mathcal{L}}(e))_i = v \qquad (s_{\mathcal{L}}(e'))_k = v$$

  with $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{L}}}(s_{\mathcal{L}(e)})$ and $1 \leq k \leq \mathsf{lg}_{V_{\mathcal{L}}}(t_{\mathcal{L}(e)})$. For every $D \in \mathbf{D}$ we can compute to get

  $$\begin{aligned}(s_{S(F(D))}(p_E^D(e)))_i &= (p^\star_{V,D}(s_{\mathcal{L}}(e)))_i = p^\star_{V,D}((s_{\mathcal{L}}(e))_i)\\ =&p_{V,D}(v) = p_{V,D}((s_{\mathcal{L}}(e'))_k) = (p^\star_{V,D}(s_{\mathcal{L}}(e')))_k = (s_{S(F(D))}(p_{E,D}(e')))_k\end{aligned}$$

  Since $S(F(D))$ is an object of **Sep** we get that, for every $D \in \mathbf{D}$, $p_{E,D}(e)$ and $p_{E,D}(e')$ must be equal. Since $(E_{\mathcal{L}}, \{p_{E,D}\}_{D\in\mathbf{D}})$ this entails that $e = e'$, which is a contradiction.

- We proceed as before, supposing that e and e' are distinct elements of $\mathcal{L}$ for which there are $1 \leq i \leq \mathsf{lg}_{V_{\mathcal{L}}}(s_{\mathcal{L}(e)})$, $1 \leq k \leq \mathsf{lg}_{V_{\mathcal{L}}}(t_{\mathcal{L}(e)})$ and a node $v \in V_{\mathcal{L}}$ such that

  $$(t_{\mathcal{L}}(e))_i = v \qquad (t_{\mathcal{L}}(e'))_k = v$$

  Then, given $D \in \mathbf{D}$ we have

  $$\begin{aligned}(t_{S(F(D))}(p_E^D(e)))_i &= (p^\star_{V,D}(t_{\mathcal{L}}(e)))_i = p^\star_{V,D}((t_{\mathcal{L}}(e))_i)\\ =&p_{V,D}(v) = p_{V,D}((t_{\mathcal{L}}(e'))_k) = (p^\star_{V,D}(t_{\mathcal{L}}(e')))_k = (t_{S(F(D))}(p_{E,D}(e')))_k\end{aligned}$$

  Thus $p_{E,D}(e)$ and $p_{E,D}(e')$ coincide for every $D \in \mathbf{D}$, hence $e = e'$ and again we get a contradiction. $\square$

Lemma 3.54. **pSep** *is equivalent to* $\mathbf{Set}/(\mathbb{N}\times\mathbb{N})$.

Proof. Let $\mathcal{G}$ be a pruned and separated hypergraph. We can define $F(\mathcal{G})$ as $(\mathsf{lg}_{V_{\mathcal{G}}} \circ s_{\mathcal{G}}, \mathsf{lg}_{V_{\mathcal{G}}} \circ t_{\mathcal{G}})\colon E_{\mathcal{G}} \to \mathbb{N}\times\mathbb{N}$. Given a morphism $(h, k)\colon \mathcal{G} \to \mathcal{H}$ and using Remark 3.4 we have

$$(\mathsf{lg}_{V_{\mathcal{H}}} \circ s_{\mathcal{H}}, \mathsf{lg}_{V_{\mathcal{H}}} \circ t_{\mathcal{H}}) \circ h = (\mathsf{lg}_{V_{\mathcal{H}}} \circ s_{\mathcal{H}} \circ h, \mathsf{lg}_{V_{\mathcal{H}}} \circ t_{\mathcal{H}} \circ h) = (\mathsf{lg}_{V_{\mathcal{H}}} \circ k^\star \circ s_{\mathcal{G}}, \mathsf{lg}_{V_{\mathcal{H}}} \circ k^\star \circ t_{\mathcal{G}}) = (\mathsf{lg}_{V_{\mathcal{G}}} \circ s_{\mathcal{G}}, \mathsf{lg}_{V_{\mathcal{G}}} \circ t_{\mathcal{G}})$$

Thus we can define $F(h, k)$ as h to get a functor $\mathbf{pSep} \to \mathbf{Set}/(\mathbb{N}\times\mathbb{N})$. Let us show that it is an equivalence.

- F is faithful. This follows at once from Remark 3.48.
- F is full. Let $k\colon E_{\mathcal{G}} \to E_{\mathcal{H}}$ be an arrow such that

$$(\mathsf{lg}_{V_{\mathcal{H}}} \circ s_{\mathcal{H}}, \mathsf{lg}_{V_{\mathcal{H}}} \circ t_{\mathcal{H}}) \circ f = (\mathsf{lg}_{V_{\mathcal{G}}} \circ s_{\mathcal{G}}, \mathsf{lg}_{V_{\mathcal{G}}} \circ t_{\mathcal{G}})$$

Now, for every $e \in E_{\mathcal{G}}$, let $\iota^s_e\colon V^s_e \rightarrowtail V_{\mathcal{G}}$ and $\iota^t_e\colon V^t_e \rightarrowtail V_{\mathcal{G}}$ be the inclusions of

$$V^s_e = \{v \in V_{\mathcal{G}} \mid \mathsf{in}_{\mathcal{G}}(v,e) = 1\} \qquad V^t_e = \{v \in V_{\mathcal{G}} \mid \mathsf{out}_{\mathcal{G}}(v,e) = 1\}$$

Since $\mathcal{G}$ is pruned and separated, $(V_{\mathcal{G}}, \{\iota^s_e\}_{e\in E_{\mathcal{G}}} + \{\iota^t_e\}_{e\in E_{\mathcal{G}}})$ is a coproduct. Given $v \in V^s_e$ and $w \in V^t_e$, let i_v and k_w be the unique indexes such that $\iota^s_e(v) = s_{\mathcal{G}}(e)_{i_v}$ and $\iota^t_e(w) = t_{\mathcal{G}}(e)_{k_w}$ and define

$$\begin{tikzcd} & V^s_{\mathcal{G}} \arrow[dl, tail, "\iota^s_e"'] \arrow[dr, "g^s_e"] & \\ V_{\mathcal{G}} \arrow[rr, dotted, "g"] & & V_{\mathcal{H}} \end{tikzcd}$$

$$\begin{tikzcd} & V^t_{\mathcal{G}} \arrow[dl, tail, "\iota^t_e"'] \arrow[dr, "g^t_e"] & \\ V_{\mathcal{G}} \arrow[rr, dotted, "g"] & & V_{\mathcal{H}} \end{tikzcd}$$

$$\begin{aligned} g^s_e\colon V^s_e &\to V_{\mathcal{H}} \\ v &\mapsto s_{\mathcal{H}}(f(e))_{i_v} \end{aligned} \qquad \begin{aligned} g^t_e\colon V^t_e &\to V_{\mathcal{H}} \\ w &\mapsto t_{\mathcal{H}}(f(e))_{k_w} \end{aligned}$$

so that there is an arrow $g\colon V_{\mathcal{G}} \to \mathcal{H}$ fitting in the diagram aside.
Given $e \in E_{\mathcal{G}}$, for every $1 \le i \le \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ and $1 \le k \le \mathsf{lg}_{V_{\mathcal{G}}}(t_{\mathcal{G}}(e))$ we can compute to get

$$\begin{aligned} (g^\star(s_{\mathcal{G}}(e)))_i &= (g^\star(s_{\mathcal{G}}(e)))_i = g(s_{\mathcal{G}}(e)_i) = s_{\mathcal{H}}(f(e))_i \\ (g^\star(t_{\mathcal{G}}(e)))_k &= (g^\star(t_{\mathcal{G}}(e)))_k = g(t_{\mathcal{G}}(e)_k) = t_{\mathcal{H}}(f(e))_k \end{aligned}$$

Hence (f,g) is a morphism $\mathcal{G} \to \mathcal{H}$ such that $F(f,g) = f$.

- F is essentially surjective. Consider a function $(f_1, f_2)\colon X \to \mathbb{N} \times \mathbb{N}$. Given $x \in X$, we can consider $f_1(x)$ and $f_2(x)$ as ordinal sets and take the coproduct $(V, \{\iota^1_x\}_{x\in X} + \{\iota^2_x\}_{x\in X})$ of the family $\{f_1(x)\}_{x\in X} + \{f_2(x)\}_{x\in X}$. Define

$$\begin{aligned} s\colon X &\to V^\star \\ x &\mapsto \prod_{n=0}^{f_1(x)-1} \iota^1_x(n) \end{aligned} \qquad \begin{aligned} t\colon X &\to V^\star \\ x &\mapsto \prod_{n=0}^{f_2(x)-1} \iota^2_x(n) \end{aligned}$$

Let $\mathcal{G}$ be (X, V, s, t), by construction $F(\mathcal{G}) = (f_1, f_2)$ and we can conclude. □

Proposition 3.56. **pSep** *is closed in* **Hyp** *under colimits and pullbacks.*

Proof. Let $F\colon \mathbf{D} \to \mathbf{pSep}$ be a diagram and suppose that $(\mathcal{C}, \{(q^D_E, q^D_V)\}_{D\in\mathbf{D}})$ is a colimiting cocone for F. By Proposition 3.50 we know that $P \circ S_p$ preserves colimits, but then the vertex of every colimiting cocone for $P \circ S_p \circ F$ is isomorphic to $\mathcal{C}$ and thus it must be pruned and separated.

We can now turn to pullbacks. Let us consider a pullback square in **Hyp** as the one aside in which $\mathcal{G}$, $\mathcal{H}$ and $\mathcal{K}$ are pruned and separated. We already know, by Lemma 3.51 that $\mathcal{P}$ is separated, so it is enough to show that $\mathcal{P}$ is pruned.

$$\begin{tikzcd} \mathcal{P} \arrow[r, "(p_1,p_2)"'] \arrow[d, "(q_1,q_2)"'] & \mathcal{G} \arrow[d, "(f_1,f_2)"] \\ \mathcal{H} \arrow[r, "(g_1,g_2)"] & \mathcal{K} \end{tikzcd}$$

Let v be a node in $V_{\mathcal{P}}$. Since $\mathcal{G}$ and $\mathcal{H}$ are pruned we know that $p_2(v)$ and $q_2(v)$ must occur in the source or target of some hyperedge. We have four cases.

- There exist $e \in E_{\mathcal{G}}$ and $h \in E_{\mathcal{H}}$ such that $p_2(v)$ occurs in the source of e and $q_2(v)$ in that of h. Let $1 \le i \le \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{G}}(e))$ and $1 \le j \le \mathsf{lg}_{V_{\mathcal{G}}}(s_{\mathcal{H}}(h))$ be indexes such that:

$$s_{\mathcal{G}}(e)_i = p_2(v) \qquad s_{\mathcal{H}}(h)_j = q_2(v)$$

Hence we have:

$$\begin{aligned} s_{\mathcal{K}}(f_1(e))_i &= (f_2^\star(s_{\mathcal{G}}(e)))_i = f_2(s_{\mathcal{G}}(e)_i) = f_2(p_2(v)) \\ &= g_2(q_2(v)) = g_2(s_{\mathcal{H}}(h)_j) = (g_2^\star(s_{\mathcal{H}}(h)))_k = s_{\mathcal{H}}(g_1(h))_j \end{aligned}$$

Since $\mathcal{K}$ is separated this entails that $i = j$ and $f_1(e) = g(e)$. So there exists $a \in E_\mathcal{P}$ such that $p_1(a) = e$ and $q_1(a) = h$. Computing we have:

$$p_2(s_\mathcal{P}(a)_i) = (p_2^\star(s_\mathcal{P}(a)))_i = s_\mathcal{G}(p_1(a))_i = s_\mathcal{G}(e)_i = p_2(v)$$
$$q_2(s_\mathcal{P}(a)_i) = (q_2^\star(s_\mathcal{P}(a)))_i = s_\mathcal{G}(q_1(a))_i = s_\mathcal{G}(h)_i = q_2(v)$$

and so $s_\mathcal{P}(a)_i = v$.

- There exist $e \in E_\mathcal{G}$ and $h \in E_\mathcal{H}$ such that $p_2(v)$ occurs in $t_\mathcal{G}(e)$ and $q_2(v)$ in $t_\mathcal{H}(h)$. As before, let $1 \leq i \leq \mathsf{lg}_{V_\mathcal{G}}(t_\mathcal{G}(e))$ and $1 \leq j \leq \mathsf{lg}_{V_\mathcal{G}}(t_\mathcal{H}(h))$ be indexes such that:

  $$t_\mathcal{G}(e)_i = p_2(v) \qquad t_\mathcal{H}(h)_j = q_2(v)$$

  so that

  $$t_\mathcal{K}(f_1(e))_i = (f_2^\star(t_\mathcal{G}(e)))_i = f_2(t_\mathcal{G}(e)_i) = f_2(p_2(v))$$
  $$= g_2(q_2(v)) = g_2(t_\mathcal{H}(h)_j) = (g_2^\star(t_\mathcal{H}(h)))_k = t_\mathcal{H}(g_1(h))_j$$

  The fact that $\mathcal{K}$ is separated assures that $i = j$ and $f_1(e) = g(e)$. Let then $a \in E_\mathcal{P}$ be such that $p_1(a) = e$ and $q_1(a) = h$. Computing we have:

  $$p_2(t_\mathcal{P}(a)_i) = (p_2^\star(t_\mathcal{P}(a)))_i = t_\mathcal{G}(p_1(a))_i = t_\mathcal{G}(e)_i = p_2(v)$$
  $$q_2(t_\mathcal{P}(a)_i) = (q_2^\star(t_\mathcal{P}(a)))_i = t_\mathcal{G}(q_1(a))_i = t_\mathcal{G}(h)_i = q_2(v)$$

  allowing us to deduce that $t_\mathcal{P}(a)_i = v$.

- Suppose that there are indexes $1 \leq i \leq \mathsf{lg}_{V_\mathcal{G}}(s_\mathcal{G}(e))$ and $1 \leq j \leq \mathsf{lg}_{V_\mathcal{G}}(t_\mathcal{H}(h))$ such that:

  $$s_\mathcal{G}(e)_i = p_2(v) \qquad t_\mathcal{H}(h)_j = q_2(v)$$

  Computing we get:

  $$s_\mathcal{K}(f_1(e))_i = (f_2^\star(s_\mathcal{G}(e)))_i = f_2(s_\mathcal{G}(e)_i) = f_2(p_2(v))$$
  $$= g_2(q_2(v)) = g_2(t_\mathcal{H}(h)_j) = (g_2^\star(t_\mathcal{H}(h)))_k = t_\mathcal{H}(g_1(h))_j$$

  This contradicts the fact that $\mathcal{K}$ is separated.

- Consider now indexes $1 \leq i \leq \mathsf{lg}_{V_\mathcal{G}}(t_\mathcal{G}(e))$ and $1 \leq j \leq \mathsf{lg}_{V_\mathcal{G}}(s_\mathcal{H}(h))$ such that:

  $$t_\mathcal{G}(e)_i = p_2(v) \qquad s_\mathcal{H}(h)_j = q_2(v)$$

  Proceeding as in the previous point we have:

  $$t_\mathcal{K}(f_1(e))_i = (f_2^\star(t_\mathcal{G}(e)))_i = f_2(t_\mathcal{G}(e)_i) = f_2(p_2(v))$$
  $$= g_2(q_2(v)) = g_2(s_\mathcal{H}(h)_j) = (g_2^\star(s_\mathcal{H}(h)))_k = s_\mathcal{H}(g_1(h))_j$$

  This again stands in contradiction with $\mathcal{K}$ being an object of **Sep**. □

A.3. Proofs for Section 4

Proposition 4.4. *Consider the forgetful functor $T \colon \mathbf{Hyp}_{\mathsf{eq}} \to \mathbf{Hyp}$. Then the following facts hold*

1. *T is faithful;*
2. *T has a left adjoint;*
3. *T has a right adjoint.*

Proof. 1. This follows at once from Remark 4.3.

2. Let $\mathcal{H}$ be a hypergraph, and define $L(\mathcal{H}) := (E_{\mathcal{H}}, V_{\mathcal{H}}, V_{\mathcal{H}}, s_{\mathcal{H}}, t_{\mathcal{H}}, \mathsf{id}_{V_{\mathcal{H}}})$. By construction we have $T(L(\mathcal{H})) = \mathcal{H}$, thus we can define $\eta_{\mathcal{H}} \colon \mathcal{H} \to T(L(\mathcal{H}))$ as the identity $\mathsf{id}_{\mathcal{H}}$.
   To see that in this way we get a unit, take an arrow $(h, k) \colon \mathcal{H} \to T(\mathcal{G})$ for some $\mathcal{G}$ in $\mathbf{Hyp}_{\mathsf{eq}}$. Then $(h, k, q_{\mathcal{G}} \circ k)$ is an arrow $L(\mathcal{H}) \to \mathcal{G}$ and the unique one such that $T(h, k, q_{\mathcal{G}} \circ k) \circ \eta_{\mathcal{H}} = (h, k)$
3. For every hypergraph $\mathcal{H}$ define $R(\mathcal{H})$ as $(E_{\mathcal{H}}, V_{\mathcal{H}}, 1, s_{\mathcal{H}}, t_{\mathcal{H}}, !_{V_{\mathcal{H}}})$, so that $T(R(\mathcal{H}))$ is again $\mathcal{H}$. Now, let $\epsilon_{\mathcal{H}} \colon T(R(\mathcal{H})) \to \mathcal{H}$ be the identity and take an arrow $(h, k) \colon T(\mathcal{G}) \to \mathcal{H}$ for some $\mathcal{G}$ in $\mathbf{Hyp}_{\mathsf{eq}}$. Notice that $!_{Q_{\mathcal{G}}} \circ q_{\mathcal{G}} = !_{V_{\mathcal{H}}} \circ k$ so that $(h, k, !_{Q_{\mathcal{G}}})$ is an arrow $\mathcal{G} \to R(\mathcal{H})$ of $\mathbf{Hyp}_{\mathsf{eq}}$ such that $\epsilon_{\mathcal{H}} \circ T(h, k, !_{Q_{\mathcal{G}}}) = (h, k)$.
   Uniqueness of such an arrow follows at once from the first point and the fact that 1 is terminal. □

Remark A.6. Before proceeding further, let us recall the following result about the classes of regular epis (i.e. surjections) and of monos in **Set**. In particular, we need the fact that they form a *factorization system* [26] on **Set**. This amounts to ask that

1. every arrow $f \colon X \to Y$ can be factored as $m \circ e$, where $e \colon X \twoheadrightarrow Q$ is a regular epi and $m \colon Q \rightarrowtail Y$ is a mono;
2. for every commuting square as the one on the right, with $e \colon X \twoheadrightarrow E$ a surjection and $m \colon N \rightarrowtail Y$ a mono, there exists a unique $k \colon E \to N$ making it commutative.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, two heads, "e"'] & N \arrow[d, tail, "m"] \\ E \arrow[r, "g"'] \arrow[ur, dotted, "k"] & Y \end{tikzcd}$$

From these properties, one can deduce that if $f = m \circ e$ and $f = m' \circ e'$ are two factorizations of f then there is a bijection ϕ such that $m = m' \circ \phi$ and $\phi \circ e = e'$.

We will also need the following well-known fact about regular epis.

Lemma A.7. *Let $F, G \colon \mathbf{D} \rightrightarrows \mathbf{X}$ be two diagrams and suppose that $\mathbf{X}$ has all colimits of shape $\mathbf{D}$. Let $(X, \{x_d\}_{d \in \mathbf{D}})$ and $(Y, \{y_d\}_{d \in D})$ be the colimits of F and G, respectively. If $\phi \colon F \to G$ is a natural transformation whose components are regular epis, then the arrow induced by ϕ from X to Y is a regular epi.*

Lemma 4.8. *Consider a diagram $F \colon \mathbf{D} \to \mathbf{Hyp}_{\mathsf{eq}}$ and let $(E_D, V_D, Q_D, s_D, t_D, q_D)$ be the image of an object D. Then the following facts hold*

1. *F has a colimit, which is preserved by K;*
2. *consider a cone $(L, \{l_D\}_{D \in \mathbf{D}})$ limiting for $K \circ F$ and let $((E, V), \{(\pi_E^D, \pi_V^D)\}_{D \in \mathbf{D}})$ be one for $T \circ F$, then F has a limit (E, V, Q, s, t, q) and there is a mono $m \colon Q \rightarrowtail L$ such that the diagram on the right commutes for every $D \in \mathbf{D}$.*

$$\begin{tikzcd} V \arrow[r, "\pi_V^D"'] \arrow[d, two heads, "q"'] & V_D \arrow[dd, two heads, "q_D"] \\ Q \arrow[d, dotted, tail, "m"'] & \\ L \arrow[r, "l_D"] & Q_D \end{tikzcd}$$

Proof. 1. We know by Corollary 3.18 that **Hyp** is cocomplete. Thus, let (E, V, s, t) together with $\{(\kappa_E^D, \kappa_V^D)\}_{D \in \mathbf{D}}$ be a colimit for $T \circ F$. By Lemma B.3 we also know that $(V, \{\kappa_V^D\}_{D \in \mathbf{D}})$ is a colimit for $U_{\mathsf{eq}} \circ F$.
Moreover, since **Set** is cocomplete too we can also take a colimit $(C, \{c_d\}_{d \in \mathbf{D}})$ for $K \circ F$. Now, let α be an arrow $D \to D'$ in $\mathbf{D}$, and suppose that $F(\alpha)$ is (h_1, h_2, h_3). By definition of morphisms in $\mathbf{Hyp}_{\mathsf{eq}}$, the square on the right commutes.

$$\begin{tikzcd} V_d \arrow[r, "h_2"] \arrow[d, two heads, "q_D"'] & V_{D'} \arrow[d, two heads, "q_{D'}"] \\ Q_D \arrow[r, "h_3"] & Q_{D'} \end{tikzcd}$$

Thus the family $\{q_D\}_{D \in \mathbf{D}}$ form a natural transformation $U_{\mathsf{eq}} \circ F \to K \circ F$. By Lemma A.7, the induced arrow $q : V \to C$ between the colimits is a surjection. We can then consider the object (E, V, C, s, t, q) of $\mathbf{Hyp}_{\mathsf{eq}}$, together with the family $\{(\kappa_E^D, \kappa_V^D, c_D)\}_{D \in \mathbf{D}}$, which by construction is a cocone on F. Let $((E_{\mathcal{G}}, V_{\mathcal{G}}, Q_{\mathcal{G}}, s_{\mathcal{G}}, t_{\mathcal{G}}, q_{\mathcal{G}}), \{(h_E^D, h_V^D, h_Q^D)\}_{D \in \mathbf{D}})$ be a cocone, then $((E_{\mathcal{G}}, V_{\mathcal{G}}, s_{\mathcal{G}}, t_{\mathcal{G}}), \{(h_E^D, h_V^D)\}_{D \in \mathbf{D}})$ and $(Q_{\mathcal{G}}, \{h_Q^D\}_{D \in \mathbf{D}})$ are cocone on, respectively, $T \circ F$ and $K \circ D$, giving arrows (h_E, h_V) in **Hyp** and h_Q in **Set** such that

$$(h_E, h_V) \circ (\kappa_E^D, \kappa_V^D) = (h_E^D, h_V^D) \qquad h_Q \circ c_D = h_Q^D$$

Now, to show that (h_E, h_V, h_Q) is an arrow of $\mathbf{Hyp}_{\mathsf{eq}}$ we can compute

$$h_Q \circ q \circ \kappa_V^D = h_Q \circ c_D \circ q_D = h_Q^D \circ q_D = q_{\mathcal{G}} \circ h_V^D = q_{\mathcal{G}} \circ h_V \circ \kappa_V^D$$

Uniqueness of such arrow follows form the fact that T is faithful and Remark 4.3.

2. **Hyp** is complete, again by Corollary 3.18, we can then consider a limiting cone $((E, V, s, t), \{(\pi_E^D, \pi_V^D)\}_{D\in\mathbf{D}})$ over of $T \circ F$. Now, $(V, \{q_D \circ \pi_V^D\}_{D\in\mathbf{D}})$, is a cone for $K \circ F$: indeed, if $\alpha \colon D \to D'$ is an arrow of $\mathbf{D}$, and $F(\alpha) = (h_1, h_2, h_3)$, then we have

$$h_3 \circ q_D \circ \pi_V^D = q_{D'} \circ h_2 \circ \pi_V^D = q_{D'} \circ \pi_V^{D'}$$

Thus there is an arrow $l \colon V \to L$ such that $l_D \circ l = q_D \circ \pi_V^D$. By Remark A.6, we know that there exist $m \colon Q \rightarrowtail L$ and $q \colon X \twoheadrightarrow Q$ such that $m \circ q = l$. Since the identity is mono, Remark A.6 yield a unique arrow π_Q^D fitting in the rectangle aside.

$$\begin{tikzcd} V \arrow[r, "\pi_V^D"] \arrow[d, two heads, "q"'] & V_D \arrow[r, two heads, "q_D"] & Q_D \arrow[d, "\mathsf{id}_{Q_D}"] \\ Q \arrow[r, tail, "m"] \arrow[urr, dotted, "\pi_Q^D"] & L \arrow[r, "l_D"] & Q_D \end{tikzcd}$$

Let α be an arrow $D \to D'$ in $\mathbf{D}$. Then we have

$$T(F\alpha \circ (\pi_E^D, \pi_V^D, \pi_Q^D)) = T(F(\alpha)) \circ (\pi_E^D, \pi_V^D) = (\pi_E^{D'}, \pi_V^{D'}) = T(\pi_E^{D'}, \pi_V^{D'}, \pi_Q^{D'})$$

Thus, by faitfhulness of T, $((E, V, C, s, t, q), \{(\pi_E^D, \pi_V^D, \pi_Q^D)\}_{D\in\mathbf{D}})$ is a cone over F. To see that it is terminal, let $((E_\mathcal{G}, V_\mathcal{G}, Q_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G}, q_\mathcal{G}), \{(h_E^D, h_V^D, h_Q^D)\}_{D\in\mathbf{D}})$ be another cone. In particular there is an arrow $(h_E, h_V) \colon (E_\mathcal{G}, V_\mathcal{G}, s_\mathcal{G}, t_\mathcal{G}) \to (E, V, s, t)$ in **Hyp** such that

$$\pi_E^D \circ h_E = h_E^D \qquad \pi_V^D \circ h_V = h_V^D$$

Moreover, applying K we get that $(Q_\mathcal{G}, \{h_Q^D\}_{D\in\mathbf{D}})$ is a cone over $K \circ F$, so that there is an arrow $h \colon Q_\mathcal{G} \to L$ such that $l_D \circ h = h_Q^D$. Thus

$$l_D \circ h \circ q_\mathcal{G} = h_Q^D \circ q_\mathcal{G} = q_D \circ h_V^D = q_D \circ \pi^D \circ h_V = l_D \circ l \circ h_V$$

$$\begin{tikzcd} V_\mathcal{G} \arrow[r, "h_V"] \arrow[d, two heads, "q_\mathcal{G}"'] & V \arrow[r, two heads, "q"] & Q \arrow[d, tail, "m"] \\ Q_\mathcal{G} \arrow[rr, "h"] \arrow[urr, dotted, "k_Q"] & & L \end{tikzcd}$$

Hence the outer boundary of the square on the left commutes, yielding the dotted diagonal arrow $k_Q \colon Q_\mathcal{G} \to Q$. We have therefore built an arrow (h_E, h_V, k_Q) of $\mathbf{Hyp}_{\mathsf{eq}}$ such that

$$(\pi_E^D, \pi_V^D, \pi_Q^D) \circ (h_E, h_V, k_Q) = (h_E^D, h_V^D, h_Q^D)$$

As in the point above, uniqueness is guaranteed by faithfulness of T and Remark 4.3. ☐

Corollary 4.9. *An arrow $(h_E, h_V, h_Q) : \mathcal{G} \to \mathcal{H}$ in $\mathbf{Hyp}_{\mathsf{eq}}$ is a regular mono if and only if all its components are injective functions.*

Proof. $(\Rightarrow)$ If (h_E, h_V, h_Q) is mono, from Corollary 4.6 we have that h_E and h_V are monos.

Let now $(f_E, f_V, f_Q), (g_E, g_V, g_Q) : \mathcal{H} \rightrightarrows \mathcal{K}$ be arrows equalized by (h_E, h_V, h_Q). Let $e \colon E_Q \rightarrowtail Q_\mathcal{H}$ be the equalizer in **Set** of $f_Q, g_Q \colon Q_\mathcal{H} \rightrightarrows Q_\mathcal{K}$. From the third point of Proposition 3.1 and Corollary B.4 we know that $h_V \colon V_\mathcal{G} \rightarrowtail V_\mathcal{H}$ is the equalizer of $f_V, g_V \colon V_\mathcal{H} \rightrightarrows V_\mathcal{K}$, thus by the second point of Lemma 4.8 we know that there is a mono $m \colon Q_\mathcal{G} \rightarrowtail E_Q$ fitting in the diagram aside. Now, notice that:

$$\begin{tikzcd} V_\mathcal{G} \arrow[r, tail, "h_V"'] \arrow[d, two heads, "q_\mathcal{G}"'] & V_\mathcal{H} \arrow[dd, two heads, "q_\mathcal{H}"] \\ Q_\mathcal{G} \arrow[d, tail, dotted, "m"'] & \\ E_Q \arrow[r, tail, "e"] & Q_\mathcal{H} \end{tikzcd}$$

$$e \circ m \circ q_\mathcal{G} = q_\mathcal{H} \circ h_V = h_Q \circ q_\mathcal{G}$$

Since $q_\mathcal{G}$ is epi we conclude that $e \circ m = h_Q$, from which the claim follows.

$(\Leftarrow)$ Suppose that h_E, h_V, and h_Q are monos. We can build the cokernel pair of the three arrows, taking the pushout of h_E, h_V and h_Q along itself, obtaining the three diagrams below, which by Proposition 2.6 are also pullbacks.

$$\begin{tikzcd} E_\mathcal{G} \arrow[r, tail, "h_E"] \arrow[d, tail, "h_E"'] & E_\mathcal{H} \arrow[d, tail, "g_E"] \\ E_\mathcal{H} \arrow[r, tail, "f_E"'] & E_\mathcal{K} \end{tikzcd} \qquad \begin{tikzcd} V_\mathcal{G} \arrow[r, tail, "h_V"] \arrow[d, tail, "h_V"'] & V_\mathcal{H} \arrow[d, tail, "g_V"] \\ V_H \arrow[r, tail, "f_V"'] & V_\mathcal{K} \end{tikzcd} \qquad \begin{tikzcd} Q_\mathcal{G} \arrow[r, tail, "h_Q"] \arrow[d, tail, "h_Q"'] & Q_\mathcal{H} \arrow[d, tail, "g_Q"] \\ Q_\mathcal{H} \arrow[r, tail, "f_Q"'] & Q_\mathcal{K} \end{tikzcd}$$

Now, we know by Proposition 3.7 and Lemma B.3 that there exists arrows $s, t \colon E_\mathcal{K} \rightrightarrows V_\mathcal{K}$ such that the resulting hypergraph is the cokernel pair of (h_E, h_V) in **Hyp**. Moreover, by Lemma A.7 we know that the unique arrow $q\colon V_{\mathcal{K}} \to Q_{\mathcal{K}}$ induced by $q_Q \circ q_{\mathcal{H}}$ and $f_Q \circ q_{\mathcal{H}}$ is a regular epi. Let thus $\mathcal{K}$ be the resulting hypergraph with equivalence.

By construction we have two arrows $(f_E, f_V, f_Q), (g_E, g_V, g_Q) : \mathcal{H} \rightrightarrows \mathcal{K}$ such that

$$(f_E, f_V, f_Q) \circ (h_E, h_V, h_Q) = (g_E, g_V, g_Q) \circ (h_E, h_V, h_Q)$$

Now, in **Set** every mono is regular [30], so by the dual of Proposition 2.11 we deduce that h_E, h_V and h_Q are the equalizers of the arrows built above. By Proposition 3.7 and corollary B.4 we also know that (h_E, h_V) is the equalizer in **Hyp** of (f_E, f_V) and (g_E, g_V). Hence, if $(z_E, z_V, z_Q)\colon \mathcal{Z} \to \mathcal{H}$ is an arrow such that

$$(f_E, f_V, f_Q) \circ (z_E, z_V, z_Q) = (g_E, g_V, g_Q) \circ (z_E, z_V, z_Q)$$

then we get a unique morphism $(a_E, a_V)\colon (E_{\mathcal{Z}}, V_{\mathcal{Z}}) \to (E_{\mathcal{G}}, V_{\mathcal{G}})$ such that $(h_E, h_V) \circ (a_E, a_V) = (z_E, z_V)$. Moreover, since $g_Q \circ z_Q = f_Q \circ z_Q$ there is a unique $a_Q\colon Q_{\mathcal{Z}} \to Q_{\mathcal{G}}$ such that $h_Q \circ a_Q = z_Q$. Thus

$$h_Q \circ a_Q \circ q_{\mathcal{Z}} = z_Q \circ q_{\mathcal{Z}} = q_{\mathcal{H}} \circ z_V = q_{\mathcal{H}} \circ h_V \circ a_V = h_Q \circ q_{\mathcal{G}} \circ a_V$$

Since h_Q is mono we conclude that $a_Q \circ q_{\mathcal{Z}} = q_{\mathcal{G}} \circ a_V$ and we conclude. $\square$

Lemma 4.11. *The class* Pb *contains all isomorphisms, it is closed under composition, decomposition and it is stable under pullbacks and pushouts.*

Proof. Closure under composition and decomposition follows from Lemma A.1, while stability under pushout follows from the first point of Lemma 2.14 because pushouts along injections are also pullbacks by Proposition 2.6. We are left with stability under pullbacks. Let $h\colon \mathcal{G}_1 \to \mathcal{G}_3$ be an arrow in Pb and $k\colon \mathcal{G}_2 \to \mathcal{G}_3$ be another morphism of $\mathbf{Hyp}_{\mathsf{eq}}$, with $\mathcal{G}_i = (E_i, V_i, Q_i, s_i, t_i, q_i)$.

Consider the diagrams below, in which the square on the left is a pullback in $\mathbf{Hyp}_{\mathsf{eq}}$, with $\mathcal{P} = (E_{\mathcal{P}, V_{\mathcal{P}}}, Q_{\mathcal{P}}, s_{\mathcal{P}}, t_{\mathcal{P}}, q_{\mathcal{P}})$, and in which the right half of the right rectangle is a pullback too, so that the existence of $m_{\mathcal{P}}$ is guaranteed by Lemma 4.8.

Again by Lemma 4.8 the two halves of the central rectangle are pullbacks and so by Lemma A.1 the whole rectangle is a pullback too. This, in turn entails that the whole right rectangle is a pullback and so, also its left half is a pullback (again by Lemma A.1)

$$\begin{tikzcd} \mathcal{P} \arrow[r, "p"] \arrow[d, "r"'] & \mathcal{G}_1 \arrow[d, "h"] \\ \mathcal{G}_2 \arrow[r, "k"'] & \mathcal{G}_3 \end{tikzcd} \qquad \begin{tikzcd} V_{\mathcal{P}} \arrow[r, "p_V"'] \arrow[d, tail, "r_V"'] \arrow[rr, bend left, "p_Q \circ q_{\mathcal{P}}"] & V_1 \arrow[r, two heads, "q_1"'] \arrow[d, tail, "h_V"'] & Q_1 \arrow[d, tail, "h_Q"] \\ V_2 \arrow[r, "k_V"] \arrow[rr, bend right, "k_Q \circ q_2"'] & V_3 \arrow[r, two heads, "q_3"] & Q_3 \end{tikzcd} \qquad \begin{tikzcd} V_{\mathcal{P}} \arrow[r, two heads, "q_{\mathcal{P}}"'] \arrow[d, tail, "r_V"'] & Q_{\mathcal{P}} \arrow[r, tail, "m_{\mathcal{P}}"'] \arrow[dr, tail, "r_Q"'] \arrow[rr, bend left, "p_Q"] & T \arrow[r, "t_1"'] \arrow[d, tail, "t_2"'] & Q_1 \arrow[d, tail, "h_Q"] \\ V_2 \arrow[rr, two heads, "q_2"] & & Q_2 \arrow[r, "k_Q"] & Q_3 \end{tikzcd}$$

Let now $z_1\colon Z \to Q_{\mathcal{P}}$ and $z_2\colon Z \to V_2$ be two arrows such that $q_2 \circ z_2 = r_Q \circ z_1$. Then there exists a unique arrow $z\colon Z \to V_{\mathcal{P}}$ such that

$$m_{\mathcal{P}} \circ q_{\mathcal{P}} \circ z = m_{\mathcal{P}} \circ z_1 \qquad r_V \circ z = z_2$$

Since $m_{\mathcal{P}}$ is mono we conclude that $q_{\mathcal{P}} \circ z = z_1$ as wanted.

Uniqueness follows at once since r_V, being the pullback of h_V is a mono. $\square$

Lemma 4.12. *In* $\mathbf{Hyp}_{\mathsf{eq}}$, Pb*-pushouts are stable.*

Proof. Let $\mathcal{G}_i = (A_i, B_i, Q_i, s_i, t_i, q_i)$, $\mathcal{G}'_i = (A'_i, B'_i, Q'_i, s'_i, t'_i, q'_i)$, for $i \in \{1, 2, 3, 4\}$, be hypergraphs with equivalence, and suppose that in the first diagram below, all the vertical faces are pullbacks, the bottom face is a pushout h is a regular mono and $k_Q\colon Q_1 \rightarrowtail Q_3$ is mono. By Lemma 4.8 the same is true for the other two cubes, by Corollary 4.9, h_E and h_V are monos and so the top faces of these cubes are pushouts

$$\begin{tikzcd}[column sep=small, row sep=small]
& \mathcal{G}'_1 \arrow[rr, "h'"] \arrow[dl, "k'"'] \arrow[dd, "a" near start] & & \mathcal{G}'_2 \arrow[dl, "z'"'] \arrow[dd, "b"] \\
\mathcal{G}'_3 \arrow[rr, "p'" near start] \arrow[dd, "c"'] & & \mathcal{G}'_4 \arrow[dd, "d" near start] & \\
& \mathcal{G}_1 \arrow[rr, "h" near start] \arrow[dl, "k"'] & & \mathcal{G}_2 \arrow[dl, "z"] \\
\mathcal{G}_3 \arrow[rr, "p"'] & & \mathcal{G}_4 &
\end{tikzcd}
\quad
\begin{tikzcd}[column sep=small, row sep=small]
& A'_1 \arrow[rr, tail, "h'_E"] \arrow[dl, "k'_E"'] \arrow[dd, "a_E" near start] & & A'_2 \arrow[dl, "z'_E"'] \arrow[dd, "b_E"] \\
A'_3 \arrow[rr, tail, "p'_E" near start] \arrow[dd, "c_E"'] & & A'_4 \arrow[dd, "d_E" near start] & \\
& A_1 \arrow[rr, tail, "h_E" near start] \arrow[dl, "k_E"'] & & A_2 \arrow[dl, "z_E"] \\
A_3 \arrow[rr, tail, "p_E"'] & & A_4 &
\end{tikzcd}
\quad
\begin{tikzcd}[column sep=small, row sep=small]
& B'_1 \arrow[rr, tail, "h'_V"] \arrow[dl, "k'_V"'] \arrow[dd, "a_V" near start] & & B'_2 \arrow[dl, "z'_V"'] \arrow[dd, "b_V"] \\
B'_3 \arrow[rr, tail, "p'_V" near start] \arrow[dd, "c_V"'] & & B'_4 \arrow[dd, "d_V" near start] & \\
& B_1 \arrow[rr, tail, "h_V" near start] \arrow[dl, "k_V"'] & & B_2 \arrow[dl, "z_V"] \\
B_3 \arrow[rr, tail, "p_V"'] & & B_4 &
\end{tikzcd}$$

Now, if $d = (d_E, d_V, d_Q)$, we can pull back the third component to get the solid part of the cube aside. Notice moreover that the commutativity of the solid diagram yields the existence of the dotted $w\colon T \to Y$. By the usual composition and decomposition properties of pullbacks (cf. Lemma A.1) the left face is a pullback. By the first point of Lemma 4.8 the bottom face is a pushout and h_Q is a mono by Corollary 4.9, so the top face is a pushout too.

$$\begin{tikzcd}[column sep=small, row sep=small]
& T \arrow[rr, tail, "x_1"] \arrow[dl, dotted, "w"'] \arrow[dd, "x_2" near start] & & U \arrow[dl, "u_1"'] \arrow[dd, "u_2"] \\
Y \arrow[rr, tail, "y_1" near start] \arrow[dd, "y_2"'] & & Q'_4 \arrow[dd, "d_Q" near start] & \\
& Q_1 \arrow[rr, tail, "h_Q" near start] \arrow[dl, tail, "k_Q"'] & & Q_2 \arrow[dl, "z_Q"] \\
Q_3 \arrow[rr, tail, "p_C"'] & & Q_4 &
\end{tikzcd}$$

By the second point of Lemma 4.8 we know that there are monos $m_2\colon Q'_2 \rightarrowtail U$ and $m_3\colon Q'_3 \rightarrowtail Y$ fitting in the first two diagrams below:

$$\begin{tikzcd}
B'_3 \arrow[r, tail, "p'_V"] \arrow[d, two heads, "q'_3"] \arrow[dd, bend right, "c_V"'] & B'_4 \arrow[dr, bend left, "q'_4"] & \\
Q'_3 \arrow[r, tail, dotted, "m_3"] & Y \arrow[r, tail, "y_1"] \arrow[d, "y_2"] & Q'_4 \arrow[d, "d_C"] \\
B_3 \arrow[r, two heads, "q_3"'] & Q_3 \arrow[r, tail, "p_Q"'] & Q_4
\end{tikzcd}
\quad
\begin{tikzcd}
B'_2 \arrow[r, "z_V"] \arrow[d, two heads, "q'_2"] \arrow[dd, bend right, "b_V"'] & B'_4 \arrow[dr, bend left, "q'_4"] & \\
Q'_2 \arrow[r, tail, dotted, "m_2"] & U \arrow[r, "u_1"] \arrow[d, "u_2"] & Q'_4 \arrow[d, "d_C"] \\
B_2 \arrow[r, two heads, "q_3"'] & Q_2 \arrow[r, "z_Q"'] & Q_4
\end{tikzcd}
\quad
\begin{tikzcd}
B'_1 \arrow[r, two heads, "q'_1"'] \arrow[rr, bend left, "a_V"] \arrow[d, tail, "h'_V"'] & Q'_1 \arrow[d, tail, dotted, "m_1"] & B_1 \arrow[dr, two heads, "q_1"] & \\
B'_2 \arrow[dr, two heads, "q'_2"'] & S \arrow[r, tail, "s_2"] \arrow[d, "s_1"'] & T \arrow[r, "x_2"] \arrow[d, tail, "x_1"] & Q_1 \arrow[d, tail, "h_Q"] \\
& Q'_2 \arrow[r, tail, "m_2"] & U \arrow[r, "u_2"] & Q_2
\end{tikzcd}$$

For Q'_1, we can make a similar argument. Taking (S, s_1, s_2) as the pullback of m_2 along x_1, we can use again the composition properties of pullbacks and the second point of Lemma 4.8 to guarantee the existence of a monomorphism $m_1\colon Q'_1 \to S$ that makes the last diagram above commutative.

We have to show that the top face of the cube with which we have begun is a pushout. Let $\mathcal{H}$ be (E, V, Q, s, t, q) and suppose that there exist $o\colon \mathcal{G}'_2 \to \mathcal{H}$ and $w\colon \mathcal{G}'_3 \to \mathcal{H}$ such that $o \circ h' = w \circ k'$. Since T preserves colimits by Proposition 4.4, we know that there exists a morphism $(v_E, v_V)\colon T(\mathcal{G}'_4) \to T(\mathcal{H})$ such that

$$v_E \circ p'_E = w_E \quad v_V \circ p'_V = w_V \quad v_E \circ z'_E = o_e \quad v_V \circ z'_V = o_V$$

We want to extend such a morphism to one of $\mathbf{Hyp}_{\mathsf{eq}}$ between $\mathcal{G}'_4 \to \mathcal{H}$. If we are able to do so we can conclude because uniqueness is guaranteed by Remark 4.3.

Now, consider the cube aside, h is in Pb so by Lemma A.1 we know that its back face is a pullback. We can then apply the second point of Lemma 2.14 to deduce that the square on the right is a pushout.

$$\begin{tikzcd}[column sep=small, row sep=small]
& B'_1 \arrow[rr, tail, "h'_V"] \arrow[dl, "k'_V"'] \arrow[dd, two heads, "s_2 \circ m_1 \circ q'_1"' near start] & & B'_2 \arrow[dl, "z'_V"'] \arrow[dd, "m_2 \circ q'_2"] \\
B'_3 \arrow[rr, tail, "p'_V" near start] \arrow[dd, "m_3 \circ q'_3"'] & & B'_4 \arrow[dd, two heads, "q'_4" near start] & \\
& T \arrow[rr, tail, "x_1" near start] \arrow[dl, "w"'] & & U \arrow[dl, "u_1"] \\
Y \arrow[rr, tail, "y_1"'] & & Q'_4 &
\end{tikzcd}$$

By construction we have that:

$$m_3 \circ q'_3 \circ \pi^1_{m_3 \circ q'_3} = m_3 \circ q'_3 \circ \pi^2_{m_3 \circ q'_3} \qquad m_2 \circ q'_2 \circ \pi^1_{m_2 \circ q'_2} = m_2 \circ q'_2 \circ \pi^2_{m_2 \circ q'_2}$$

$$\begin{tikzcd}
K_{s_2 \circ m_1 \circ q'_1} \arrow[r, tail, "k_{h'_V}"'] \arrow[d, "k_{k'_V}"] & K_{m_2 \circ q'_2} \arrow[d, "k_{z'_V}"'] \\
K_{m_3 \circ q'_3} \arrow[r, tail, "k_{p'_V}"] & K_{q'_4}
\end{tikzcd}$$

but since m_3 and m_2 are monos this implies

$$q'_3 \circ \pi^1_{m_3 \circ q'_3} = q'_3 \circ \pi^2_{m_3 \circ q'_3} \qquad q'_2 \circ \pi^1_{m_2 \circ q'_2} = q'_2 \circ \pi^2_{m_2 \circ q'_2}$$

By computing, we obtain

$$
\begin{aligned}
q \circ v_V \circ \pi^1_{q'_4} \circ k_{p'_V} &= q \circ v_V \circ p'_V \circ \pi^1_{m_3 \circ q'_3} = q \circ w_V \circ \pi^1_{m_3 \circ q'_3} = w_Q \circ q'_3 \circ \pi^1_{m_3 \circ q'_3} \\
= w_Q \circ q'_3 \circ \pi^2_{m_3 \circ q_3} &= q \circ w_V \circ \pi^2_{m_3 \circ q'_3} = q \circ v_V \circ p'_V \circ \pi^2_{m_3 \circ q'_3} = q \circ v_V \circ \pi^2_{q'_4} \circ k_{p'_V} \\
q \circ v_V \circ \pi^1_{q'_4} \circ k_{z'_V} &= q \circ v_V \circ z'_V \circ \pi^1_{m_2 \circ q'_2} = q \circ o_V \circ \pi^1_{m_2 \circ q'_2} = o_Q \circ q'_2 \circ \pi^1_{m_2 \circ q'_2} \\
= o_Q \circ q'_2 \circ \pi^2_{m_2 \circ q_2} &= q \circ o_V \circ \pi^2_{m_2 \circ q'_2} = q \circ v_V \circ t'_V \circ \pi^2_{m_2 \circ q'_2} = q \circ v_V \circ \pi^2_{q'_4} \circ k_{z'_V}
\end{aligned}
$$

As already noticed the square above is a pushout, hence we can conclude that

$$q \circ v_V \circ \pi^1_{q'_4} = q \circ v_V \circ \pi^2_{q'_4}$$

Now, q'_4 is a regular epi and so it is the coequalizer of its kernel pair, hence there exists $v_Q \colon Q'_4 \to Q$ such that $v_Q \circ q'_4 = q \circ v_V$ and we can conclude. $\square$

Lemma 4.14. *In* $\mathbf{Hyp}_{\mathsf{eq}}$*, pushouts along arrows in* Pb *are* $\mathsf{Reg}(\mathbf{Hyp}_{\mathsf{eq}})$*-Van Kampen.*

Proof. Consider a cube as the one below on the left, with regular monos as vertical arrows, pullbacks as back faces, pushouts as bottom and top faces and such that h is in Pb. Given Lemma 4.12, if we show that the front faces are pullbacks too we can conclude.

To fix notation, let $\mathcal{G}_i = (A_i, B_i, Q_i, s_i, t_i, q_i)$, $\mathcal{G}' = (A'_i, B'_i, Q'_i, s'_i, t'_i, q'_i)$, for $i = 1, 2, 3, 4$. By Lemma 4.8 and Corollary 4.9 we know that the central and right cube below have pushouts as bottom faces and pullbacks as back faces, thus their front faces are pullbacks

$$
\begin{tikzcd}[row sep=small, column sep=small]
& \mathcal{G}'_1 \arrow[rr, "h'"] \arrow[dl, "k'"'] \arrow[dd, "a"', near end] & & \mathcal{G}'_2 \arrow[dl, "z'"'] \arrow[dd, "b"] \\
\mathcal{G}'_3 \arrow[rr, "p'", near start, crossing over] \arrow[dd, "c"'] & & \mathcal{G}'_4 \\
& \mathcal{G}_1 \arrow[rr, "h", near end] \arrow[dl, "k"] & & \mathcal{G}_2 \arrow[dl, "z"] \\
\mathcal{G}_3 \arrow[rr, "p"'] & & \mathcal{G}_4 \arrow[from=uu, "d", near start, crossing over]
\end{tikzcd}
\qquad
\begin{tikzcd}[row sep=small, column sep=small]
& A'_1 \arrow[rr, tail, "h'_E"] \arrow[dl, "k'_E"'] \arrow[dd, "a_E"', near end] & & A'_2 \arrow[dl, "z'_E"'] \arrow[dd, "b_E"] \\
A'_3 \arrow[rr, tail, "p'_E", near start, crossing over] \arrow[dd, tail, "c_E"'] & & A'_4 \\
& A_1 \arrow[rr, tail, "h_E", near end] \arrow[dl, "k_E"] & & A_2 \arrow[dl, "z_E"] \\
A_3 \arrow[rr, tail, "p_E"'] & & A_4 \arrow[from=uu, "d_E", near start, crossing over]
\end{tikzcd}
\qquad
\begin{tikzcd}[row sep=small, column sep=small]
& B'_1 \arrow[rr, tail, "h'_V"] \arrow[dl, "k'_V"'] \arrow[dd, "a_V"', near end] & & B'_2 \arrow[dl, "z'_V"'] \arrow[dd, "b_V"] \\
B'_3 \arrow[rr, tail, "p'_V", near start, crossing over] \arrow[dd, tail, "c_V"'] & & B'_4 \\
& B_1 \arrow[rr, tail, "h_V", near end] \arrow[dl, "k_V"] & & B_2 \arrow[dl, "z_V"] \\
B_3 \arrow[rr, tail, "p_V"'] & & B_4 \arrow[from=uu, "d_V", near start, crossing over]
\end{tikzcd}
$$

Let us now consider the diagrams below, in which the inner squares are pullbacks. Since the outer diagrams commute, by definition of morphism of $\mathbf{Hyp}_{\mathsf{eq}}$, then we have the existence of $m_2 \colon Q'_2 \to U$, $m_3 \colon Q'_3 \to Y$, $a_3 \colon B'_3 \to Y$ and $a_2 \colon B'_2 \to Y$.

Now, since c_Q and b_2 are injective functions, then m_3 and m_2 are monomorphisms. By the proof of Lemma 4.8, to conclude it is enough to show that

$$m_3 \circ q'_3 = a_3 \qquad m_2 \circ q'_2 = a_2$$

Now, from the previous two equations we can deduce that Q'_3 and Q'_2 are images for a_3 and a_2 and therefore the claim follows from Lemma 4.8.

$$
\begin{tikzcd}
Q'_3 \arrow[r, dotted, tail, "m_3"'] \arrow[rr, tail, bend left, "p'_Q"] \arrow[dr, bend right, "c_Q"'] & Y \arrow[r, tail, "y_1"'] \arrow[d, tail, "y_2"'] & Q'_4 \arrow[d, tail, "d_Q"] \\
& Q_3 \arrow[r, tail, "p_3"] & Q_4
\end{tikzcd}
\qquad
\begin{tikzcd}
Q'_2 \arrow[r, dotted, "m_2"'] \arrow[rr, bend left, "z'_Q"] \arrow[dr, bend right, "b_Q"'] & U \arrow[r, "u_1"'] \arrow[d, tail, "u_2"'] & Q'_4 \arrow[d, tail, "d_Q"] \\
& Q_2 \arrow[r, "z_Q"] & Q_4
\end{tikzcd}
$$

$$
\begin{tikzcd}
B'_3 \arrow[r, tail, "p'_V"'] \arrow[d, two heads, "q'_3"'] \arrow[dr, dotted, "a_3"] & B'_4 \arrow[dr, "q'_4"] & \\
Q'_3 \arrow[dr, tail, "c_Q"'] & Y \arrow[r, tail, "y_1"] \arrow[d, tail, "y_2"'] & Q'_4 \arrow[d, tail, "d_Q"] \\
& Q_3 \arrow[r, tail, "p_Q"] & Q_4
\end{tikzcd}
\qquad
\begin{tikzcd}
B'_2 \arrow[r, "z'_2"'] \arrow[d, two heads, "q'_2"'] \arrow[dr, dotted, "a_2"] & B'_4 \arrow[dr, "q'_4"] & \\
Q'_2 \arrow[dr, tail, "c_Q"'] & U \arrow[r, "u_1"] \arrow[d, tail, "u_2"'] & Q'_4 \arrow[d, tail, "d_Q"] \\
& Q_2 \arrow[r, "z_Q"] & Q_4
\end{tikzcd}
$$

Computing we have:

$$
\begin{aligned}
y_1 \circ a_3 &= q'_4 \circ p'_2 = p'_3 \circ q'_3 = y_1 \circ m_3 \circ q'_3 & \qquad y_2 \circ a_3 &= d_3 \circ q'_3 = y_2 \circ m_3 \circ q'_3 \\
u_1 \circ a_2 &= q'_4 \circ t'_2 = t'_3 \circ q'_3 = u_1 \circ m_2 \circ q'_2 & \qquad u_2 \circ a_2 &= d_2 \circ q'_2 = u_2 \circ m_2 \circ q'_2
\end{aligned}
$$

And we are done. $\square$

Lemma 4.18. *The following facts hold*

1. $\mathbf{LMH}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under equalizers, binary products and pullbacks.*
2. $\mathbf{Sep}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under connected limits;*
3. $\mathbf{pSep}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under pullbacks.*

Proof. 1. Let $F\colon \mathbf{D} \to \mathbf{LMH}_{\mathsf{eq}}$ be a functor with binary product, pullback or equalizer shape. Let $(E_{\mathcal{L}}, V_{\mathcal{L}}, Q_{\mathcal{L}}, s_{\mathcal{L}}, t_{\mathcal{L}}, q_{\mathcal{L}})$ be a vertex for a limiting cone for $M_{\mathsf{eq}} \circ F$. By Lemma 4.8 we know that $\mathcal{L} := (E_{\mathcal{L}}, V_{\mathcal{L}}, s_{\mathcal{L}}, t_{\mathcal{L}})$ is the vertex of a limiting cone for $T \circ M_{\mathsf{eq}} \circ F$ and so by Corollaries 3.27 and 3.29 and Lemma 3.28 it is left-monogamous.
2. Let $\mathbf{D}$ be a connected category and $F\colon \mathbf{D} \to \mathbf{Sep}_{\mathsf{eq}}$ a functor. If $(E_{\mathcal{L}}, V_{\mathcal{L}}, Q_{\mathcal{L}}, s_{\mathcal{L}}, t_{\mathcal{L}}, q_{\mathcal{L}})$ is a vertex for a limiting cone for $S_{\mathsf{eq}} \circ F$ then, by Lemmas 3.51 and 4.8, $(E_{\mathcal{L}}, V_{\mathcal{L}}, s_{\mathcal{L}}, t_{\mathcal{L}})$ is separated, proving the claim.
3. This is done as int he previous two points using Proposition 3.56. $\square$

Lemma 4.19. *The following facts hold*

1. $\mathbf{LMH}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under pushouts along morphisms whose image through* T *is downward-closed.*
2. $\mathbf{pSep}_{\mathsf{eq}}$ *is closed in* $\mathbf{Hyp}_{\mathsf{eq}}$ *under colimits.*

Proof. 1. Consider two arrows $(f,g)\colon \mathcal{G} \to \mathcal{H}$ and $(h,k)\colon \mathcal{G} \to \mathcal{K}$ in $\mathbf{Hyp}_{\mathsf{eq}}$ with the same domain. Let $\mathcal{P}$ be the vertex of a pushout diagram for them. By Lemma 4.8 we know that $T(\mathcal{P})$ is a pushout for $T(f,g)$ along $T(h,k)$. Thus if $T(\mathcal{G})$ and $T(\mathcal{H})$ are left-monogamous and $T(f,g)$ is downward-closed, by Corollary 3.33 we deduce that $T(\mathcal{P})$ is left-monogamous and so $\mathcal{P}$ is in $\mathbf{LMH}_{\mathsf{eq}}$.
2. Let $F\colon \mathbf{D} \to \mathbf{pSep}_{\mathsf{eq}}$ be a functor. Let $\mathcal{C}$ be the vertex of a colimiting cocone for $S_{p,\mathsf{eq}} \circ F$. by the first point of Lemma 4.8 we know that $T(\mathcal{C})$ is the vertex of a colimiting cocone for $T \circ S_{p,\mathsf{eq}} \circ F$. By Proposition 3.56, thus, $T(\mathcal{C})$ is pruned and separated and we can conclude. $\square$

A.4. Proofs for Section 5

Proposition 5.2. *Let* $\mathcal{G} = (E, V, Q, s, t, q)$ *be a hypergraph with equivalence and* (A, π_1^A, π_2^A), (B, π_1^B, π_2^B) *kernel pairs for, respectively,* $q^\star \circ s$ *and* $q^\star \circ t$. *Then the following facts are equivalent*

1. $\mathcal{G}$ *is an e-hypergraph;*
2. $q^\star \circ t \circ \pi_1^A = q^\star \circ t \circ \pi_2^A$*;*
3. *there exists a unique arrow* $k\colon A \to B$ *such that*

$$\pi_1^B \circ k = \pi_1^A \qquad \pi_2^B \circ k = \pi_2^A$$

Proof. Define

$$A_E := \{(e, e') \in E \mid q^\star(s(e)) = q^\star(s(e'))\}$$

If $p_1, p_2 \colon A_E \rightrightarrows E$ are the two projections, we know that (A_E, p_1, p_1) is a kernel pair for $q^\star \circ s$. Thus there exists an isomorphism $\phi\colon A_E \to A$ such that

$$\pi_1^A \circ \phi = p_1 \qquad \pi_2^A \circ \phi = p_2$$

$(1 \Rightarrow 2)$ By hypothesis $\mathcal{G}$ is an e-hypergraph, then the components of $\phi^{-1}(a)$ are q-joined for every $a \in A$ and so:

$$q^\star(t(\pi_1^A(a))) = q^\star(t(p_1(\phi^{-1}(a)))) = q^\star(t(p_2(\phi^{-1}(a)))) = q^\star(t(\pi_2^A(a)))$$

$(2 \Rightarrow 3)$ This follows at once from the definition of kernel pairs.

$(3 \Rightarrow 1)$ Let e, e' be hyperedges such that $q^\star(s(e)) = q^\star(s(e'))$. Then (e, e') belongs to A_E and thus we have:

$$\begin{aligned} q^\star(t(e)) &= q^\star(t(p_1(e,e'))) = q^\star(t(\pi_1^A(\phi(e,e')))) = q^\star(t(\pi_1^B(k(\phi(e,e'))))) \\ =&q^\star(t(\pi_2^B(k(\phi(e,e'))))) = q^\star(t(\pi_2^A(\phi(e,e')))) = q^\star(t(p_2(e,e'))) = q^\star(t(e')) \end{aligned}$$

and we can conclude. $\square$

Lemma 5.4. *Let $F\colon \mathbf{D} \to \mathbf{Hyp}_{\mathsf{eq}}$ be a connected diagram and let $F(D)$ be $(E_D, V_D, Q_D, s_D, t_D, q_D)$. Suppose that $(\mathcal{L}, \{(a_D, b_D, c_D)\}_{D\in\mathbf{D}})$ is a limiting cone for F, then for every hyperedges $e, e' \in E_{\mathcal{L}}$, if $a_D(e)$ and $a_D(e')$ are q_D-joined for every $D \in \mathbf{D}$, then e and e' are $q_{\mathcal{L}}$-joined.*

Proof. Let $\mathcal{L}$ be $(E_{\mathcal{L}}, V_{\mathcal{L}}, Q_{\mathcal{L}}, s_{\mathcal{L}}, t_{\mathcal{L}}, q_{\mathcal{L}})$ and consider $e, e' \in E_{\mathcal{L}}$. Take a limit $(L, (l_i)_{i\in\mathbf{I}})$ for $K \circ F$. By Lemma 4.8 there exists a mono $m\colon Q_{\mathcal{L}} \rightarrowtail L$ such that $c_D = l_D \circ m$. Since $\mathbf{D}$ is connected then by Proposition 3.1 we know that $(L^\star, \{l^\star_D\}_{D\in\mathbf{D}})$ is a limiting cone. Computing we get

$$\begin{aligned} &l^\star_D(m^\star(q^\star_{\mathcal{L}}(t_{\mathcal{L}}(e)))) = c^\star_D(q^\star_{\mathcal{L}}(t_{\mathcal{L}}(e))) = q^\star_D(b^\star_D(t_{\mathcal{L}}(e))) = q^\star_D(t_D(a_D(e))) \\ =&q^\star_D(t_D(a_D(e'))) = q^\star_D(b^\star_D(t_{\mathcal{L}}(e'))) = c^\star_D(q^\star_{\mathcal{L}}(t_{\mathcal{L}}(e'))) = l^\star_D(m^\star(q^\star_{\mathcal{L}}(t_{\mathcal{L}}(e')))) \end{aligned}$$

Thus we deduce that $m^\star(q^\star_{\mathcal{L}}(t_{\mathcal{L}}(e))) = m^\star(q^\star_{\mathcal{L}}(t_{\mathcal{L}}(e')))$. But by Proposition 3.1, $(-)^\star$ sends monos to monos and so we can conclude. ☐

Corollary 5.5. e-$\mathbf{Hyp}_{\mathsf{eq}}$ *has all connected limits and E creates them.*

Proof. Let $F\colon \mathbf{D} \to \mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq}}$ be a connected diagram. Let $F(D)$ be $(E_D, V_D, Q_D, s_D, t_D, q_D)$ and suppose that $(\mathcal{L}, \{(a_D, b_D, c_D)\}_{D\in\mathbf{D}})$ is a limiting cone for $E \circ F$. If we show that $\mathcal{L}$ is an e-hypergraph we are done. Let $e, e' \in E_{\mathcal{L}}$ be such that $q^\star_{\mathcal{L}}(s_{\mathcal{L}}(e)) = q^\star_{\mathcal{L}}(s_{\mathcal{L}}(e'))$. Thus for every $D \in \mathbf{D}$ we have:

$$\begin{aligned} &q^\star_D(s_D(a_D(e))) = q^\star(b^\star_D(s_{\mathcal{L}}(e))) = c^\star_D(q_{\mathcal{L}}(s_{\mathcal{L}}(e))) \\ =&c^\star_D(q_{\mathcal{L}}(s_{\mathcal{L}}(e'))) = c^\star_D(q_{\mathcal{L}}(s_{\mathcal{L}}(e'))) = q^\star(b^\star_D(s_{\mathcal{L}}(e'))) = q^\star_D(s_D(a_D(e'))) \end{aligned}$$

and so we conclude that $q^\star_D(t_D(a_D(e))) = q^\star_D(t_D(a_D(e')))$. Thus $a_D(e)$ and $a_D(e')$ are q_D-joined and we can conclude by Lemma 5.4. ☐

Lemma 5.8. *The class* Pb^+ *contains all isomorphisms, it is closed under composition, decomposition and it is stable under pullbacks and pushouts.*

Proof. As in the case of Lemma 4.11, closure under composition and decomposition follows from Lemma A.1 and stability under pushouts follows from the first point of Lemma 2.14.

Let now $h\colon \mathcal{G}_1 \to \mathcal{G}_3$ be an arrow in Pb^+ and $k\colon \mathcal{G}_2 \to \mathcal{G}_3$ be another morphism of $\mathbf{Hyp}_{\mathsf{eq}}$, with $\mathcal{G}_i = (E_i, V_i, Q_i, s_i, t_i, q_i)$.

Let $r\colon \mathcal{P} \to \mathcal{G}_2$ and $p\colon \mathcal{P} \to \mathcal{G}_1$ be the projection of a pullback of h along k. We have to show that r is in Pb^+.

By Lemma 4.11 we already know that it is in Pb. Now, let $z_1\colon Z \to E_2$, $z_2\colon Z \to V_{\mathcal{P}}$ be two arrows such that

$$s_2 \circ z_1 = r_V \circ z_2$$

Now, by Corollary 4.6 and since h belongs to Pb^+, all the square below beside the left half of the right-most diagram are pullbacks.

$$\begin{tikzcd} E_{\mathcal{P}} \arrow[r, "p_E"'] \arrow[rr, bend left, "p_V \circ s_{\mathcal{P}}"] \arrow[d, "r_E"'] & E_1 \arrow[r, "s_1"'] \arrow[d, "h_E"'] & V_1 \arrow[d, "h_V"] \\ E_2 \arrow[r, "k_E"] \arrow[rr, bend right, "k_V \circ s_2"'] & E_3 \arrow[r, "s_3"] & V_3 \end{tikzcd} \qquad \begin{tikzcd} E_{\mathcal{P}} \arrow[r, "s_{\mathcal{P}}"'] \arrow[rr, bend left, "s_1 \circ p_E"] \arrow[d, "r_E"'] & V_{\mathcal{P}} \arrow[r, "p_V"'] \arrow[d, "r_V"'] & V_1 \arrow[d, "h_V"'] \\ E_2 \arrow[r, "s_2"] \arrow[rr, bend right, "s_3 \circ k_E"'] & V_2 \arrow[r, "k_V"] & V_3 \end{tikzcd}$$

But by Lemma A.1 this last square is a pullback too, proving the thesis. ☐

Remark A.8. Let $\delta_1\colon 1 \to \mathbb{N}$ be the arrow which picks $1 \in \mathbb{N}$. Let $f\colon X \to Y$ be an arrow and consider the inclusions $v_1\colon X \to X^\star$, $u_1\colon Y \to Y^\star$ defined by the first point of Proposition 3.1. Then we can build the cube aside which, by extensivity, has pullbacks as back, front and right faces. Thus, by Lemma A.1 its left face is a pullback too.

$$\begin{tikzcd} & X \arrow[rr, "!_X"] \arrow[dl, "f"'] \arrow[dd, "v_1"' near end] & & 1 \arrow[dl, "\mathsf{id}_1"] \arrow[dd, "v_1"] \\ Y \arrow[rr, "!_Y" near start] \arrow[dd, "u_1"'] & & 1 \arrow[dd, "\delta_1" near start] & \\ & X^\star \arrow[rr, tail, "\mathsf{lg}_X" near end] \arrow[dl, tail, "f^\star"'] & & \mathbb{N} \arrow[dl, "\mathsf{id}_N"] \\ Y^\star \arrow[rr, tail, "\mathsf{lg}_Y"] & & \mathbb{N} & \end{tikzcd}$$

Lemma 5.9. *Suppose that the square on the right is a pushout in $\mathbf{Hyp}_{\mathsf{eq}}$ and that m is a mono in Pb^+. If $\mathcal{G}_1$, $\mathcal{G}_2$ and $\mathcal{G}_3$ are e-hypergraphs, then $\mathcal{P}$ is is an e-hypergraph.*

$$\begin{tikzcd} \mathcal{G}_1 \arrow[r, "h"] \arrow[d, tail, "m"'] & \mathcal{G}_2 \arrow[d, "n"] \\ \mathcal{G}_3 \arrow[r, "z"] & \mathcal{P} \end{tikzcd}$$

Proof. To fix the notation, let $\mathcal{P}$ be (E, V, Q, s, t, q) and, for $i \in \{1, 2, 3\}$ let $\mathcal{G}_i$ be $(E_i, V_i, Q_i, s_i, t_i, q_i)$. Let, moreover, (K_i, π_i^1, π_i^2) be a kernel pair of $q_i^\star \circ s_i$, again for $i \in \{1, 2, 3\}$. Finally, let (U, u_1, u_2) be the kernel pair of $q^\star \circ s$.

Consider now the cube on the right. Since m is in Pb^+ then the squares below are both pullbacks.

$$\begin{tikzcd} & E_1 \arrow[rr, "h_E"] \arrow[dl, tail, "m_E"'] \arrow[dd, "q_1^\star \circ s_1" near start] & & E_2 \arrow[dl, tail, "n_E"'] \arrow[dd, "q_2^\star \circ s_2"] \\ E_3 \arrow[rr, "z_E" near end, crossing over] \arrow[dd, "q_3^\star \circ s_3"'] & & E & \\ & Q_1^\star \arrow[rr, "h_Q^\star" near end] \arrow[dl, tail, "m_Q^\star"'] & & Q_2^\star \arrow[dl, tail, "n_Q^\star"] \\ Q_3^\star \arrow[rr, "z_Q^\star"] & & Q^\star \arrow[from=uu, "q^\star \circ s"' near start, crossing over] & \end{tikzcd}$$

$$\begin{tikzcd} E_1 \arrow[r, tail, "m_E"] \arrow[d, "s_1"'] & E_3 \arrow[d, "s_3"] \\ V_1 \arrow[r, tail, "m_V"] & V_3 \end{tikzcd} \qquad \begin{tikzcd} V_1 \arrow[r, tail, "m_V"] \arrow[d, two heads, "q_1"'] & V_3 \arrow[d, two heads, "q_3"] \\ Q_1 \arrow[r, tail, "m_Q"] & Q_3 \end{tikzcd}$$

$$\begin{tikzcd} K_1 \arrow[r, "k_{h_E}"] \arrow[d, tail, "k_{m_E}"'] & K_2 \arrow[d, tail, "k_{n_E}"] \\ K_3 \arrow[r, "k_{z_E}"] & U \end{tikzcd}$$

By Remark A.8, then, left face of the cube is a pullback too. Its bottom face is not a pushout but it is still a pullback by the first point of Proposition 2.6 and the third one of Proposition 3.1. But by hypothesis and the first point of Lemma 4.8 the top face is a pushout. Thus we can apply Lemma 2.14 to deduce that the square on the left is a pushout.

By computing we obtain

$$\begin{aligned} q^\star \circ t \circ u_1 \circ k_{n_E} &= q^\star \circ t \circ n_E \circ \pi_2^1 = q^\star \circ n_V \circ t_2 \circ \pi_2^1 = n_Q^\star \circ q_2^\star \circ t_2 \circ \pi_2^1 \\ &= n_Q^\star \circ q_2^\star \circ t_2 \circ \pi_2^2 = q^\star \circ n_V \circ t_2 \circ \pi_2^2 = q^\star \circ t \circ n_E \circ \pi_2^2 = q^\star \circ t \circ u_1 \circ k_{n_E} \\ q^\star \circ t \circ u_1 \circ k_{z_E} &= q^\star \circ t \circ z_E \circ \pi_3^1 = q^\star \circ n_V \circ t_3 \circ \pi_3^1 = z_Q^\star \circ q_3^\star \circ t_3 \circ \pi_3^1 \\ &= z_Q^\star \circ q_3^\star \circ t_3 \circ \pi_3^2 = q^\star \circ z_V \circ t_3 \circ \pi_2^3 = q^\star \circ t \circ z_E \circ \pi_3^2 = q^\star \circ t \circ u_1 \circ k_{z_E} \end{aligned}$$

We can therefore deduce that $q^\star \circ t \circ u_1 = q^\star \circ t \circ u_2$, and the claim now follows from Proposition 5.2. □

Lemma 5.16. *Let $(a, b, c)\colon \mathcal{H} \to \mathcal{G}$ be an object of $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ and (A, π_1^A, π_2^A), (B, π_1^B, π_2^B) kernel pairs for, respectively, $(q^\star \circ s_\mathcal{H}, a), (q^\star \circ t_\mathcal{H}, a)\colon E_\mathcal{H} \to Q_\mathcal{H} \times E_\mathcal{G}$. Then the following are equivalent*

1. *(a, b, c) is a labelled e-hypergraph;*
2. *$q^\star \circ t_\mathcal{H} \circ \pi_1^A = q^\star \circ t_\mathcal{H} \circ \pi_2^A$;*
3. *there exists a unique arrow $k\colon A \to B$ such that*

$$\pi_1^B \circ k = \pi_1^A \qquad \pi_2^B \circ k = \pi_2^A$$

Proof. We proceed more or less as in the proof of Proposition 5.2. Define

$$A_E := \{(e, e') \in E \mid q^\star(s(e)) = q^\star(s(e')) \text{ and } a(e) = a(e')\}$$

If $p_1, p_2\colon A_E \rightrightarrows E$ are the two projections, we know that (A_E, p_1, p_1) is a kernel pair for $(q^\star \circ s, a)$. Thus there exists an isomorphism $\phi\colon A_E \to A$ such that

$$\pi_1^A \circ \phi = p_1 \qquad \pi_2^A \circ \phi = p_2$$

$(1 \Rightarrow 2)$ By hypothesis (a, b, c) is an e-hypergraph, then the components of $\phi^{-1}(x)$ are q-joined for every $x \in A$ and so:

$$q^\star(t_\mathcal{H}(\pi_1^A(x))) = q^\star(t_\mathcal{H}(p_1(\phi^{-1}(x)))) = q^\star(t_\mathcal{H}(p_2(\phi^{-1}(x)))) = q^\star(t_\mathcal{H}(\pi_2^A(x)))$$

$(2 \Rightarrow 3)$ This follows at once from the definition of kernel pairs.

($3 \Rightarrow 1$) Let e, e' be hyperedges such that $q^\star(s(e)) = q^\star(s(e'))$ and $a(e) = a(e0)$. Then (e, e') belongs to A_E and thus we have:

$$
\begin{aligned}
q^\star(t(e)) = q^\star(t(p_1(e,e'))) = q^\star(t(\pi_1^A(\phi(e,e')))) = q^\star(t(\pi_1^B(k(\phi(e,e')))))\\
= q^\star(t(\pi_2^B(k(\phi(e,e'))))) = q^\star(t(\pi_2^A(\phi(e,e')))) = q^\star(t(p_2(e,e'))) = q^\star(t(e'))
\end{aligned}
$$

allowing us to conclude. □

Lemma 5.18. *Let $\mathcal{G}$ be a hypergraph with equivalence. Then $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ is closed in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ under connected limits and pushouts along arrows in $\mathsf{ePb}^+_{\mathcal{G}}$.*

Proof. Let us examine connected limits and pushouts separately.

1. Let $F\colon \mathbf{D} \to \mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ be a connected diagram. Let $F(D)$ be $(f_D, g_D, h_D)\colon \mathcal{H}_D \to \mathcal{G}$, with $\mathcal{H}_D := (E_D, V_D, Q_D, s_D, t_D, q_D)$. Suppose that $((f,g,h), \{(a_D, b_D, c_D)\}_{D\in\mathbf{D}})$ is a limiting cone for $E_\mathcal{G} \circ F$ with $(f,g,h)\colon \mathcal{L} \to \mathcal{G}$. If we show that (f,g,h) is a labelled e-hypergraph we are done. Let $e, e' \in E_\mathcal{L}$ be such that
$$q^\star_\mathcal{L}(s_\mathcal{L}(e)) = q^\star_\mathcal{L}(s_\mathcal{L}(e')) \qquad f(e) = f(e')$$
Thus for every $D \in \mathbf{D}$ we have:
$$
\begin{aligned}
q^\star_D(s_D(a_D(e))) = q^\star(b^\star_D(s_\mathcal{L}(e))) = c^\star_D(q_\mathcal{L}(s_\mathcal{L}(e)))\\
= c^\star_D(q_\mathcal{L}(s_\mathcal{L}(e'))) = c^\star_D(q_\mathcal{L}(s_\mathcal{L}(e'))) = q^\star(b^\star_D(s_\mathcal{L}(e'))) = q^\star_D(s_D(a_D(e')))\\
f_D(a_D(e)) = f(e) = f(e') = f_D(a_D(e'))
\end{aligned}
$$
Thus $a_D(e)$ and $a_D(e')$ are q_D-joined and we can conclude by Lemma 5.4.
2. We are going to adapt the argument exploited in the proof of Lemma 5.9.
Suppose that the square on the right is a pushout in $\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ and that m is a mono in $\mathsf{ePb}^+_{\mathcal{G}}$ and, for $i \in \{1,2,3\}$, $(a_i, b_i, c_i)\colon \mathcal{G}_i \to \mathcal{G}$ is a labelled e-hypergraph. We want to show that $(a,b,c)\colon \mathcal{P} \to \mathcal{G}$ is in $\mathsf{e}\text{-}\mathbf{Hyp}_{\mathsf{eq},\mathcal{G}}$ too.
$$\begin{tikzcd}
(a_1,b_1,c_1) \arrow[r, "h"] \arrow[d, "m"'] & (a_2,b_2,c_2) \arrow[d, "n"] \\
(a_3,b_3,c_3) \arrow[r, "z"] & (a,b,c)
\end{tikzcd}$$
To fix the notation, let $\mathcal{P}$ be (E, V, Q, s, t, q) and, for $i \in \{1,2,3\}$ let $\mathcal{G}_i$ be $(E_i, V_i, Q_i, s_i, t_i, q_i)$. Let, moreover, (K_i, π_i^1, π_i^2) be a kernel pair of $(q_i^\star \circ s_i, a_i)$, again for $i \in \{1,2,3\}$. Finally, let (U, u_1, u_2) be a kernel pair of $(q^\star \circ s, a)$.
Consider the cube on the right. By hypothesis and Remark A.8 its left face is a pullback. Moreover its bottom face is not a pushout but it is still a pullback by the first point of Proposition 2.6 and the third one of Proposition 3.1 and by the fact that, since limits commute with limits, $(-) \times E_\mathcal{G}$ preserves pullbacks.
$$\begin{tikzcd}
& E_1 \arrow[rr, "h_E"] \arrow[dl, "m_E"'] \arrow[dd, "(q_1^\star\circ s_1, a_1)" near start] & & E_2 \arrow[dl, "n_E"'] \arrow[dd, "(q_2^\star\circ s_2, a_2)"] \\
E_3 \arrow[rr, "z_E" near start] \arrow[dd, "(q_3^\star\circ s_3, a_3)"'] & & E \arrow[dd, "(q^\star\circ s, a)"' near start] & \\
& Q_1^\star \times E_\mathcal{G} \arrow[rr, "h_Q^\star\times \mathsf{id}_{E_\mathcal{G}}" near end] \arrow[dl, "m_Q^\star\times \mathsf{id}_{E_\mathcal{G}}"'] & & Q_2^\star \times E_\mathcal{G} \arrow[dl, "n_Q^\star\times \mathsf{id}_{E_\mathcal{G}}"] \\
Q_3^\star \times E_\mathcal{G} \arrow[rr, "z_Q^\star\times \mathsf{id}_{E_\mathcal{G}}"] & & Q^\star \times E_\mathcal{G} &
\end{tikzcd}$$
$$\begin{tikzcd}
K_1 \arrow[r, "k_{h_E}"] \arrow[d, "k_{m_E}"'] & K_2 \arrow[d, "k_{n_E}"] \\
K_3 \arrow[r, "k_{z_E}"] & U
\end{tikzcd}$$
By hypothesis and the first point of Lemma 4.8 the top face is a pushout. Thus we can apply Lemma 2.14 to deduce again that the square on the left is a pushout.
By computing we obtain
$$
\begin{aligned}
q^\star \circ t \circ u_1 \circ k_{n_E} = q^\star \circ t \circ n_E \circ \pi_2^1 = q^\star \circ n_V \circ t_2 \circ \pi_2^1 = n_Q^\star \circ q_2^\star \circ t_2 \circ \pi_2^1\\
= n_Q^\star \circ q_2^\star \circ t_2 \circ \pi_2^2 = q^\star \circ n_V \circ t_2 \circ \pi_2^2 = q^\star \circ t \circ n_E \circ \pi_2^2 = q^\star \circ t \circ u_1 \circ k_{n_E}\\
q^\star \circ t \circ u_1 \circ k_{z_E} = q^\star \circ t \circ z_E \circ \pi_3^1 = q^\star \circ n_V \circ t_3 \circ \pi_3^1 = z_Q^\star \circ q_3^\star \circ t_3 \circ \pi_3^1\\
= z_Q^\star \circ q_3^\star \circ t_3 \circ \pi_3^2 = q^\star \circ z_V \circ t_3 \circ \pi_2^3 = q^\star \circ t \circ z_E \circ \pi_3^2 = q^\star \circ t \circ u_1 \circ k_{z_E}
\end{aligned}
$$
From these identities we get that $q^\star \circ t \circ u_1 = q^\star \circ t \circ u_2$, and the claim follows from Proposition 5.2. □

B. Some properties of comma categories

In this section we briefly recall the definition of the comma category [29] and some of its properties.

Definition B.1. Let $L\colon \mathbf{A} \to \mathbf{X}$ and $R\colon \mathbf{B} \to \mathbf{X}$ be two functors with the same codomain, the *comma category* $L{\downarrow}R$ is the category in which:

- objects are triples (A, B, f) with $A \in \mathbf{A}$, $B \in \mathbf{B}$, and $f\colon L(A) \to R(B)$;
- a morphism $(A, B, f) \to (A', B', g)$ is a pair (h, k) with $h\colon A \to A'$ in $\mathbf{A}$ and $k\colon B \to B'$ in $\mathbf{B}$ such that the diagram aside commutes.

$$\begin{tikzcd} L(A) \arrow[r, "L(h)"'] \arrow[d, "f"'] & L(A') \arrow[d, "g"] \\ R(B) \arrow[r, "R(k)"] & R(B') \end{tikzcd}$$

We have two forgetful functors $U_L\colon L{\downarrow}R \to \mathbf{A}$ and $U_R\colon L{\downarrow}R \to \mathbf{B}$ given respectively by

$$\begin{array}{ccc} (A,B,f) & \longmapsto & A \\ {\scriptstyle (h,k)}\big\downarrow & & \big\downarrow{\scriptstyle h} \\ (A',B',g) & \longmapsto & A' \end{array} \qquad \begin{array}{ccc} (A,B,f) & \longmapsto & B \\ {\scriptstyle (h,k)}\big\downarrow & & \big\downarrow{\scriptstyle k} \\ (A',B',g) & \longmapsto & B' \end{array}$$

Given $L\colon \mathbf{A} \to \mathbf{X}$ and $R\colon \mathbf{B} \to \mathbf{X}$, we can also consider their duals $L^{op}\colon \mathbf{A}^{op} \to \mathbf{X}^{op}$ and $R^{op}\colon \mathbf{B}^{op} \to \mathbf{X}^{op}$. An arrow $f\colon L(A) \to R(B)$ in $\mathbf{X}$ is the same thing as an arrow $f\colon R^{op}(B) \to L^{op}(A)$ in $\mathbf{X}^{op}$. Thus $L{\downarrow}R$ and $R^{op}{\downarrow}L^{op}$ have the same objects. Moreover, the left square below commutes in $\mathbf{X}$ if and only if the right one commutes in $\mathbf{X}^{op}$.

$$\begin{tikzcd} L(A) \arrow[r, "L(h)"] \arrow[d, "f"'] & L(A') \arrow[d, "g"] \\ R(B) \arrow[r, "R(k)"'] & R(B') \end{tikzcd} \qquad \begin{tikzcd} R(B') \arrow[r, "R(k)"] \arrow[d, "g"'] & R(B) \arrow[d, "f"] \\ L(A') \arrow[r, "L(h)"'] & L(A) \end{tikzcd}$$

Summing up we have just proved the following fact.

Proposition B.2. *$(L{\downarrow}R)^{op}$ is equal to $R^{op}{\downarrow}L^{op}$, and $U_L^{op} = U_{L^{op}}$ and $U_R^{op} = U_{R^{op}}$.*

Lemma B.3. *Let $L\colon \mathbf{A} \to \mathbf{X}$ and $R\colon \mathbf{B} \to \mathbf{X}$ be functors and $F\colon \mathbf{D} \to L{\downarrow}R$ a diagram such that L preserves colimits along $U_L \circ F$. Then the family $\{U_L, U_R\}$ jointly creates colimits of F (see [11, Sec. 5.1.3] or [12, Sec. 2.3]).*

Proof. Suppose that $U_L \circ F$ and $U_R \circ F$ have respectively colimiting cocones $(A, \{a_D\}_{D\in\mathbf{D}})$ and $(B, \{b_D\}_{D\in\mathbf{D}})$. By hypothesis $(L(A), \{L(a_D)\}_{D\in\mathbf{D}})$ is colimiting for $L \circ U_L \circ F$. Now, let $F(D)$ be (A_D, B_D, f_D): we have arrows $R(b_D) \circ f_D\colon L(A_D) \to R(B)$ that forms a cocone on $L \circ U_L \circ F$. If $d\colon D \to D'$ is an arrow in $\mathbf{D}$, then $F(d)$ is an arrow in $L{\downarrow}R$ and so

$$\begin{aligned} &R(b_{D'}) \circ f_{D'} \circ L(U_L(F(d))) = R(b_{D'}) \circ R(U_R(F(d))) \circ f_D \\ =&R(b_{D'} \circ U_R(F(d))) \circ f_D = R(b_D) \circ f_D \end{aligned}$$

Thus there exists $f\colon L(A) \to R(B)$ fitting in the diagram on the right. Notice that f is the unique arrow in $\mathbf{X}$ which makes (a_D, b_D) an arrow $(A_D, B_D, f_D) \to (A, B, f)$ of $L{\downarrow}R$. If we show that $\big((A, B, f), \{(a_D, b_D)\}_{D\in\mathbf{D}}\big)$ is colimiting for F we are done.

$$\begin{tikzcd} L(A_D) \arrow[r, "L(a_D)"] \arrow[d, "f_D"'] & L(A) \arrow[d, "f"] \\ R(B_D) \arrow[r, "R(b_D)"'] & R(B) \end{tikzcd}$$

First of all, let us show that it is a cocone. Given $d\colon D \to D'$ in $\mathbf{D}$ we have

$$\begin{aligned} &(a_{D'}, b_{D'}) \circ F(d) = (a_{D'}, b_{D'}) \circ (U_L(F(d)), U_R(F(d))) \\ =& (a_{D'} \circ U_L(F(d)), b_{D'} \circ U_R(F(d))) = (a_D, b_D) \end{aligned}$$

For the colimiting property, let $\big((X, Y, g), \{(x_D, y_D)\}_{D\in\mathbf{D}}\big)$ be another cocone on F. In particular, $\big(X, \{x_D\}_{D\in\mathbf{D}}\big)$ and $\big(Y, \{y_D\}_{D\in\mathbf{D}}\big)$ are cocones on $U_L \circ F$ and $U_R \circ F$ respectively, so we have uniquely determined arrows $x\colon A \to X$ and $y\colon B \to Y$ such that

$$x \circ a_D = x_D \qquad y \circ b_D = y_D$$

Let us show that (x, y) is an arrow of $L{\downarrow}R$. Given $D \in \mathbf{D}$ we have

$$\begin{aligned} R(y) \circ f \circ L(a_D) &= R(y) \circ R(b_D) \circ f_D = R(y \circ b_D) \circ f_D \\ =&R(y_D) \circ f_D = g \circ L(x_D) = g \circ L(x \circ a_D) = g \circ L(x) \circ L(a_D) \end{aligned}$$

from which it follows that $g \circ L(x) = R(y) \circ f$ as wanted. □

Proposition B.2 and Lemma B.3 now yields the following.

Corollary B.4. *The family $\{U_L, U_R\}$ jointly creates limits along every diagram $F\colon \mathbf{D} \to L{\downarrow}R$ such that R preserves the limit of $U_R \circ F$.*

We can use Corollary B.4 to characterize monos in comma categories.

Corollary B.5. *If R preserves pullbacks then an arrow (h, k) in $L{\downarrow}R$ is mono if and only if both h and k are monos.*

Proof. ($\Rightarrow$) If $(h, k)\colon (A, B, f) \to (A', B', g)$ is a mono then the first rectangle below is a pullback in $L{\downarrow}R$. By Corollary B.4 then also the other two squares are pullbacks, respectively in $\mathbf{A}$ and $\mathbf{B}$, proving that both h and k are monos.

$$\begin{tikzcd} (A,B,f) \arrow[r, "\mathsf{id}_{(A,B,f)}"'] \arrow[d, "\mathsf{id}_{(A,B,f)}"'] & (A,B,f) \arrow[d, "(h,k)"] \\ (A,B,f) \arrow[r, "(h,k)"] & (A',B',g) \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, "\mathsf{id}_A"'] \arrow[d, "\mathsf{id}_A"'] & A \arrow[d, "h"] \\ A \arrow[r, "h"] & A' \end{tikzcd} \qquad \begin{tikzcd} B \arrow[r, "\mathsf{id}_B"'] \arrow[d, "\mathsf{id}_B"'] & B \arrow[d, "k"] \\ B \arrow[r, "k"] & B' \end{tikzcd}$$

$$\begin{tikzcd} A \arrow[r, "\mathsf{id}_A"'] \arrow[d, "\mathsf{id}_A"'] & A \arrow[d, "h"] \\ A \arrow[r, "h"] & A' \end{tikzcd} \qquad \begin{tikzcd} B \arrow[r, "\mathsf{id}_B"'] \arrow[d, "\mathsf{id}_B"'] & B \arrow[d, "k"] \\ B \arrow[r, "k"] & B' \end{tikzcd}$$

($\Leftarrow$) Since h and k are monos then we have the two pullback squares on the left. Thus by Corollary B.4 the pullback of (h, k) along itself has isomorphisms as projections and so (h, k) is mono. □

We end this section pointing out another useful fact, showing that in some cases we can guarantee the existence of a left adjoint to U_R.

Proposition B.6. *If $\mathbf{A}$ has initial objects and L preserves them, then the forgetful functor $U_R\colon L{\downarrow}R \to \mathbf{B}$ has a left adjoint Δ.*

Proof. For an object $B \in \mathbf{B}$ we can define $\Delta(B)$ as $(0, B, ?_{R(B)})$, for 0 is an initial object in $\mathbf{A}$ and $?_{R(B)}$ is the unique arrow $L(0) \to R(B)$. Let $\mathsf{id}_B\colon B \to U_R(\Delta(B))$ be the identity, and suppose that a $k\colon B \to U_R(A, B', f)$ in $\mathbf{B}$ is given.

$$\begin{tikzcd} L(0) \arrow[r, "L(?_A)"'] \arrow[d, "?_{R(B)}"'] & L(A) \arrow[d, "f"] \\ R(B) \arrow[r, "R(k)"] & R(B') \end{tikzcd}$$

By initiality of 0, there is only one arrow $?_A : 0 \to A$ in $\mathbf{A}$ and, since L preserves initial objects, the square aside commutes. Thus $(?_A, k)$ is the unique morphism $\Delta(B) \to (A, B', f)$ such that $U_R(?_A, k) = k$. □

Dualising we get immediately the following.

Corollary B.7. *If $\mathbf{B}$ has terminal objects preserved by R, then $U_L\colon L{\downarrow}R \to \mathbf{A}$ has a right adjoint.*

B.1. Slice categories

This section is devoted to recall some basic facts about *slice categories*.

Definition B.8. Let X be an object of a category $\mathbf{X}$. We define the following two categories.

- The *slice category over X* is the category $\mathbf{X}/X$ which has as objects arrows $f\colon Y \to X$ and in which an arrow $h\colon f \to g$ is $h\colon Y \to Y'$ in $\mathbf{X}$ such that the triangle on the right commutes.

$$\begin{tikzcd} Y \arrow[rr, "h"] \arrow[dr, "f"'] & & Y' \arrow[dl, "g"] \\ & X & \end{tikzcd}$$

- Dually, the *slice category under X* is the category $X/\mathbf{X}$ in which objects are arrows $f\colon X \to Y$ with domain X and a morphism $h\colon f \to g$ is an arrow of $\mathbf{X}$ fitting in a triangle as the one aside.

$$\begin{tikzcd} & X \arrow[dl, "f"'] \arrow[dr, "g"] & \\ Y \arrow[rr, "h"'] & & Y' \end{tikzcd}$$

Remark B.9. For every $X \in\ X$ we have forgetful functors

$$\begin{array}{rcl} \mathsf{dom}_X\colon \mathbf{X}/X & \to & \mathbf{X} \\ f & \longmapsto & \mathsf{dom}(f) \\ {\scriptstyle h}\big\downarrow & & \big\downarrow{\scriptstyle h} \\ g & \longmapsto & \mathsf{dom}(g) \end{array} \qquad \begin{array}{rcl} \mathsf{cod}_X\colon X/\mathbf{X} & \to & \mathbf{X} \\ f & \longmapsto & \mathsf{cod}(f) \\ {\scriptstyle h}\big\downarrow & & \big\downarrow{\scriptstyle h} \\ g & \longmapsto & \mathsf{cod}(g) \end{array}$$

We can obtain the slice over and under an object $X \in \mathbf{X}$ as comma categories.

Proposition B.10. *Let X be an object in a category $\mathbf{X}$. If $\delta_X\colon \mathbf{1} \to \mathbf{X}$ is the constant functor of value X from the category with only one object, then $\mathbf{X}/X$ and $X/\mathbf{X}$ are isomorphic to $\mathsf{id}_X{\downarrow}\delta_X$ and $\delta_X{\downarrow}\mathsf{id}_X$, respectively.*

Proof. Let 1 be the only object of the category $\mathbf{1}$ and define functors $F_1\colon \mathsf{id}_X{\downarrow}\delta_X \to \mathbf{X}/X$ and $G_1\colon \mathbf{X}/X \to \mathsf{id}_X{\downarrow}\delta_X$ as follows

$$\begin{array}{rcl} (Y,1,f) & \longmapsto & f \\ {\scriptstyle (h,\mathsf{id}_1)}\big\downarrow & & \big\downarrow{\scriptstyle h} \\ (Y',1,g) & \longmapsto & g \end{array} \qquad \begin{array}{rcl} f & \longmapsto & (\mathsf{dom}(f),1,f) \\ {\scriptstyle h}\big\downarrow & & \big\downarrow{\scriptstyle (h,\mathsf{id}_1)} \\ g & \longmapsto & (\mathsf{dom}(g),1,g) \end{array}$$

Similarly, we have $F_2\colon \delta_X{\downarrow}\mathsf{id}_X \to X/\mathbf{X}$ and $G_2\colon X/\mathbf{X} \to \delta_X{\downarrow}\mathsf{id}_X$

$$\begin{array}{rcl} (1,Y,f) & \longmapsto & f \\ {\scriptstyle (\mathsf{id}_1,h)}\big\downarrow & & \big\downarrow{\scriptstyle h} \\ (1,Y',g) & \longmapsto & g \end{array} \qquad \begin{array}{rcl} f & \longmapsto & (1,\mathsf{cod}(f),f) \\ {\scriptstyle h}\big\downarrow & & \big\downarrow{\scriptstyle (\mathsf{id}_1,h)} \\ g & \longmapsto & (1,\mathsf{cod}(g),g) \end{array}$$

It is now obvious to see that F_1, G_1 and F_2, G_2 are pairs of inverses. □

A straightforward application of Corollary B.4 and lemma B.3 now yields the following.

Corollary B.11. *For every object $\mathbf{X}$, $\mathbf{X}/X$ has all colimits and connected limits that $\mathbf{X}$ has. Moreover such limits and colimits are created by dom_X.*